RESEARCH AND REPORT WRITING

A Practical Handbook for University Students and Beginning Researchers

Prepared by
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2026

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This handbook has been prepared for educational and academic purposes.

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PREFACE

Research and report writing are essential skills for university students and beginning researchers. This handbook brings together foundational concepts and practical guidance for moving from a research problem to a written report. It retains the broad progression of the source manuscript while presenting the material as a standalone book.

The seven chapters address basic concepts of research; literature review and research problems; research design and methodology; sampling, data collection, analysis and presentation; research proposal and report writing; citation and referencing; and ethical considerations. Activities, chapter checklists, self-tests, an answer key, and references support active study and review.

The handbook is intended to be useful both for reading from beginning to end and for consulting individual topics during research planning and writing. Readers are encouraged to attempt the activities and self-tests before consulting the answer key.

Meseret Teshome Rumicha (PhD)

ABOUT THIS BOOK

This handbook introduces the principal stages of academic research and report writing. Chapter 1 presents foundational concepts and types of research. Chapter 2 focuses on reviewing literature, identifying research problems, formulating hypotheses, selecting a topic, and managing research time. Chapter 3 discusses methodology, research design, qualitative and quantitative approaches, validity, and reliability.

Chapter 4 addresses sampling, data collection, data analysis, and data presentation. Chapter 5 considers research proposals, academic reports, report-writing stages, and report structure. Chapter 6 explains citation and referencing across different source types. Chapter 7 addresses ethical considerations in research and report writing.

Activities, checklists, self-tests, an answer key, and a reference list accompany the main discussions. The original assessment questions are retained where available; where the source answer key is incomplete, that limitation should be checked rather than silently filled.

HOW TO USE THIS BOOK

Read each chapter introduction and learning objectives first to identify its focus. Work through the sections in sequence and complete the activities as you encounter them. Use the chapter checklist to review your understanding, then answer the self-test questions without consulting the answer key.

The answer key provides source-based answers where these were included in the original manuscript. Some questions ask you to construct examples of your own; compare such responses with the models and explanations in the relevant chapter. Use the references as a starting point for further reading, and verify bibliographic details before citing a source in your own research.

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CHAPTER 1

Basic Concepts of Research

Introduction

Research involves systematic inquiry into a question or problem, using evidence and an appropriate approach to develop and communicate knowledge. Clear research writing helps readers understand the purpose of an investigation, the reasoning behind it, and the significance of its findings.

This chapter introduces the basic concepts of research and report writing. It explains what research is, considers major research paradigms and purposes, and distinguishes among common types of research and research papers.

★ Learning Objectives

At the end of this chapter you will be able to:

- define Research.
- Describe paradigms of research
- list types of research
- describe the difference between analytic and argumentative papers.
- explain why research is conducted

.

✎ Activity 1 What is Research?

if your tutor just assigned you a research work, or an essay that requires research, or some kind of project that requires research and citing your sources, how do you handle it? Well, stay calm, know you can do it, hang in there and go step by step! This research guide will take you step by step through the process of "doing research" so that you don't get lost along the way.

1.1 Definition of research

✎ Activity 1 How do you define research?

Merriam-Webster, Inc. (1985) defines research in terms of systematic searching or investigation. In academic work, research involves a careful and organized inquiry undertaken to develop, examine, or refine knowledge about a subject.

A research project is more complicated than getting the answers from a book or writing out of your own head. It requires you to keep track of a lot of different things, do some original thinking, look for examples to support your ideas and to put it all together in a neat package to present to an audience (readers).

Some tutors will require research projects and *assume* that you know how to do them. Don't panic if you have never done one. This guide will help you as you go through all the steps. Not only that, it will help you do a really good project.

a research process is recursive, which means that at any point you may go back and redo an earlier step. This is one of the main reasons you need to hang in there and stay organized. Give yourself time at the end of every work session to re-organize yourself for the next time you work

You may be required during your time as a student to write a variety of reports as assessment tasks. A research report is one type that is often used in the natural sciences, social sciences, engineering and psychology. Here your aim is to write clearly and concisely about your research topic so that the reader can easily understand the purpose and results of your research. This chapter will give a general outline of the process and structure of a research report.

1.2 General considerations

1.2.1 Paradigms of Research

✎ Activity 2 What do you think is the meaning of the word paradigms in research?

How many kinds of research paradigms do you know?

The term paradigm denotes pattern or schools of thought, each of which is self-contained with its own methods and theories. In science the term denotes general ways of seeing the world, which dictate what kind of scientific work should be done and what kinds of theory are acceptable. There are three basic research paradigms -- positivism (quantitative, scientific approach), interpretivism, and critical science

1.2.1.1 Positivism

/the scientific approach/ is a doctrine in the philosophy of science, it is characterized mainly by an insistence that science can only deal with observable entities known directly to experience. The positivist aims to construct general laws, or theories, which express relationships between phenomena. Observation and experiment will then show whether the phenomena do or do not fit the theory; explanation of phenomena consists in showing that they are instances of the general laws or regularities

1.2.1. 2 Critical science, or the critical approach, explores the social world, critiques it, and seeks to empower the individual to overcome problems in the social world. Critical science enables people to understand how society functions and methods by which unsatisfactory aspects can be changed. We do not cover critical science in this book.

1.2.1.3 Interpretivism

or the qualitative approach, is a way to gain insights through discovering meanings by improving our comprehension of the whole. Qualitative research explores the richness, depth, and complexity of phenomena. Qualitative research, broadly defined, means "any kind of research that produces findings not arrived at by means of statistical procedures or other means of quantification" (Strauss & Corbin, 1990).

The underlying assumption of interpretivism is that the whole needs to be examined in order to understand a phenomenon. Interpretivism is critical of the positivism because it seeks to collect and analyze data from parts of a phenomena and, in so doing, positivism can miss important aspects of a comprehensive understanding of the whole.

It is this fundamental difference that underpins the approach of researchers in the interpretive tradition and which leads them to use qualitative methodologies and methods. To enable them to enter the everyday social world, (a world where social reality is the product of the way in which individuals negotiate and construct meanings for actions, events and situations) to better understand, describe, make sense of, and develop into theories which are viewed as the socially constructed meanings employed by people in day to day life.

Interpretivism proposes that there are multiple realities, not single realities of phenomena, and that these realities can differ across time and place.

Although acceptance of interpretivism is increasing within human movement sciences, positivism remains the dominant paradigm, as it does in other social science fields

So far we have discussed the paradigms of research. Let us now discuss types of research. Research in general is divided into two main types. These are qualitative and quantitative.

1.2.2 Purpose of research

✎ Activity 3: Why does a researcher conduct research?

A researcher deals with a wide range of associations, from concrete day-to-day activities and problems to a philosophical level of search for truth. The purposes of research are:

- training.
- solving problem and
- generating truth or knowledge.

1.2.2.1 Training in Research

Executing a research in a scientific manner needs training in research methods. This book on research methodology provides the theoretical background of conducting research. This chapter is supplemented by interactive contact activities, primarily to strengthen the conceptual aspect of research methodology. This will also facilitate the right choice of research question, designs, tools and techniques. Hence, training in research methodology is an important purpose of research in online learning.

1.2.2.2 Problem Solving

The second purpose of research is diagnosing and solving the problems in the system. In this case, the system is online learning. While we talk of problem solving, it is necessary to develop a more comprehensive view. For instance, it may also mean a realistic understanding of a situation on the basis of data and statistics. Let us take the case of media utilization for online learning. Many countries invest considerable amount of money on educational mass media, television in particular. The studies that assess the extent of media utilization and also show the differential performance of those learners who use media and those who do not, provide a revised understanding of the role of media in online learning. Thus a series of studies will go into diagnosis and develop better understanding of the practices in a system. This may also include study of socio-economic background of students.

Another set of studies may actually get into experimenting with an innovative solution. For example, a university can model its contact programmes into a two-week or three-week module, tutorial or lecture modes, group interactive mode or group assignment mode and so on. At the end of such

exercises, the college may assess the differential performance of the readers exposed to differential treatment. This will provide the university an empirical basis for designing the contact programmes. Similarly, there are several alternative formats in the medium of television, e.g. lecture, interview, documentation, and drama. The programmes can be developed in different formats and their impact on readers assessed. There can be experiments in the administrative area as well. For example, most universitys send printed instructional material by post, a common practice all over the world. Some of the universities in the western world offer it on-line so that a reader can either read it from the monitor or download the text at his/her work station. the university delivers printed instructional material to the study centers by trucks. Students are advised to come and collect the material from the study centers. The analysis of this practice has shown that it causes minimum loss of mater and is also quite cost-effective.

The important Point is to recognize that research can contribute to solving day to day problems. Most of the researches in online learning can actually take the purpose of problem solving.

It is important to recognize that there is no clash of interests between purposes of research for training and problem solving. Both can go hand-in- hand. This is particularly true for researches where most of the participants are not trained in research methodology but many of them are well experienced in online learning and are very familiar with the problems in the system.

1.2.2.3 Generating Knowledge

Generating knowledge is the biggest challenge in research. Research in search of truth is often classified as pure research in the otherwise controversial pure-applied research continuum. The search for truth, from the angle of research, is the ability to generalize and thus, create knowledge. Such generalizations are derived from occurrences in repeated instances. Generalizations are drawn primarily on two bases, namely, repeated observation in various possible situations and application of statistical designs and analyses where the variables are statistically controlled.

It is quite possible for you to undertake a study that could come close to generalization through application of statistical models and methodologies. Such research, however, will be based purely on quantitative techniques. There are qualitative methods of research as well. Such methods can also be applied in search of truth. Important emphasis, however, is on the use of scientific approach to research.

The scientific methods can be explained in a series of steps, the exact formulation of which varies from one researcher to another; the steps are as follows:

- certain phenomena are observed,
- a problem situation which develops therein, is noted and clarified,
- crude relationships are tentatively identified and elaborated,
- a more or less formal hypothesis is derived,
- a design is developed to test the hypothesis,
- the hypothesis is verified or refuted, and
- the results are subjected to further tests and refinements.

Finally, the conclusions of research are integrated with the existing knowledge of the subject. The process involves such subsidiary steps as:

- review of relevant experience,
- manipulation of factors,
- measurement of the quantities, defining of variables, and
- analysis and interpretation of data.

Scientific enquiry starts with facts and then moves towards theorizing. The primary purpose of the scientific method is to develop a mechanism of organizing the facts, as they accumulate, and become meaningful from the standpoint of their objectives. Through empirical investigations, scientists gather many facts. As these facts accumulate, there is a need for integration, organization, and classification in order to make the isolated findings meaningful.

A theory is a set of related concepts and propositions that offers an organized account of relationships among phenomena and can help explain or predict them (Kerlinger, 1986).

Every Research should be an attempt to solve some intellectual problem. No research paper is successful unless it contributes to the solution of a problem and adds to the knowledge of both the writer and the reader.

1.3 Types of research

✎ Activity 4 how many types of research are you familiar with?

What is fundamental research?

What is applied research?

We can think of two major types of researches: fundamental or basic research, and applied research. This classification is based on the goal or objective of the research.

1.3.1 Fundamental Research

This type of research aims at obtaining the empirical data that can be used to formulate and /or evaluate a theory.

This type of study is not oriented in design or purpose towards the solution of practical problems. Its essential aim is to discover the frontiers of knowledge. Of course, the findings may be eventually applied to practical problems that have social value. For example, advances in the practice of medicine are dependent upon basic research in biochemistry and Microbiology. Likewise, progress in educational practice has been related to progress in the discovery of general laws of learning through basic research, however, is the creation of knowledge solely for the sake of knowledge. Its design is not hampered by considerations of special usefulness of the findings.

1.3.2 Applied Research

Applied research is directed towards the solution of an immediate, specific and practical problem. This is a research performed in relation to actual problems and under conditions in which they are found in practice. Through applied research, educators are often able to solve their problems at the appropriate level of complexity. For instance, in the classroom teaching-learning situation, we may depend on basic research for discovering the more general laws of learning, but applied research must be conducted in order to determine how these laws operate in the classroom. This approach is essential if scientific changes in teaching practices are to be effected.

It needs to be noted that there is not always a sharp line of demarcation between basic and applied research. Certainly, applications of theory help in solving practical problems. We apply theories of learning in the classroom. On the other hand, basic research may depend upon the findings of applied research to complete its theoretical formulations. An experiment in classroom learning could shed some light on a learning theory. Furthermore, observations in a practical situation serve to test theories and may lead to the formulation of new theories.

Educational research is a systematic inquiry into questions and problems concerning education. It may examine teaching, learning, institutions, policy, or other educational processes, using methods appropriate to the research question.

Within the educational researches, there are differences in approach to a research among different disciplines. For example, unlike physical and biological sciences, education and social sciences deal with living humans.

Although problems of discovering principles of human behavior are difficult, they are not impossible. Social scientists will need to carry out observations as carefully as are done in physical sciences. Subjective, qualitative judgments need to be supplemented by more exact, quantitative measurements which are not easy to achieve in the case of human beings. This lack of 'quantifying' and 'generalizing' of data, quite often becomes a drawback in educational research as well. Whereas exact sciences tend to become increasingly quantitative in their units, measures and the terminologies, in social sciences most of the matter is qualitative and does not approve of quantitative statements. We may talk of growing indiscipline, but unless we can measure it, we cannot generalize the concept.

Research in education adopts both quantitative and qualitative techniques. Social sciences have not been able to establish generalizations equivalent to theories of the natural sciences or, to predict events accurately. Perhaps, social sciences will never realize the objective of science as completely as natural sciences do. In fact, there are several limitations involved in the application of the scientific approach in education and the other disciplines of social sciences.

Any of the above two types of research again can be classified as scientific/experimental/ and literary/ analytical/. In the scientific or experimental type it is necessary to get first hand facts, at their source, and actively to go about doing certain things to stimulate the production of the desired information. Whereas, the analytic researcher deals with on either first hand information (Primary sources) or secondary sources. Primary source is the original writings of a person whose thinking, investigation, discovery, life, or imagination is the basis for the material, for example, research reports, philosophical discourses, autobiography, fiction, poetry, analysis of an issue. A secondary source is a work written about someone else's thinking, investigation, discovery, life, or imagination, usually a response to someone else's writing, for example, analysis of a poem, interpretation or explanation of a research project by someone other than the researcher, biography, literary criticism, book review

Scientific research attempts to explain the relationship among phenomena carefully and rules out metaphysical (Ideal) explanations. Fore example why are people poor? In metaphysical explanation, people are poor because God wills it. But for scientific explanation, one has to conduct research to answer this question.

In the scientific research, the researcher must first provide him/herself with a working hypothesis or guess as to the probable results. The researcher then works to get enough facts to prove or disprove his/her hypothesis. S/he sets up experimental procedures which s/he thinks will manipulate the persons or the materials concerned so as to bring forth the desired information. Experiments must be

designed in such a way that there is only one uncontrolled variable. There must also be control groups in which the same variable factor is influenced in different ways or not at all.

The facts are added up at the end of the experiment and the sum is scrutinized to see whether it presents any total picture. Most students are familiar with this method from their chemistry classes; for example, every time two parts of hydrogen are combined with one part of oxygen, under the proper conditions concerning temperature and pressure, the product is water.

now it is time to discuss experimental method, even if we are studying subjects which do not require its frequent use.

Literary or analytical research is a little simpler and it is the type which undergraduates use most in gathering material for reports required in their college classes. This method is more passive since we usually concentrate ourselves on facts or information already available, instead of making our own as we do in the experimental kind. The value of the investigation depends upon the quantity of pertinent material which we are able to dig out. We must analyze these facts and make a critical evaluation of the material. The method is still inductive; that is, we do not arrive at conclusions by means of intuition, faith, or deductive logic, but only on the basis of a careful inspection of the facts available. Our measuring devices are apt to be less concrete than those of the experimental scientist, however, since we make use of such abstractions as values, ethics, and aesthetics. But the measuring instruments which we use should be the best, the most concrete and specific in the field in which we are doing our work. This does not mean either that we never use firsthand information. As will be pointed out in detail a little later, material procured at its original source is always the best and most reliable for the purposes of research.

regardless of the type of research paper you're writing, your finished paper should be a presentation of your own thinking backed up by the ideas or information of others in the field. However, whether your paper is ANALYTICAL (uses evidence to analyze facets of an issue) or ARGUMENTATIVE (uses evidence to attempt to convince the reader of your particular stance on a debatable topic), is definitely going to have a bearing on your strategy. In fact, it will determine your paper's **purpose**. So here is a more thorough discussion of the difference between the two, followed by a concrete example that directly compares the two.

Analytical Papers

✎ Activity 5 What is analytical paper?

"To analyze means to break a topic or concept down into its parts in order to inspect and understand it, and to restructure those parts in a way that makes sense to you. In an analytical research paper, you do research to become an expert on a topic so that you can restructure and present the parts of the topic from your own perspective."

In this brand of research paper, therefore, you go into the researching stage with a specific topic about which you have not made any kind of conclusions. Often you will hear this called your research question**.** Your task is to survey the information and views already out there--both *before* and *once* you become familiar with the topic. That will require critical thinking and reading, plus evaluation of the resources you handle. By the end of the paper you will be able to contribute your own thoughts to the academic discussion by drawing some conclusions about the topic you have just analyzed.

Your task is to survey the information and views already out there- both before and once you become familiar with the topic. That will require critical thinking and reading, plus the evaluation of the resources.

✎ Activity 6 what exactly does critical thinking mean?

A term "critical thinking" has a broad concept that encompasses a lot about college or university academic expectations. But for our purposes it's enough to say that in a *research* or reading context it means not considering any view as "truth" simply because a source has been published or seems to be an expert. It requires you to maintain some objectivity and ask questions to yourself as you read (or watch or listen). In short, a critical eye teaches you to regard anything-- *especially* if it's published or in other media-- as if you're doing a peer edit or with the attitude your tutor will be adopting while marking your paper. No matter what your knowledge level, as a student with fresh eyes and unique experiences, you always have inquisitiveness as a skill; this is how students enter a research community with some authority of their own.

In addition to the concept of critical thinking (which any paper at the university level will demand of you), another widely-used term at the college level which you may or may not be familiar within its academic context, is the term argument. This is the basis of the persuasive kind of research paper.

Argumentative (or Persuasive) Papers

✎ ***Activity 7*** *Dear* learner, What does argumentative (persuasive) paper mean?

How is it different from others?

The term argumen**t** can be defined as "a series of generalizations or propositions, supported by evidence or reasoning and connected in a logical manner, that lead to a justified conclusion. You must sustain your argument by giving evidence and reasons." And persuasion is the art and skill of convincing someone of the credibility of your argument

In direct contrast to the analytical paper, your approach here is to take a stand on an issue and use evidence to back-up your stance, not to explore or flesh out an unresolved topic. We have included an entire step just on this aspect of the research paper writing process, but it's probably worth your while *now* to know that this stance, this debatable statement or interpretation is known as your report.

Argumentative or persuasive papers, as these names suggest, attempts to convince the reader of a debatable or controversial point of view.

Example

For an **analytical** research paper, let's say you have decided to explore **"the purpose of Secondary School Leaving Examination (SSLE)."** In order to investigate the purpose of SSLE, you can conduct a research to locate the fact.

The body of the paper would analyze or break down the topic into three or four "parts" which will later become the main paragraphs of your draft. Perhaps your research helps you discover several purposes to SSLE, with your paper devoting a paragraph to considering each. Or perhaps there's debate among scholars as to the main purpose of SSLE, so you decide to present some of these varying opinions. However you choose to explore the topic, in the body of your paper you'd be using evidence from **primary sources** and expert opinions (**secondary sources**).

Your concluding paragraph(s) would finally incorporate some of *your* critical interpretations of both sources. Here, you'd include a critical evaluation and discussion of your overall findings as well as some conclusions based on the patterns you've researched or detected yourself to make some final comments about the purpose SSLE.

In an argumentative paper you would lay out exactly what you consider to be the purpose of SSLE in a declarative sentence right in the introduction--the statement of the problem. Thus, the form of writing would change accordingly to "the purpose of SSLE is _____ (Identify students who can join higher institutions? to evaluate students' academic performance? and so on and so forth)." look, it ceases to be just a topic and has become instead an interpretation. The course of the paper will develop why you believe--and importantly, why the reader should believe--what you do.

This time, you'll select only that evidence which *directly supports your thesis*. The body of your paper turns into a site for laying out the proof you've collected rather than a canvas for delineating a topic.

Remember that, any form of research paper, be it analytical or argumentative, vary greatly in length and the kind of data it contains. Briefly, the sources you use will be determined by the aims and scope of your report. You may gather data yourself, for example through carrying out interviews or experiments. You will also be looking for relevant secondary data, information that someone else has gathered or produced and that you will find in, for example, books, journals, newspapers, and other reports. Ensure that the information you use is relevant and that you always reference its source.

For example, business firms frequently report to the letter form, one or two pages in length. Banks, insurance companies, and other financial institutions are fond of the short, balance-sheet type of tabulation for their annual reports to their customers and stockholders. Mathematicians prefer to write the results of their investigations in the form of algebraic notation. Chemists report their results in symbols and formulae. Students of language and literature usually prefer to write long and exact papers presenting the critical analysis of some writer, period, or genre, with a liberal sprinkling of quotations from the works of the author under discussion. In the field of education and psychology the favorite form is the report on the results of experimentation accompanied by the detailed statistical tabulations. Clinical psychologists and social pathologists frequently find it necessary to make use of the case-history form.

Have you ever read newspapers? News items in the daily papers are also forms of report writing. They represent firsthand, on-the-scene accounts of the events described or compilations of interviews with persons who were on the scene. Newspaper stories are usually written in a special way, however. First comes the "lead" which summarizes all the information contained in the whole report. In the first paragraph, the most important information is given in detail. The succeeding paragraphs contain material which is progressively less and less important. This procedure is followed for two reasons: the first is that the reader can devote as much time to the article as his particular interests in the subject warrants, but even if s/he reads only the first few sentences s/he will have all the essential information; the second reason is that this form enables the copyreader or the editor to cut the article from the bottom up, depending on the limitations of space, without doing any damage to the structure of the report.

Book reviews which appear in magazines, the Sunday supplements of newspapers, or in the daily book column are another kind of report. They analyze the content of the book and report on the author's intentions, his success or failure in achieving his aims, his, language, his style, scholarship, bias, or point of view. Reviews also tell something about the scope and limitations of the book, the

central theme, and the special excellences and faults of the work. Moreover, it is customary to say something concerning a comparison of this particular book with other books on the same or similar subjects and with other books by the same author.

● Conclusion

Research is a formal paper in which the researcher draws on many sources to present new information. A research project is more complicated than getting the answers from a book or writing out of your own head. Therefore, a research process is recursive, which means that at any point you may go back and redo an earlier step.

Research can be conducted based on different scientific inquiries. Scientific enquiry starts with facts and then moves towards theorizing. The primary purpose of the scientific method is to develop a mechanism of organizing the facts, as they accumulate, and become meaningful from the standpoint of their objectives. Through empirical investigations, scientists gather many facts. As these facts accumulate, there is a need for integration, organization, and classification in order to make the isolated findings meaningful

The scientific methods, in research, are implemented through research paradigms.

The term paradigm denotes pattern or schools of thought, each of which is self-contained with its own methods and theories. In science the term denotes general ways of seeing the world, which dictate what kind of scientific work should be done and what kinds of theory are acceptable. There are three basic research paradigms -- positivism (quantitative, scientific approach), interpretivism, and critical science

A researcher deals with a wide range of associations, from concrete day-to-day activities and problems to a philosophical level of search for truth. The purposes of research are:

- training.
- solving problem and
- generating truth or knowledge

We can think of two major types of researches: fundamental or basic research, and applied research. This classification is based on the goal or objective of the research.

Fundamental Research aims at obtaining the empirical data that can be used to formulate and /or evaluate a theory. Applied research is directed towards the solution of an immediate, specific and practical problem. This is a research performed in relation to actual problems and under conditions in which they are found in practice.

Any of the above two types of research again can be classified as scientific**/**experimental**/** and literary/ analytical**/**. In the scientific research, the researcher must first provide him/herself with a working hypothesis or guess as to the probable results. The researcher then works to get enough facts to prove or disprove his/her hypothesis.

A researcher can present the facts in analytical or argumentative ways. To analyze means to break a topic or concept down into its parts in order to inspect and understand it, and to restructure those parts in a way that makes sense to you. In an analytical research paper, you do research to become an expert on a topic so that you can restructure and present the parts of the topic from your own perspective. The term argument can be defined as "a series of generalizations or propositions,

supported by evidence or reasoning and connected in a logical manner, that lead to a justified conclusion. You must sustain your argument by giving evidence and reasons." And persuasion is the art and skill of convincing someone of the credibility of your argument

In direct contrast to the analytical paper, your approach here is to take a stand on an issue and use evidence to back-up your stance, not to explore or flesh out an unresolved topic.

Summary

Research is like a crossword or a jigsaw puzzle. It is meaningless until the various pieces begin to fall neatly into place. When the over-all pattern or picture becomes visible, certain creative satisfactions result.

A capable writer will use every means at his disposal to maintain interest and to simplify terminology. Illustrations and examples are most effective when they are based on common experiences. Avoid abstract terminology and technical jargon. Make what you have to say as simply as possible.

Research is not a substitute for your own careful thought and analysis. It is a formal paper in which draws on many sources to present new and interesting information. Based on the purpose of the research, there are two main kinds of research: analytic and argumentative. While conducting research, you, as a researcher, can face a problem on how to use sources and still write something new. You might try to judge or challenge old ideas, or concentrating on a controversy between sources. As much as possible try to Use both primary and secondary sources to show that you know what has been said and give your opinion about it.

Evidence comes from either primary or secondary sources. The primary source is the text on which you are commenting, or documents that deal directly with your topic. Secondary sources are opinions or interpretation of others on the topic (your research finding itself becomes a secondary source, should anyone wish to quote it).

✓ Chapter Checklist

Put a ‘ ’ mark against each of the following statements that you are able to answer.

I can:-

- Define research
- Describe about research paradigms.
- List types of research
- Describe the difference between analytic and argumentative research.
- Define Research.
- Describe research paradigms.
- Explain why we conduct research?
- Explain primary and secondary sources

◆ Self-Test

Part One:

Instruction: write a complete answer for the following questions.

1. What is research?

2. Why do we conduct research?

3. What are the purposes of research?

4. Write the main types of research.

5. What are the differences among main types of researches?

6. What is primary source?

7. What is secondary source?

8. What is the primary purpose of scientific research?

9. What is positivism?

10. What is an argument?

Part Two: Write true in the statement is correct and false if the statement is wrong.

1. In an analytical research paper, you do research to become an expert on a topic.

2. Scientific research attempts to explain the relationship among phenomena carefully and rules out metaphysical (Ideal) explanations

3. For a researcher it is advisable to use only primary sources so as to avoid confusion.

4. The scientific approach is a doctrine in the philosophy of science, it is characterized mainly by an insistence that science can only deal with observable entities known directly to experience.

5. Basic research is directed towards the solution of an immediate, specific and practical problem.

CHAPTER 2

Literature Review and Research Problems

Introduction

After examining the basic concepts and types of research, the next step is to identify a worthwhile problem and understand what is already known about it. A focused literature review helps researchers locate relevant evidence, refine their questions, and justify the need for a study.

This chapter discusses how to review the literature, select and narrow a research topic, formulate a research problem, and develop research questions or hypotheses.

★ Learning Objectives

At the end of this topic you will be able to:

- Write criteria about researchable topic
- Explain procedures in conducting research.
- Identify the basic consideration for research topic selection.

2.1 Review of the literature

✎ **Activity 8** What is review of literature?

Why do you (as a researcher) need to review literature?

How do you select researchable topic?

Review of the literature is a survey of the scholarly work in a particular subject area; also called "literature review". Searching the literature is the method used to learn new facts and principles through the study of documents, records and literature of the field. This research method is valuable in all fields whenever knowledge and insights into events of the past are required. Since human knowledge is a structure which grows by the addition of new material which has previously been gained, here is little chance of making a worthwhile new contribution if an investigator is completely ignorant of what is already known about his problem. Therefore, before beginning an investigation, he must find out what has been written thus far about the problem. Library search can help in at least two broad ways:

a) To find out if the information which is the object of the proposed research is already available.

b) To acquire a broad general background in the area of research.

Clues leading to the desired library source

1. The card catalogue - by either author, title or subject

2. Literature guides and sources of published materials – these are accumulation of bibliographies in different subjects.

This may appear in the form of abstracts.

- Yearbooks, journals and monographs (detailed papers on specific subjects) are often important sources.
- Computerized information services- this resource makes available quickly and efficiently the recorded documentation of the published literature of almost every field.

When one stats his library investigation it is often advisable to proceed from the more general source (such as an encyclopedia or a hand book) to a more specific source like a monograph, which pinpoints to the most relevant source which may enable one to easily pick up reference and facts.

A thorough understanding of known facts and ideas in the field of your interest as a researcher constitute the first and the most important step in selecting a problem for your study.

A thorough knowledge of the research studies conducted in the field provides you with details about the problems which have remained unresolved. A survey of suggestions for further research given at the end of research reports and reviews of research would help you to get an idea about the gaps which exist in the knowledge pertaining to your field of educational research. Periodicals and bibliographies of educational research are helpful in keeping you informed about the research going on in the field in which you are interested and show competence.

The choice and identification of a suitable problem is usually difficult. For a beginner to identify a research problem is always a very difficult task. This may be due to his/her limited knowledge of research processes or his/her unpreparedness for identifying the problem. He/she may not be familiar with the problems in the field which need solution through research. Even an experienced researcher finds it difficult to list all the problems that need to be research. Each researcher selects a problem because of his/her own needs and purposes. The following, are, however, some important sources which would help you identify a suitable and significant problem.

Professional experience

Professional experience of a researcher is one of the most fruitful resources which could suggest research problems. For example, if you are a classroom teacher, you may encounter a number of problems in your day-to-day classroom activities. In the classroom, there is a dynamic interaction between you as the teacher and your learners, between learners themselves, and between learners and learning materials. Such interaction provides you with a variety of problems to be solved through research. You may also be confronted with a number of behavioral problems in and outside the classroom. You have to make decisions about the probable effects of classroom instruction on your learners' behaviors so as to establish a relationship between instructional objectives, learning experiences and learning outcomes in terms of behavioral changes in the learners. The learner behavior may be analyzed in terms of their academic achievement, interests, attitudes, motivation, values and some personality characteristics. You have to take decisions about the effectiveness of certain innovative teaching methods and techniques so as to base future classroom instructions upon empirical evidence rather than upon your subjective opinion.

The teacher may wish to get answers to certain important and significant questions pertaining to certain components of the teaching learning process. What are the organizational or management techniques that are employed? How are learning materials presented? How do students, teachers and parents feel about the use of certain innovations in the classroom? What out –of-school activities affect students and the teaching learning process? How does the isolation of a distance student affect his/her progress in a curse?

The educational administrators and planners may find subjects of research in the areas concerning decision-making, scheduling, teacher recruitment and placement, instructional supervision, and several other matters with which they are concerned.

Contacts and discussions with research oriented people are also helpful to researchers in identifying crucial problems and issues concerning education. Participation in conferences, seminars and workshops and listening to the learned speakers are very helpful in locating research problems the Council in India has identified some areas of research. They also provide financial aid for doing research in those areas. Such organizations are a good source for finding a research problem.

Professional Literature

The study of professional literature will not only expose you to pressing research problems but will also suggest the way in which research is conducted. Research reports, bibliographies of books and articles, periodicals, research abstracts, yearbooks, dictionaries, research guides etc, would suggest areas in which research is needed. Some specialized sources are:

- Encyclopedia of Educational Research,
- Dissertation Abstracts International
- Psychological abstracts and similar publications are rich sources of research problems.

Some other publications are:

- Research Needs in the Study of Education (1968)
- Second Survey of Research in Education
- Fourth Survey of Research in Education (1991), and
- Fifth Survey of Research in Education (1997)

The above publications are exclusively devoted to identifying and bringing into sharp focus the varied research needs in the different areas and aspects of Education. There is another type of resource in print-those that review research on specific areas, example, Research on Open and. Besides these computerized databases are important sources of educational research. Most prominent amongst them is the Educational Resources Information Centre (ERIC) database. International Council for Distance Learning (ICDL) database provides information about research literature on Open and.

All published research reports generally conclude by making suggestions about further research. These suggestions are helpful to researchers for making decisions about methods and procedures employed by other researchers in similar studies with different samples.

Many research studies are criticized for weaknesses in design, treatment and analysis of data, contradictions and inconsistencies in the results, and so on. Thus, researchers can make significant contributions not by repeating these studies, but by making necessary modifications in the design and procedures so as to correct the inconsistencies found in earlier works.

Inference from Theories and Other Sources

Another important source of identifying the research problems lies in the generalizations that are drawn from various theories pertaining to education, psychology, sociology, etc., known to the researcher.

Purpose of Theory

There are several purposes to be served by a theory in the development of science. We shall briefly consider three of them here. First, theory summarizes and puts in order the existing knowledge in a particular area. It permits deeper understanding of data and translates empirical findings into a more easily retainable and adaptable form. The theory of oxidation, for instance, places into focus many of the chemical reactions common to everyday life.

Secondly, theory provides a provisional explanation for observed events and relationships. It identifies the variables that are related and the nature of their relationship. A theory of learning, for example, could explain the relationship between the speed and efficiency of learning and such other variables as motivation, reward and practice.

Lastly, theory permits the prediction of the occurrence of phenomena and enables the investigator to postulate and, eventually, to discover hitherto unknown phenomena. At the time when the 'Periodic Table' was being completed, for instance, certain gaps were noted in the sequence of the elements. Since theory provided that, there should have been no gaps, scientists were spurred on to look for the other missing elements. In time, these were found, anticipated by theory. Theory, therefore, stimulates the development of new knowledge by providing the lead for further inquiry.

For example, the application of general principles involved in various theories to specific problems of makes an excellent starting point for research in this area. Various theories of personality, intelligence, motivation, etc. are helpful in identifying problems pertaining to classroom situations and practices. It is only through research that we can profitably test the validity, scope, and practicability of various theories in educational situations.

Technological and social changes and curricular developments constantly give rise to new problems and issues for research. Such innovations as teaching by television, programmed instructions, computers, and other hardware and software techniques need to be carefully evaluated through the research process.

2.2 Research problems and hypothesis

✎ Activity 9. What is hypothesis?

How do you formulate research hypothesis?

What is research problem?

Hypothesis
A hypothesis is a testable statement. Research hypotheses fall into two categories, the experimental hypothesis and the correlational hypothesis. An experimental hypothesis (H1) is used in experiments! It is thus a testable statement of cause and effect between two variables called the independent variable (IV) and dependent variable (DV). An example of a research/experimental hypothesis is H1: 'That gender has an influence on perceived intelligence.' Another would be HA: 'That alcohol consumption affects driver's reaction time.' Etc.

A hypothesis, when accepted, explains a small number of facts and the relationship between them. Generalization, as the term denotes, is a hypothesis based on broader phenomena.

Characteristics of a Good Hypothesis

There are some important aspects to be looked into judge the worth of a hypothesis in research. A good hypothesis must be:

i) Consistent with know facts and theories, and might be even expected to predict or previously unknown data,

ii) Able to explain the data in simpler term,

iii) Stated in the simplest possible terms, depending upon the complexity of the concepts involved in the research problem, and

iv) Stated in a way that it can be tested for its being probably true or probably false, in order to arrive at conclusion in the form of empirical or operational statements.

Formulation of Hypothesis

Hypotheses are formulated to explain observed facts, conditions, or behaviour and to serve as a guide in the research process. The statements or tentative generalizations which constitute hypotheses are partly based on facts and explanations, and partly conceptual.

Your information search at this stage involves getting a "bird's eye" view of possible topics, browsing for ideas and finding out what kind of sources might be available to you on various topics

2.2.1 Considerations for research topic selection

✎ Activity 10. What should be taken into consideration to select a topic?

identifying something to write about is the first step in writing a report. In some situations, you might be assigned a topic or given a range of topics from which to choose. In other situations, the choice may be completely up to you.

The best way to begin searching for a topic is to conduct some initial research. One way to begin is to review your course textbook(s) for topics that are of interest to you, but might not be covered in great detail during the semester. The table of contents and subject index can help you identify components of a topic. The author index can help you identify leading thinkers in the field. Next, extend this initial topic search by conducting a more complete assessment of the different topics you have in mind with a more complete search using references available for you. As you begin your search for a topic, keep the following in mind:

A. Select a topic that is of interest to you.

Dear Colleague, Nothing is more tedious than researching and writing about a topic that is of little interest or relevance to you. Keep in mind that you'll spend a large amount of time on the topic you select. If you've been assigned a topic, find a way to make it interesting and relevant by applying it to areas such as your career goals, political points of view, hobbies, etc.

B. Select a topic for which you have some background knowledge.

Learning about a topic or subject in which you have no prior knowledge often requires learning the terminology, jargon, and/or lexicon of the subject. While it is interesting to learn about new things, make sure that you have the background knowledge necessary to comprehend the articles and books you will be reading. When you decide to pick a topic for research you must see the Feasibility of the study.

C. Select a topic for which you will be able to find references.

If a topic is too current, you may have difficulty finding the references you need to write the report. If the topic is outdated, you may have difficulty finding references that are current. Therefore, as you conduct your initial topic search, make sure you will have access to enough references to allow you to complete the report.

✎ Activity 11 To cheek your understanding of this topic Write a sentence or two about what your topic will be and why you are choosing it.

D. **Understand the Audience**

Along with identifying a topic for your report, you should begin to learn more about the audience who will read your report. In some instances, your tutor may be the only member of the audience. In other situations you might be asked to write for a specific audience. Use the following questions to help you understand the needs of your audience. Who will read the report? Will the report be read by adults, children, members of a certain profession, other students, a potential employer, a civic group?

For what reasons will different individuals read the report? Will they read it for pleasure, or do they require the information to help them accomplish a goal?

What are the expectations of the readers? Do they expect this paper to provide them with all the information they will need, or will they read this paper in conjunction with other resources?

How much do the readers of the report already know about the topic? To what level of detail should the report be written so that it is not too basic or too advanced for the audience?

What are audience members' prevailing attitudes about the topic? Does the audience generally agree with the position that you will argue or do they support opposing positions?

E. Cost

All research involves some expenses even if they consist mostly of your time and effort.

Even when the research is done for your employer, it is important to know the likely cost. If you expect them to cover costs, you must tell them what the likely expenses will be. If you hope to have an external agency support your research, they will need quite specific cost information.

As costs are identified, the research design may need to be altered to make the project more affordable. When the study is completed, you will probably need to be able to report the actual expenditures and compare them with the anticipated costs [your budget].

When to consider costs

During the planning stage, consider the cost of professional time, clerical support time, and consultants.

During the setup stage, consider the cost of research design, literature review, creating the problem statement, creating forms, and the like.

During the data collection stage, consider professional time, clerical or student assistant time, travel, meals, postage, printing, telephone, and computer-related expenses.

During the data reduction/analysis stage, consider professional, clerical or student assistant time plus computer time and statistical assistance.

During the reporting stage, consider those expenses related to writing, revising, printing, and distributing. In a few cases, PAGE CHARGES may also need to be considered.

✏️ Activity 12: Think about the points above. Choose a topic and identify its intended audience. Then write a few sentences explaining how you would adapt your communication to that audience.

2.2. 2 Selecting a topic

The proper selection of a topic of your research task requires sound thinking and, a little intuition. Some phases which will uncover a field of investigation not adequately surveyed previously. In order to do so, you must first examine carefully all the available bibliographies of work already complete or still in progress to make sure that the topic you have in mind has not been satisfactorily handled by someone else. By so doing you avoid needless duplication and waste of effort.

Collecting information with no particular plan or purpose and then hoping to transform it into a research project is not likely to succeed. Often, the data collected is the result of some sort of convenience sample and may not represent a typical or representative situation.

Taking an available collection of data and trying to identify a research question that it might answer may seem practical, but this approach can still be problematic. Ideally, we begin with a problem (the dependent variable) and go from there to consider likely causal variables (the independent variables). Using locally available data may solve the several problems related to data collection, but this approach works well only if the data is related to a problem that you care about.

A more common pitfall associated with senior essay, thesis and dissertation research is to select a genuinely important topic, but one that is so large or amorphous that it would require a life's work to make sense of it. With the senior essay, thesis and the dissertation, the main point is to develop one's research skills and experience and the second point is to create a publishable paper. Smaller, bit-size research is much more likely to be successful. Build on success with your next research project.

Research assignments are given to you as much for the inherent training and practice as for the actual achievement. In many cases, as a reader, you will be given a list of suggested topics to choose from. You will, nevertheless, have to limit, expand, or otherwise alter the suggested topic in order to meet the requirements. When the choice is left to you, try to tie the general subject in with some phase of your particular interests. In the final selection of a specific topic you must set its boundaries so that you can complete a full, thorough, and accurate research project within the limits of the time available and the requirements concerning length. Don't try to do too much!

For instance, if your topic concerns philosophy, don't make your final choice "Eighteenth-century Philosophy." You might need a couple of years and four or five hundred pages to make a thorough investigation and to write an accurate report on that subject. If, later on, you find that you cannot dig up sufficient information on that phase of your subject, expand your topic as necessary may require.

Remember that you are only selecting a topic in this first instance. The final title of your report probably will not be the same as the original designation of your topic. The selecting and wording of the title are among the last things you will do. Titles, in so far as possible, should be chosen so as to present the exact nature and extent of the contents of your report.

The title of a research should be clear and concise. It should communicate to potential readers what information they can expect to find. Dear Colleague, remember that titles which promise more than they deliver deceive the reader and justifiably are severely criticized.

Creating a title before the report is written maybe used as a guide in the writing of the report. Since a research project may investigate several problems simultaneously, selecting a title in advance of writing the report assists in separating the relevant from the irrelevant. Frequently, researchers create a tentative title before writing the report and then revise it afterward.

Titles vary with the medium of the report and the intended audience. An examination of reports written on topics similar to yours and with a similar audience will provide clues as to proper title construction and format. Basically, there are two types of audiences: academic and nonacademic. Titles vary with the type of audience you are trying to catch.

If your audience is academic, you should create a title which generalizes the nature of the problem you investigated and the findings you have obtained. This will assist potential readers in locating your research among that of thousands of others which are competing for their attention. It will also assist prospective readers in bridging the connection between your research and others on related topics.

If your audience is nonacademic, you should avoid abstract professional language both in the title and elsewhere in the report. There are two reasons for this. First, such language is more likely to serve as an obstacle than an aid to communication to readers outside the profession. Second, nonacademic readers are more likely to be interested in the specific problem and findings of the research than in its implications for theory or broader problems.

Narrowing your research Topic

The first step in writing an essay is finding something to write about. Whether you are working from a list of assigned topics or selecting your own, try to find something which sparks your interest: because your commitment will also help you write a more convincing report. Some preliminary reading may help determine how deep your interest goes, as well as letting you know what kind of material will be available as you write. Once you have selected a topic, the next thing you have to do is narrowing it so as to make it manageable.

A common problem of beginning writers is choosing a topic that is too broad for their purposes. General words such as "media," war," "life," or "nature" are often incorrectly used as if they were topics. However, students often begin to write with nothing more in mind than a general concept, and the result is a vague and generalized. If you start with a broad area, concentrate on narrowing your subject„it will also help you deal with your topic within the purpose of the research and the time you have been given to complete it.

You can narrow your topic by considering a particular approach to the subject, or a sub-topic within it. You might ask yourself key questions, such as the following:

War

- Am I writing of one particular war or of war in general?
- Which war do I wish to write about? WWI? WWII? The Gulf War? "War" taken more metaphorically, between the sexes, siblings, or members of different races?
- Am I concentrating on the history of the war itself, or its causes or outcome?

- What specific events or examples will illustrate my points?

In deriving a workable topic from your subject, be careful not to narrow it too far; your topic must provide scope to develop a sustained presentation and argument.

General subject: Media

Narrowed topic: news papers

Still narrowed topic: private news papers

Specific topic: How commercials manipulate their audience

Look at the following example where a broad topic is narrowed.

- **Broad**: The Ethiopian Secondary School Education has many problems.
- Narrow: The primary problem of Ethiopian Secondary School Education is the lack of well educated teachers.

You can revise your statement whenever you want to while you are writing your essay. Writers often discover what their real purpose and point is in the process of putting their thoughts into words and then reading what they've written.

2.2.3 Stating the research problem

✎ Activity 13 Do you know how to state research problem(s)?

A research problem may be stated as a hypothesis or in a number of research questions.

A hypothesis is a conjecture or proposition about the solution to a problem, the relationship of two or more variables, or the nature of some phenomenon Kerlinger(1986)

Hypotheses are usually used in quantitative/scientific type research.

There are two general types of hypotheses.

The first type of Research hypotheses are conjectures or **propositions** about the solution to a research problem. They define what has to be investigated.

Example: Laboratory instruction enhances understanding of scientific processes over an instructional approach limited to theoretical problem solution.

The second type is Statistical hypotheses: they are stated in **statistical** terms. It is a statement about one or two measures of the population under study.

Example: The mean test score of the third-year students at College “A” equals to the mean test score of the third-year students at College “B”.

Dear Colleague, The latter is an example of a hypotheses stated in null form. That is no difference in the means is hypothesized. It is a hypothesis of no difference or no relationship. Statistical hypotheses usually take this form.

If your review has been thorough, a statement of the problem should flow naturally from the previous discussion. Readers should be given a succinct, but clear idea of the research problem as you perceived and structured it. Many of the research problems will be solved if you have a direct, clear and specific statement of the problem to be investigated. A good statement of the problem will help

you to organize your report and give it direction; it is the central idea around which the rest of the research work is built.

Formulation of Research Questions

One of the key questions for researchers, regardless of their area of specialization, is to formulate a relevant **research question** and to select the appropriate research tool(s) for the specific question they are investigating. How can one do this? Let us look first at the problem of formulating research questions.

What do you want to know? Black (1993) considers that research questions sometimes are "too vague and do not provide sufficient direction for the research effort". He cautions that this type of research questions often get generated out to oversee research in very general terms like..."the Cause of Poverty...or Mathematics Teaching in secondary schools".

Kerlinger (1986) states that one can attempt to evaluate the quality of a research question by considering these three points:

- The research question expresses a relationship between variables.
- The research question is stated in an unambiguous form and
- The research question might be tested empirically

Although these three points raised by Kerlinger(1986) might be contested on certain grounds within the qualitative research tradition, particularly the point on the need to express "a priori" relationships between variables these three points still provide a good starting ground for thinking about ways of formulating and eventually evaluating research questions because even within the qualitative tradition of research, it is not possible to pose relevant research questions which can be empirically examined within field-work without having a broad contextual understanding of the way in which other researchers have examined the same or related research questions. In other words it is not possible to formulate "new" and relevant research questions" without a strong familiarity with the scholars within the broad research area and within that particular field. Therefore "old" research influences the belief structures of researchers formulating "new" research questions.

Usually researchers in the social sciences (regardless of their methodological tradition) state their research questions within a context that provides the justification for the research to be carried out, or for the research question to be stated in the particular terms in which it is stated Literature reviews provide the major sources for contextual justification of the research question. One of the functions of the literature review is to prepare the reader to understand the purpose of the proposed research and within that function; the researcher use a literature review that persuades the readers that there is a need for the proposed research to be carried out. Therefore it is important when evaluating or proposing research questions to look not only at the question itself, but look at the justification presented as a context for the research. In many cases this might involve looking at other research results quoted in the literature review in order to have a firm understanding of the grounds for the justification of the research question. In other words look at the relevance of the research question in relation to the context and the field of study.

When a researcher expresses the possible relationships between variables before or during fieldwork, in both cases, the interpretation of the data provides the justification for relationships to be expressed and analyzed. Qualitative researchers prefer to enter fieldwork without preconceived ideas about possible relationships. Relationships will be uncovered/discovered as part of the field-

work (Flick, 1998). None-the-less it is possible within the qualitative research tradition to approach the formulation of research questions in a somewhat systematic fashion. For example, Mason (1996) sees the formulation of research questions as the statement of "intellectual puzzles". She states that: *"....different theoretical and intellectual traditions in the social sciences are preoccupied with different kinds of intellectual puzzles and consequently different kinds of social explanations."*

She encourages students and practitioners of qualitative research to seek research questions using a typology of puzzles, i.e. developmental, mechanical or causal puzzles. She argues in favor of this typology as an aid to research question formulation because *"thinking in terms of puzzles"* focuses the mind on *research questions* and explanations.

Typology of "Intellectual Puzzles" Mason (1996)

Type of Puzzle	**Example of Research Question**
Developmental	"*How and why did **x** or **y** develop?"*
Mechanical	*"How does **x** or **y** work?"* or *"Why does it work this way?"*
Causal	*"What influences does **x** have on **y**?"* or *-"What causes **x** or **y**?"*

If your project involves testing hypotheses, they will be formally stated in this section. You must formally define the dependent, independent, and control variables. In addition, it must be clear to the reader why the predicted relationships are expected and why you are controlling for the included control variables. A good review of the literature will have provided a firm foundation for these tasks.

For example, your review of the literature might have reported that there are several well established causes of voting. But for some reason(s) (e.g., hunch, logical reasoning, deduction from some theory, partial results in some previous study, etc.) you believe yet another variable is related to whether or not people vote. But, before your research project testing whether this additional variable is related to voting is taken seriously, you must control for the well know causes covered in the review of the literature. Thus, your statement of the problem must indicate how the new variable you are testing fits into the relationships that have already been established by previous research. Is the general causal model additive or interactive, direct or indirect? By carefully specifying what all the relationships are thought to be, you will greatly simplify your task.

2.3 Time management in conducting Research

Research reports should be the product of painstaking, accurate, inductive work. Intuition has only one place in such labors, but it is an important place- and that is in the tentative selection of fruitful field for investigation.

When you try to determine how long your research paper is going to be, remember that, it has to be “long enough to cover the subject, but short enough to maintain interest.”

Everyone who is engaged in preparing a research paper should make some rough allotment of the time available for the various phases of his work. Percentages are all that is necessary at first; later on they can be translated into the actual number of days you are going to set aside for each portion of your work. Some phases of the task require only a few moments to complete; two or three others will consume the bulk of your time.

The value of small but consistent units of work over an extended period of time cannot be overemphasized. Although we live in an age when hydrogen bombes destroy cities and civilization in the twinkling of an eye, we should not lose sight of the fact that it still remains true that small drops of water, falling continually over long periods of time, will wear away the hardest rocks. Many a great book has been composed in the vagrant half-hours snatched from other duties.

Here is a convenient guide line which may be of some use to you. Divide your time so that roughly two thirds will be devoted to the obtaining of information and the other third to the actual writing of the report. But, of course make minor readjustments in your schedule from time to time to suit your needs as your work progresses.

Part I. Gathering the Material

a. Selection of a topic

b. Background reading

c. Preparation of a tentative bibliography

d. Preparation of a tentative outline

e. Organization of notes

Part II. Writing the Report

a. Logical analysis of the subject matter

b. Preparation of the final outline

c. Preparation of the rough draft

d. Rewriting and polishing.

e. Preparation of the final bibliography

f. Writing the final draft

Once you have selected your research topic, the second step is to consult all the available bibliographical aids which cover your field. If you are building up a background in some subject about which you have little previous information, it is suggested that you keep on reading and reading so that you can enrich your paper.

Preparing the preliminary Outline When you have the important features of your topic firmly fixed in mind and a list of materials which will probably provide the kind of information you are seeking, it is time to prepare a rough outline.

As the main ideas come to your mind, note on paper the main ideas which you plan to discuss. When you are satisfied that you have captured them all, think through your whole problem very carefully and then rearrange your subject headings on the basis of logical organization. Put each heading in the best possible place, taking into consideration its relationship to all the other headings.

After you have done all these things, it is a good idea to arrange for a conference with your tutor. He/she probably will want to discuss with you the entries in your tentative bibliography, out line and to make suggestions which will aid you in your work.

The following is a list of points to be checked when you are evaluating material:

1. Accuracy and dependability

2. Up-to-dateness
3. Suitability for specific requirements
4. Authority
5. Scope
6. Treatment and style
7. Arrangement
8. Illustrations, tables, and diagrams
9. Author
10. Publisher

Using these and other criteria you can ask yourself questions concerning the work at hand. The answers will provide your evaluation of that particular material.

Dependability can be checked by looking up some particular fact or thing to which you feel you already have the correct answer and by then comparing the information contained in the book with your own. If you discover one discrepancy, the chances are that there are several, and if there are several, the odds are that the work is generally unreliable.

Obviously, the copyright date gives some clue as to the up-to-dateness of the information in a book. None of the information can be more recent than the latest copyright date. However, mere up-to-dateness is not the only consideration in judging the merits of a book. Dear Colleague, Bear in mind the fact that many an old work, regarded as a standard in its field, may be more accurate and may provide a clearer presentation of the fundamental elements of the subject than a newer one; it may also be a better source, for the material which it does include, than a recent book which contains all the new information.

The information which is given on the title page of a book can frequently be of considerable useful to you. It usually summarizes the contents of the book or gives you a more exact interpretation of the scope of the work. However, be careful that you are not deceived by titles which include the words “American,” “New,” “Dictionary,” or “Encyclopedia.” Check to see that they are what they say they are. These words are frequently misused. One more thing which may be indicated on the title page is the author’s or the editor’s background.

Introductions, forewords, and prefaces are important because they indicate an author’s intentions, the plan he has followed and the limits of the inclusiveness of the work. They also explain why certain types of information have been excluded. Sometimes they give a brief summary of the contents, acknowledge indebtedness to others, and summarize the principal sources of information on which the present work is based.

Indexes are also extremely important; a good, minutely analyzed index will enable you to put your finger on desired information at the moment you want it. Appendices often contain valuable information in the form of reprints of documents or other source materials, information and statistics in tabular form, and many additional types of information which do not fit easily into the logical pattern of the main body of the book.

The importance of good bibliographies cannot be overemphasized. The best are those which are selective and annotated and give complete information.

The most recent information usually comes to us in the form of periodical articles, monographs, and pamphlets. Since good books are in preparation for such long periods of time, the reasons for this are fairly obvious. Reports of experiments, documents, letters, diaries, official reports, and daily newspapers are among the most important and rewarding sources of material. It is wise to read the current periodicals covering the field in which you are working and to keep a clipping file. Newspapers are very fond of the latter method and all the large ones maintain extensive clipping files which, in newspaper parlance, are called “morgues.”

.

Unprinted sources are also vital. Most persons are inclined to overlook the fact that a great deal of knowledge is not recorded in any printed form at all, but is locked up in someone’s head. There are two important ways to get at such information. These are the questionnaire and the personal interviews. Both are used frequently in qualitative research types.

To sum up, you have to design a procedure which helps you as a guide to the final findings. Once the direction is set properly, then you can conduct your research step by step and will not worry much about the validity and reliability of your research.

Searching the literature is the method used to learn new facts and principles through the study of documents, records and literature of the field. The study of professional literature will not only expose you to pressing research problems but will also suggest the way in which research is conducted.

. Library search can help in at least two broad ways:

- To find out if the information which is the object of the proposed research is already available.
- To acquire a broad general background in the area of research.

The choice and identification of a suitable research topic is usually difficult. For a beginner to identify a research problem is always a very difficult task. This may be due to his/her limited knowledge of research processes or lack of experiences for identifying the problem.

Researchers can use different sources so as to help them easily identify a research topic. Professional experience of a researcher is one of the most fruitful resources which could suggest research problems. Contacts and discussions with research oriented people are also helpful to researchers in identifying crucial problems and issues concerning education. Another important source of identifying the research problems lies in the generalizations that are drawn from various theories pertaining to education, psychology, sociology, etc., known to the researcher.

The best way to begin searching for a topic is to conduct some initial research. One way to begin is to review your course textbook(s) for topics that are of interest to you, but might not be covered in great detail during the semester.

As you begin your search for a topic, keep the following in mind:

- Select a topic that is of interest to you.
- Select a topic for which you have some background knowledge.
- Select a topic for which you will be able to find references.
- Understand the Audience

➢ Cost

A hypothesis is a testable statement. Research hypotheses fall into two categories, the experimental hypothesis and the correlational hypothesis. Hypotheses are formulated to explain observed facts, conditions, or behavior and to serve as a guide in the research process.

The first type of Research hypotheses are conjectures or propositions about the solution to a research problem. They define what has to be investigated.

The second type is Statistical hypotheses: they are stated in statistical terms. It is a statement about one or two measures of the population under study.

- The research question expresses a relationship between variables.
- The research question is stated in an unambiguous form and
- The research question might be tested empirically

One of the functions of the literature review is to prepare the reader to understand the purpose of the proposed research and within that function; the researcher use a literature review that persuades the readers that there is a need for the proposed research to be carried out

A sound literature review helps researchers understand existing knowledge, identify gaps, and refine a feasible research problem. A clear topic, focused research questions, and—where appropriate—testable hypotheses provide direction for the study. These decisions should be grounded in relevant literature and aligned with the purpose of the proposed research.

- Conclusion

Summary

Once you have a topic, you will have to decide what the main point of your paper will be. This point, the "controlling idea," becomes the core of your research statement and it is the unifying idea to which you will relate all your sub-topics. You can then turn this "controlling idea" into a purpose statement about what you intend to do in your paper.

- Look for patterns in your evidence
- Compose a purpose statement

Your research statement should be precise, focused, and contestable; it should predict the sub-topics or blocks of information that you will use to write your report.

In order to help you to select a research topic look at the following:

Selection Criteria

- Your genuine enthusiasm for the problem
- The degree to which research on this problem benefits the profession and society
- The degree to which research on this problem will assist your professional goals and career objectives
- The degree to which this research will interest professionals and other readers in the field

- The degree to which the research builds on your experience and knowledge
- Ease of access to the population to be studied and the likelihood that they will be cooperative [if human]

Affordability / feasibility/

- Relationship to theories or accepted generalizations in the field
- Degree to which ethical problems are involved [usually privacy issues]
- Degree to which research is unique or fills a notable gap in the literature
- Degree to which the research builds on and extends existing knowledge

In the beginning, the research statement was a tool to help you sharpen your focus, limit material and establish the paper's purpose. When your paper is finished, however, the statement becomes a tool for your reader. It tells the reader what you have learned about your topic and what evidence led you to your conclusion. It keeps the reader on track--well able to understand and appreciate your argument.

✓ Chapter Checklist

Put a ‘ ’ mark against each of the following statements that you are able to answer.

I can:-

- Write criteria about researchable topic
- Explain procedures in conducting research.
- Identify the basic consideration for research topic selection.

◆ Self-Test

Part One Instruction: write a complete answer for the following questions.

1. What are some of the procedures in conducting research?
2. Write criteria about researchable topic.
3. Searching the literature is the method used to learn new facts and principles through the study of _____,_____and_____
4. What are the uses of library search?
5. What are the possible sources, for researchers, to select topics?
6. What is research hypothesis?
7. What should a researcher take into consideration when he/she wants to select a Research topic?
8. What is a statistical hypothesis?

9. What is the purpose of review of literature?

10. What do you think is the common problem of beginner researchers in stating a research topic?

Part two: Write true if the statement is correct and false if the statement is wrong.

1. The study of professional literature will not only expose you to pressing research problems but will also suggest the way in which research is conducted.

2. Hypotheses are formulated to explain observed facts, conditions, or behavior and to serve as a guide in the research process.

3. The best way to begin searching for a topic is to conduct some initial research.

4. You can narrow your topic by considering a particular approach to the subject, or a sub-topic within it.

5. It is important when evaluating or proposing research questions to look not only at the question itself, but look at the justification presented as a context for the research

CHAPTER 3

Research Design and Methodology

Introduction

A research study requires a coherent plan for answering its questions and gathering credible evidence. Research design provides the overall structure of a study, while methodology explains and justifies the approaches and procedures used to collect and interpret data.

This chapter introduces research design and methodology, including qualitative and quantitative approaches and the related concepts of validity and reliability.

★ Learning Objectives

By the end of this chapter you will be able to:

- write some basic concepts about research method and designs.
- list some components of research design.
- define research design.
- identify the differences between qualitative and quantitative researches.
- explain research validity and reliability?

✎ Activity 13 what is Research design?

Research design is the plan, structure, strategy of investing action in order to get answers to the research question. Therefore, it includes the outline of everything the researcher/investigator will perform from writing the hypothesis and the operational implication to the final analysis of the data". The purpose of research designs is to answer a research question(s) objectively, accurately and economically as possible. In other word a research design shows us in a sense what observation to be made: how to make them, how to categorize, how to analysis the quantitative analyses of the information.

Plane Research design analysis

Organization (or reorganization) is a continuous process„it goes on simultaneously with other activities, such as narrowing your topic, forming your thesis statement, and conducting your research. However, formal organization generally involves two components: determining a method of organization for the research, and drawing up an outline which applies your ideas to that method.

As you begin to plan your research, give some thought to the methods you will use to organize the evidences that will support your report. You choose methods which are most suitable to your subject and the type of research you have chosen..

3.1 Methodology

✎ **Activity 14** how do you design your research work? What constitutes an appropriate method?

The choice of methods in research is related to your research question. This might seem like a chicken and egg situation but some methods might be clearly inappropriate to your research question.

Consider for example the case of a researcher interested in studying the influence of terrorism on racial attacks in Ethiopia. This question might be studied by sending a questionnaire or making interviews to researchers and historians. This would be an empirical approach to the question.

Alternatively one could read and analyze the writings of those historians and come-up with some new or innovative interpretation on the roots of racisms. This would not be an empirically-based approach because one is not generating data, one is analyzing and studying data produced by others and based on that data one produces a **theory** or a model.

In the above example we notice that the methods will allow the researcher to trace the influence of terrorism on racism but they will not allow him/her to investigate the second part of the question which is racial attacks in Ethiopia. Field research through either participant or non-participant **observation** might be a more appropriate method to answer this angle of the research question.

Let us now consider a different example. Let us assume that you are interested in studying the effect of prolonged exposure in battle-trenches on the health of survivors of battles during the First World War. Although in theory you could study this research question by interviewing survivors, in practical terms this method is hardly possible because most survivors are now so old that recollections are practically impossible, or most survivors of particular battles are now dead. If you had studied that question 50 years ago, you could have in fact interviewed people, now however, that method is hardly appropriate. An alternative method to interviews in this particular case would be the analysis of other sources of data, like diaries of survivors, letters of survivors, letters and records of medical personnel that treated wounded soldiers etc.

When you chose a research method you have to realize that your chosen method is also your way of generating data. You must therefore consider the quality of the data you will be generating via one method or via another method. Mason (1996) advises the generation of tables and lists with which to summarize the possible sources of data and methods within a research question and to include a section on the justification for each method.

3.2 Research Design

Remember that the broad concept of research design includes all of the topics included here in the methods section whereas the narrow conception of research design refers only to logical models of proof.

Once you have determined your method or methods of development, put together a working outline. This plan can range from a brief sketch of main points to a detailed point-by-point outline complete with paragraphs and topic sentences. The idea is to provide yourself with a rough map of where the report will go, making a diagram of your thoughts to sharpen and define your purpose. At this point you can also give your report a working title. The outline shows where to begin and breaks the research report into manageable parts.

Using an outline can help you organize your material and can also help you discover connections between pieces of information that you weren't aware of when you first conceived the plan of your paper. It can also make you aware of material that is not really relevant to the purposes of your paper or material that you have covered before and should therefore be removed.

A Working Outline might be only an informal list of topics and subtopics which you are thinking of covering in your paper. Sometimes, however, your research advisor might require that a working outline be submitted at the beginning of your work; then your advisor might suggest ways in which the

work needs to be further developed or cut back. Your advisor might also see that you're trying to accomplish too much or too little for the scope of the research topic he or she has in mind. The working outline can be revised as you discover new material and get new ideas that ought to go into your paper.

A Final Outline should enhance the organization and coherence of your research paper. Advisors sometimes require that a final outline be submitted along with the final version of your paper. Material that is not relevant to the purpose of your paper as revealed in your outline should be excised from the paper; if portions of your outline seem weak in comparison to others, more research may be required to create a sense of balance in your argument and presentation.

Outlines can be organized according to your purposes. Are you attempting to show the chronology of some historical development, the cause-and-effect relationship between one phenomenon and another, the process by which something is accomplished, or the logic of some position? Are you defining or analyzing something? Are you Comparing or contrasting one thing to another? Presenting an argument (one side or both)?

What is the logical model being employed? Is it a qualitative Design? Is it a quantitative Design? Is it both quantitative and qualitative Design? Is it an experimental design? Or what? A knowledgeable reader will see the strengths and weaknesses of the design you are using. However, other readers can use some guidance as to both strengths and weaknesses and why you are using the design-- it is because the only possible design for the problem given the resources at hand. These and other questions must be answered by the research design you have already chosen. There for, as a researcher you have to see every possible ways to decide on your research design.

3.3 Modes of Research

3.3.1 Qualitative Research

✎ Activity15 What is qualitative Research?

What does it do?

How does it interpret data?

What instruments does it use to collect data?

Qualitative studies are tools used in understanding and describing the world of human experience. Since we maintain our humanity throughout the research process, it is largely impossible to escape the subjective experience, even for the most seasoned of researchers. As we proceed through the research process, our humanness informs us and often directs us through such subtleties as intuition or 'aha' moments. Speaking about the world of human experience requires an extensive commitment in terms of time and dedication to process; however, this world is often dismissed as 'subjective' and regarded with suspicion. This paper acknowledges that small qualitative studies are not generalizable in the traditional sense, yet have redeeming qualities that set them above that requirement.

Qualitative research generally examines people’s words or actions in narrative or descriptive ways more closely representing the experiences of the people involved, it focuses on understandings, meanings and it takes seriously lay accounts and concepts

The ultimate aim of qualitative research is to offer a perspective of a situation and provide well-written research reports that reflect the researcher's ability to illustrate or describe the corresponding

phenomenon. One of the greatest strengths of the qualitative approach is the richness and depth of explorations and descriptions.

A major strength of the qualitative approach is the depth to which explorations are conducted and descriptions are written, usually resulting in sufficient details for the reader to grasp the idiosyncrasies of the situation.

Unlike quantitative research, there is no overarching framework for how qualitative research should be conducted; rather each type of qualitative research is guided by particular philosophical stances that are taken in relation by the research to each phenomenon.

Main Types of Qualitative Research

	Historical	**Descriptive**
Problem	- isolated person , period, or phenomena	- isolated and defined conditions, situations , or relationship for which norms or standard are desired.
Working Hypothesis	- theoretical development or means of arriving at a solution to be expressed in terms of causations or probabilities.	- theoretical development or means or arriving a solution to be expressed in terms of norms or standards.
Source of data	Written documents, criticisms synthetic effects upon period society or other phenomena.	- testing (written or oral observation, interview , questionnaires together with documents etc.
Analysis	Interpretive (in light of data and documents)	Interpretive (in light of data and theoretical approaches) -
Generalization	Assertion on probabilities and causation	- confirmations or disconfirmation of theoretical facts in the light of findings or interpretations.

In addition to the above mentioned main types of qualitative research

, there are others like:

Case study Attempts to shed light on a phenomena by studying in-depth a single case example of the phenomena. The case can be an individual person, an event, a group, or an institution.

Ethnography Focuses on the sociology of meaning through close field observation of socio-cultural phenomena. Typically, the ethnographer focuses on a community

3.3.2 Quantitative Research

✎ **Activity 16** What is quantitative Research?

What does it do?

How does it interpret data?

What instruments does it use to collect data?

Quantitative research generally converts observations into discrete units that can be compared to other units by statistical analysis; it focuses on explanation, prediction and proof. Like qualitative research, Quantitative research is commonly used to investigate research questions. There is a potentially infinite array of possible quantitative research designs, and in the human sciences particularly, it can be difficult to do pure, experimental research. Thus a great many adaptations of experimental designs, called quasi-experimental and non-experimental designs have been developed.

Let us see some features of experimental research which is of quantities research type:

Statement of the Problem focuses on isolated and defined factor about the operation and effect or causal relationships of which there is doubt.

The working hypothesis of experimental research is - theoretical development or means of arriving at a solution to be expressed in terms of laws or principle.

The source of data is testing (laboratory)or /and sample analysis identification measurement and observation of factors of behaviors

Data analysis is more of interpretive (in light of data and theoretical laws and principles)

Generalization or conclusion is most of the time statements of proof or disproof of the examined laws or principles.

In general it is important to consider a range of possible quantitative research designs and their strengths and weaknesses, before adopting any particular design.

3.4 Features of Qualitative and Quantitative Research

Qualitative	**Quantitative**
"All research ultimately has a qualitative grounding" - Donald Campbell	"There's no such thing as qualitative data. Everything is either 1 or 0" - Fred Kerlinger
The aim of **qualitative** analysis is a complete, detailed description.	In **quantitative** research we classify features, count them, and construct statistical models in an attempt to explain what is observed.
Recommended during earlier phases of research projects.	Recommended during latter phases of research projects.
Researcher may only know roughly in advance what he/she is looking for.	Researcher knows clearly in advance what he/she is looking for.

The design emerges as the study unfolds.	All aspects of the study are carefully designed before data is collected.
Researcher is the data gathering instrument.	Researcher uses tools, such as questionnaires or equipment to collect numerical data.
Data is in the form of words, pictures or objects.	Data is in the form of numbers and statistics.
Qualitative data is more 'rich', time consuming, and less able to be generalized.	Quantitative data is more efficient, able to test hypotheses, but may miss contextual detail.
Researcher tends to become subjectively immersed in the subject matter.	Researcher tends to remain objectively separated from the subject matter.

(Source: Miles & Huberman (1994, p. 40). *Qualitative Data Analysis*)

3.5 Research Reliability and Validity

✎ Activity 17 What is research Validity?

What is research reliability?

Why do you, as a researcher, worry about reliability and validity?

3.5.1 Validity

Validity concerns the extent to which the methods, evidence, and interpretations support the claims made in a study. In quantitative research, validity may concern the quality of measurement, the credibility of causal conclusions, and the extent to which findings can be generalized to other populations or settings. Internal validity concerns whether a study supports the conclusion that an observed relationship or effect is attributable to the factors proposed, rather than to plausible alternative explanations. External validity concerns the extent to which findings may apply beyond the particular study. In qualitative research, researchers often discuss trustworthiness through criteria such as credibility, transferability, dependability, and confirmability. The appropriate criteria depend on the research design and questions.

It therefore deals with the appropriateness of the method to the research question. In Mason's (1996) words, you should be asking yourself *"how well matched is the logic of the method to the kinds of research questions you are asking and the kind of social explanation you are intending to develop"*. This is the case for justifying the appropriateness of your method to your research question.

Validity also concerns the soundness of interpretations. Researchers should explain how they moved from collected evidence to their interpretations, provide sufficient evidence for their claims, consider alternative explanations, and acknowledge limitations. In qualitative studies, strategies such as keeping an audit trail, discussing interpretations with colleagues, examining negative cases, or inviting participants to comment on interpretations may be useful. No single strategy guarantees validity or trustworthiness.

Showing participants selected excerpts or interpretations of their interviews may help a researcher check whether an interpretation is recognizable to them. This practice is sometimes called member checking or participant feedback. It is one possible strategy, not a guarantee of validity; participants may disagree, interpret events differently, or have limited opportunity to comment. Any feedback should be considered and reported transparently. Where covert observation has been proposed, researchers must address the ethical and legal implications and obtain appropriate review or authorization before proceeding.

3.5.2 Reliability

Reliability generally concerns the consistency or dependability of a measurement, procedure, or research process. In quantitative research, researchers may examine whether an instrument produces consistent results under appropriate conditions. Reliability does not by itself establish that a measure is valid: a measure can be consistent while measuring the wrong thing. In qualitative research, researchers may instead explain their procedures, maintain records of decisions, and demonstrate how interpretations were developed so that the process is transparent and dependable.

Using more than one data source, method, investigator, or theoretical perspective is commonly called triangulation. Triangulation can provide a fuller understanding of a research problem and help researchers examine whether different forms of evidence converge or diverge. Agreement does not automatically prove an interpretation, and disagreement should not simply be discarded; differences may reveal context, complexity, or limitations in the evidence.

Validity and reliability are important because they help readers assess the quality of the evidence, the appropriateness of the methods, and the strength of the conclusions. Researchers should consider these issues when planning a study, collecting data, analyzing evidence, and writing the report—not only after the research has been completed.

You have to bear in mind that, research is not an entirely precise science. Therefore, when one studies human interactions, the question for researchers and consumers of research (i.e. students, policy makers, administrators, other researchers) is to be able to evaluate the veracity of results and the soundness of the research conclusions based on the appropriateness of the methodology and the quality of the data upon which the conclusions are based.

Look at the following example.

Imagine that a researcher is interested in studying the effects of exposure in battle-trenches during the world 1st war; let us assume that after reading a number of diaries from survivors several diaries point out that a particular type of skin-infection was developed on survivor's feet after prolonged exposure to humidity and mud. In this case, the diaries of men complaining about their feet could be considered one indicator of skin problems due to prolonged exposure to humidity and mud. If one would want to corroborate that indicator, one could look for medical records of doctors who treated men at that time and look for medical descriptions of skin problems on the feet of service-men. If these records were found, then the medical records would be corroborating the data from the diaries. The researcher would therefore be more confident in reporting that those types of skin disorders were indeed prevalent in the majority of soldiers.

Within a conventional research background one speaks of the reliability of the "research instrument". In other words, how certain are you that any person using the same interview scripts (the research instrument) will not sway the answers of the interviewees? Researchers in the qualitative tradition sometimes refuse to speak of "research instruments" but it is possible to do field-work within a

qualitative tradition and design "data recording sheets" to guide analysis of in-depth interviewing or of participant observation events. It is also possible to design "data sheets" to facilitate document analysis, so it is possible to address reliability issues regarding the consistency of your methods for recording data. These methods will influence the type of data you have access to, and the way you can interpret and analyze the data. For example, you might decide to record observation events with a video camera. This will give you access to images. You then need to figure out the way in which you are going to turn those images into "data" and the way you are going to interpret those images. Then you will need to consider how consistent your interpretations will be.

Experts in the field of research believe that most of the time data interpretations are subjective. This is entirely acceptable within the qualitative research tradition as you cannot separate your own feelings and opinions from the research question and the data. Some sources of recording data allow for specific ways of increasing the reliability of interpretations, for example, if you are doing document analysis, you might give a selected portion of text to other researchers within you team and ask them to interpret the text. You can then count the degree of agreement of the interpretation by different researchers within your team. This can also be done withinterpretations of images. However not all qualitative methods are conducive to this type of double checking, and in many cases, the analysis based on observations will be entirely suit generic. In this case, reliability issues should address demonstrating to your audience that you have not invented or misinterpreted your data and that you have not been careless in your data recording.

Look at the following Example.

Imagine that a researcher is interested to study racists. Let us assume he/she chose to "infiltrate" a racist group and follow within a participant observer fore one year of field-research. The researcher in fact would be pretending to be a new-racist in order to do his/her field research appropriately. It is believed that this type of research is highly risky. There for the researcher would be in no position to ask for corroboration of his/her interpretations, because his/her life might be in danger if he-she disclosed his/her real identity. Similarly, if as part of his/her field research he/she befriended "research subjects", he/she might be in a difficult position by asking them to corroborate his/her results, as they might feel exploited in their trust. In this case, you will need to evaluate whether you can indeed ask for corroboration from his/her research subjects, but this is not a quick-fix solution, because subjects might disagree with his/her interpretation not because it is wrong, but because it places them in a difficult position, etc.

The above example shows that how a reliability of data collection can be affected by factors which are beyond the control of the researcher. Therefore the researcher has to take care when he/she designs the research.

Although reliability and validity issues apply mostly to research results and conclusions, one must consider those issues at the time of the design of the research, i.e. at the earliest stages of the research process, because if you consider them only at the end, it will be too late to gather data on a research question that is of any relevance and quality at all.

• Conclusion

Research design is the overall plan that connects a research question with the evidence to be collected and the procedures for analyzing it. Research methodology refers to the rationale and principles informing the choice and use of methods, while research methods are the specific techniques used to collect and analyze data. These terms are related, but they are not interchangeable. A well-justified

design and appropriate methods help researchers address their questions systematically and transparently.

Qualitative studies are tools used in understanding and describing the world of human experience. The ultimate aim of qualitative research is to offer a perspective of a situation and provide well-written research reports that reflect the researcher's ability to illustrate or describe the corresponding phenomenon.

Qualitative research generally examines people's words or actions in narrative or descriptive ways more closely representing the experiences of the people involved, it focuses on understandings, meanings and it takes seriously lay accounts and concepts. Quantitative research generally converts observations into discrete units that can be compared to other units by statistical analysis

Unlike quantitative research, there is no overarching framework for how qualitative research should be conducted; rather each type of qualitative research is guided by particular philosophical stances that are taken in relation by the research to each phenomenon.

In quantitative research, validity may concern measurement, causal inference, and the applicability of findings beyond the study. In qualitative research, researchers should explain how they established the credibility and dependability of their interpretations and provide enough contextual information for readers to consider transferability.

Researchers should make the process of interpretation transparent and support claims with evidence. Participant feedback, peer discussion, triangulation, and examination of alternative explanations may contribute to this process, but none guarantees validity on its own.

Reliability concerns the consistency or dependability of measures and procedures; it should not be defined simply as accuracy.

Triangulation uses multiple sources, methods, investigators, or perspectives to examine a research question. It can strengthen an analysis, but convergence is not proof and divergence may itself be informative.

Summary

Research design, methodology, and methods are related but distinct. The design provides the overall plan; methodology explains the reasoning behind the approach; and methods are the practical techniques used to collect and analyze evidence. Choosing and justifying these elements in relation to the research questions is essential to producing credible, transparent findings.

Here are some principles of organization:

Chronological order:

Paragraphs separate the process or series of events into major stages.

Classification:

Paragraphs divide the material into major categories and distinguish between them.

Increasing importance:

Paragraphs are arranged so that the most important point comes last, thus building the essay's strength.

Cause and effect

Indicates causal relationships between things and events. Be careful not to make mistake coincidence with causality, nor to disregard other possible causes.

Comparison and contrast

Involves lining up related ideas for a detailed account of similarities and differences. In this kind of research it is important to decide whether you will be concentrating on similarities or differences. In general, the more similar things are, the more you concentrate on the differences, and vice versa. If you are comparing two works by the same writer, what will most interest you will be the differences between them.

Although one pattern should serve as the overall organizing framework, your argument can benefit from a combination of these strategies. For example, while the paragraphs may be arranged in ascending order of importance, within the paragraphs it is likely that you will incorporate comparisons, causes, classification or chronology. These principles apply to both the greater structure of the research and each individual idea.

research design and methodology section mainly focuses on:

- Variables used
- Population and selection of sample
- Tools used for data gathering
- Data gathering procedures
- Methods of data analysis:- Quantitative, qualitative, etc

Quantitative Mode	Qualitative mode

Assumptions • Social facts have measurable reality • Primacy of method • Variable can be identified and relationships measured • Etic (outsider's perspectives)	***Assumptions*** • Reality is socially constructed • Primacy of subjects • Variables are complex, difficult to measure • Emic (insider's perspective)
Purpose • Generalization • Prediction • Causality	***Purpose*** • Contextualization • Interpretation • Understanding actors' perspectives
Approach • Begins with hypothesis and theory • Manipulation and control • Uses formal instruments • Experimentation • Deductive • Component analysis • Seeks consensus • Reduce data to numerical indices • Abstract language in write – up	***Approach*** • Ends with hypothesis and grounded theory • Emergency and portrayal • Researcher as instrument • Naturalistic • Inductive • Synthesis-search for patterns • Seeks pluralism, complexity • Minor use of numerical • Descriptive write - up
Researcher's Role • Detachment and impartiality • Learning Objectives portrayal	***Researcher's Role*** • Personal involvement and partiality • Emphatic understanding

Adapted from: Miles & Huberman (1994) *Qualitative Data Analysis*.

Regardless of the type of research you apply, you have to think about the quality of your findings. The quality of your research can be affected because of different reasons which results in the questions of reliability and validity.

✓ Chapter Checklist

Put a ' ' mark against each of the following statements that you are able to answer.

I can:-

- Write some basic concepts about research method and designs.
- List some components of research design.

- Define research design.
- Write the differences between qualitative and quantitative researches.
- Explain about research validity and reliability?

◆ Self-Test

Part one Instruction: write a complete answer for the following questions.

1. What is research design?

2. What elements are included in research design and methodology?

3. Write some features of qualitative and quantitative researches?

4. How can you increase the validity of your research work?

5. What is qualitative research?

6. What is quantitative research?

Part Two Write true if the statement is correct and false if the statement is wrong.

1. The purpose of research design is to answer a research question(s) objectively, accurately and economically as possible.

2. When you chose a research method you have to realize that your chosen method is also your way of generating data.

3. Quantitative studies are tools used in understanding and describing the world of human experience.

4. The ultimate aim of qualitative research is to offer a perspective of a situation and provide well-written research reports that reflect the researcher's ability to illustrate or describe the corresponding phenomenon.

5. Validity is the plan, structure, strategy of investing action in order to get answers to the research question.

6. One way of increasing the validity of your research consists of showing your research subjects excerpts of your interpretation of their interviews.

7. Reliability addresses how accurate your research methods and techniques produce data.

8. The use of several data sources and different methods is called Reliability**.**

CHAPTER 4

Sampling, Data Collection, Analysis and Presentation

Introduction

Reliable research depends on appropriate sampling, systematic data collection, careful analysis, and clear presentation of evidence. Researchers need to select suitable participants or cases, use data-collection methods that fit their questions, and organize the resulting information without losing sight of the study's purpose.

★ Learning Objectives

By the end of this chapter you will be able to:

- Explain the purpose and main stages of data collection.
- Describe appropriate ways of taking and organizing research notes.
- Define data collection and explain why systematic procedures are important.
- Explain how researchers can protect the integrity and quality of research data.
- Define sampling
- Write the purpose of sampling.
- Explain the main types of sampling.
- Distinguish sampling error from sampling bias and non-sampling error.
- Describe procedures for collecting reliable and relevant data.
- Explain ways of recording and organizing research information.
- Explain the main stages of data analysis.
- Describe appropriate ways of presenting research findings.
- Explain the purpose and main stages of data analysis.
- Explain ways of analyzing and presenting data.

4.1 Sampling

✎ Activity 18 What is a sample?

In research it is often impossible to investigate everyone in a target population. Hence the reason we use a sample from a target population instead. If your sample is representative of the target population, and if it is unbiased, then you can generalize your results and conclusions onto the target population. A target population is the one from which a sample of participants in research is drawn. It is the population to which a researcher wants to generalize their results. This is only possible if the sample selected is representative of the target population. Therefore it can be said that, a sample is a finite part of a statistical population whose properties are studied to gain information about the whole (Merriam-Webster, Inc., 1985). When dealing with people, it can be defined as a set of respondents (people) selected from a larger population for the purpose of a

survey. A population is a group of persons, objects, or items from which samples are taken for measurement for example a population of teachers or students.

What is sampling? Sampling is the act, process, or technique of selecting a suitable population, or a representative part of a population for the purpose of determining parameters or characteristics of the whole population.

✎ Activity 16 what do you think is the purpose of sampling?

When we conduct a research, we have to gather information. And to work with the whole population is tiresome, time consuming and some times impossible. Therefore, we must use inferential statistics to draw conclusions about populations from samples, which enables us to determine a population's characteristics by directly observing only a portion (or sample) of the population. We obtain a sample rather than a complete enumeration (a census) of the population for many reasons. Obviously, it is cheaper to observe a part rather than the whole, but we should prepare ourselves to cope with the dangers of using samples. In this chapter, we will investigate various kinds of sampling procedures. Some are better than others but all may yield samples that are inaccurate and unreliable. We will learn how to minimize these dangers, but some potential error is the price we must pay for the convenience and savings the samples provide.

There would be no need for statistical theory if a census rather than a sample was always used to obtain information about populations. But a census may not be practical and is almost never economical. There are six main reasons for sampling instead of doing a census. These are; -Economy -Timeliness -The large size of many populations -Inaccessibility of some of the population -Destructiveness of the observation -accuracy

The economic advantage of using a sample in research obviously, taking a sample requires fewer resources than a census. ***For example***, let us assume that you are one of the very curious students around. You have heard so much about the famous students at some school say "X" and now that you are there, you want to hear from the insiders. You want to know what all the students at "X" think about the quality of teaching they receive, you know that all the students are different so they are likely to have different perceptions and you believe you must get all these perceptions so you decide because you want an in-depth view of every student, you will conduct personal interviews with each one of them and you want the results in 20 days only, let us assume this particular time you are doing your research has only 2,000 students and those who are helping are so fast at the interviewing art that together you can interview at least 10 students per person per day in addition to your 18 credit hours of course work. You will require 100 research assistants for 20 days and since you are paying them minimum wage of 30.00 birr per hour for ten hours (300.00 birr) per person per day, you will require birr 600,000.00 just to complete the interviews, analysis will just be impossible. You may decide to hire additional assistants to help with the analysis at another birr 600,000.00 and so on assuming you have that amount money.

As unrealistic as this example is, it does illustrate the very high cost of census. For the type of information desired, a small wisely selected sample of "X" students can serve the purpose. You don't even have to hire a single assistant. You can complete the interviews and analysis on your own. Rarely does a circumstance require a census of the population, and even more rarely does one justify the expense.

The time factor.

A sample may provide you with needed information quickly. ***For example*** imagine a contagious disease has broken out in a village within your area of jurisdiction and it is killing people and animals within hours. Nobody knows what it is. You are required to conduct quick tests to help save the situation. If you try a census of those affected, they may be long dead when you arrive with your results. In such a case just a few of those already infected could be used to provide the required information.

The very large populations

Many populations about which inferences must be made are quite large. ***For example***, consider the number of students in senior secondary schools in Ethiopia is 14,000,000. If the Ministry of Education (MOE) has to plan for how they will be absorbed into the different departments of governmental and private Universities. The employers would like to have specific knowledge about the student's plans in order to make compatible plans to absorb them during the coming year. But the big size of the population makes it physically impossible to conduct a census. In such a case, selecting a representative sample may be the only way to get the information required from high school seniors.

The partly accessible populations

Some times, based on the nature of your research, there are some populations that are so difficult to get access so that only a sample can be used. Like people in prison, like crashed aeroplanes in the deep seas, presidents, etc. The inaccessibility may be economic or time related. Like a particular study population may be so costly to reach like the population of planets that only a sample can be used. In other cases, a population of some events may be taking too long to occur that only sample information can be relied on. For example natural disasters like flood which occur every 100 years.The destructive nature of the observation sometimes the very act of observing the desired characteristic of a unit of the population destroys it for the intended use. Good examples of this occur in quality control. To obtain a census of the desired characteristics of a unit of population, you have to destroy all of them. This is contrary to the purpose served by quality-control testing. In this case, only a sample should be used.

4.1.1 Accuracy and sampling

A sample may be more accurate than a census. A sloppily conducted census can provide less reliable information than a carefully obtained sample.

Bias and error in sampling

A sample is expected to mirror the population from which it comes; however, there is no guarantee that any sample will be precisely representative of the population from which it comes. Chance may dictate that a disproportionate number of untypical observations will be made.

4.1.1.1 Sampling error

✎ Activity 19. What can make a sample unrepresentative of its population?

Sampling error is the difference between a sample statistic and the corresponding population value that occurs because only some members of the population are selected. It is associated with the particular sample drawn and is expected in sample-based estimates. In probability sampling, its likely size can be estimated using appropriate statistical methods. Increasing sample size can reduce sampling error under suitable conditions, but it does not correct systematic bias.

Example, suppose that a sample of 100 women are measured and are all found to be taller than 1.75 centimeter. It is very clear even without any statistical prove that this would be a highly unrepresentative sample leading to invalid conclusions. This is a very unlikely occurrence because naturally such rare cases are widely distributed among the population. But it can occur. Luckily, this is a very obvious error and can be detected very easily.

Sampling variability occurs because different samples from the same population may produce different estimates. This variation is influenced by sample size, population variability, and the sampling design. A larger probability sample will generally have less sampling variability than a smaller one, all else being equal.

Sampling bias occurs when the selection process systematically favours some members or characteristics of the population over others. It may arise from an incomplete sampling frame, inappropriate recruitment, nonresponse, or other features of the study procedure. Unlike random sampling error, bias is not necessarily reduced by increasing the sample size.

For example, take a hypothetical case where a survey was conducted recently by the university to find out the level of stress that graduate students were going through. A mail questionnaire was sent to 100 randomly selected graduate students. Only 52 responded and the results students were not under stress at that time. Actually it was the highest time of stress for all students except those who were writing their senior essay at their own pace. Apparently, this is the group that had the time to respond. The researcher who was conducting the study went back to the questionnaire to find out what the problem was and found that all those who had responded were third and fourth year students. Bias can be very costly and has to be guarded against as much as possible; it costs the researcher's time, money and other resources, therefore s/he has to be able to planned ahead on how to avoid it.

In addition to the above points, a means of selecting the units of analysis must be designed to avoid the more obvious forms of bias. ***Another example*** would be where you would like to know the average income of some community and you decide to use the telephone numbers to select a sample of the total population in a locality where only the rich and middle class households have telephone lines. You will end up with high average income which will lead to the wrong policy decisions.

Non-sampling errors arise from sources other than the random selection of a sample. They may include coverage errors, nonresponse, inaccurate measurement, interviewer or respondent effects, recording mistakes, coding errors, and data-processing errors. Such errors can occur in both sample surveys and censuses.

The other main cause of unrepresentative samples is non-sampling error. A non-sampling error is an error that results solely from the manner in which the observations are made. Like sampling error, non-sampling error may either be produced by participants in the statistical study or be an innocent by-product of the sampling plans and procedures.

The simplest example of non-sampling error is inaccurate measurements due to malfunctioning instruments or poor procedures. For example, consider the observation of human weights. If persons are asked to state their own weights themselves, no two answers will be of equally reliability. The people will have weighed themselves on different scales in various states of poor calibration. An individual's weight fluctuates diurnally by several kilograms, so that the time of weighing will affect the answer. The scale reading will also vary with the person's state of undress.

Responses therefore will not be of comparable validity unless all persons are weighed under the same circumstances.

Remember that biased observations due to inaccurate measurement can be innocent but very devastating. A story is told of a French astronomer who once proposed a new theory based on spectroscopic measurements of light emitted by a particular star. When his friends discovered that the measuring instrument had been contaminated by cigarette smoke, they rejected his findings.

In general, in surveys of personal characteristics, unintended errors may result from: -The manner in which the response is elicited -The social desirability of the persons surveyed -The purpose of the study -The personal biases of the interviewer or survey writer, etc

The interviewer's effect

✎ Activity 18 do think that different interviewers can collect the same data on the same topic?

Yes, no two interviewers are alike and the same person may provide different answers to different interviewers. The manner in which a question is formulated can also result in inaccurate responses. Individuals tend to provide false answers to particular questions. ***For example***, some people want to feel younger or older for some reason known to themselves. If you ask such a person their age in years, it is easier for the individual just to lie to you by over stating their age by one or more years; but if you asked which year they were born since it will require a bit of quick arithmetic to give a false date and a date of birth will definitely be more accurate.

The respondent effect

Respondents sometimes might also give incorrect answers to impress the interviewer. This type of error is the most difficult to prevent because it results from out right deceit on the part of the responder. ***An example*** of this is what a reporter witnessed in his recent visit in which he was asking farmers how much quintal of maize they harvested last year. In most cases, the men tended to lie. The responses from men looked so uniform that he became suspicious. He compared with the responses of the wives of these men and their responses were all different. To decide which one was right, whenever possible he could in a tactful way verify with an older son or daughter. It is important to acknowledge that certain psychological factors induce incorrect responses and great care must be taken to design a study that minimizes their effect.

Knowing the purpose of the study

If respondents know the purpose of a study conducted they some times may give incorrect response. ***A classic example is*** the question: What is your income? If a government agency is asking, a different figure may be provided than the respondent would give on an application for a home mortgage. One way to guard against such bias is to camouflage the study's goals; another remedy is to make the questions very specific, allowing no room for personal interpretation. ***For example***, "Where are you employed?" could be followed by "What is your salary?" and "Do you have any extra jobs?" A sequence of such questions may produce more accurate information.

Induced bias

Finally, it should be noted that the personal prejudices of either the designer of the study or the data collector may tend to induce bias. In designing a questionnaire, questions may be slanted in such a way that a particular response will be obtained even though it is inaccurate. ***For example,*** an

agronomist may apply fertilizer to certain key plots, knowing that they will provide more favorable yields than others. To protect against induced bias, advice of an individual trained in statistics should be sought in the design and someone else aware of search pitfalls should serve in an auditing capacity.

4.1.2 Types of samples

✎ ***Activity 20***, how many kinds of samples do you know?

Sampling approaches are commonly grouped into probability and nonprobability sampling. In probability sampling, units are selected using a random procedure and each eligible unit has a known, non-zero chance of selection. Common methods include simple random, systematic, stratified, cluster, and multistage sampling. In nonprobability sampling, selection probabilities are not known; common methods include convenience, purposive (judgment), quota, and snowball sampling. The choice depends on the research question, population, access, and intended form of inference.

The convenient sample

A convenience sample results when the more convenient elementary units are chosen from a population for observation.

The judgment sample

A judgment sample is obtained according to the discretion of someone who is familiar with the relevant characteristics of the population.

The random sample

In probability sampling, selection is based on a random procedure and the selection probabilities can be specified. Probability sampling supports statistical inference to a defined population when the sampling design and assumptions are appropriate. It does not automatically eliminate all forms of bias or guarantee a representative sample.

Types of random samples

A simple random sample

✎ **Activity 21, what is simple random sample?**

A simple random sample is selected so that each eligible unit has an equal chance of selection, with the precise procedure depending on whether sampling is conducted with or without replacement. Random selection helps control selection bias, but it does not guarantee that the resulting sample will match every characteristic of the population. Sampling variability and non-sampling errors may still occur.

A systematic random sample

A systematic random sample is selected by choosing a random starting point and then selecting units at a fixed interval from an ordered sampling frame. For example, if 100 students are listed and a sample of 20 is required, the interval is 5. A starting number is selected at random from 1 to 5, and every fifth student thereafter is selected. Researchers should check whether the ordering of the list contains a pattern that could bias the sample.

A stratified sample

A stratified sample is obtained by dividing the population into relevant subgroups (strata) and selecting a sample from each stratum, usually by a probability method. Stratification can ensure that important subgroups are represented and may improve the precision of estimates. The allocation may be proportional or may use another justified plan.

A cluster sample

A cluster sample is obtained by dividing the population into naturally occurring groups, or clusters, and randomly selecting some of those clusters. Researchers may study all eligible units in selected clusters (one-stage cluster sampling) or select a further sample of units within them (two-stage or multistage sampling). Cluster sampling can reduce travel and administrative costs, but the analysis should account for the sampling design.

4.1.3 Sample size

Before deciding how large a sample should be, you have to define your study population. For example, all children below age three in Ethiopia. Then determine your sampling frame which could be a list of all the children below three as recorded. You can then struggle with the sample size.

Sample size should be justified in relation to the research purpose, design, population, expected variability or information richness, desired precision, planned analysis, and available resources. Dividing a budget equally between data collection and analysis is not a general rule for determining sample size.

In general, sample size depends on the nature of the analysis to be performed, the desired precision of the estimates one wishes to achieve, the kind and number of comparisons that will be made, the number of variables that have to be examined simultaneously and how heterogeneous a universe is sampled. ***For example,*** if the key analysis of a randomized experiment consists of computing averages for experimental and controls in a project and comparing differences, then a sample under 100 might be adequate, assuming that other statistical assumptions hold.

In non-experimental research, most often, relevant variables have to be controlled statistically because groups differ by factors other than chance. More technical considerations suggest that the required sample size is a function of the precision of the estimates one wishes to achieve, the variability or variance, one expects to find in the population and the statistical level of confidence one wishes to use.

In qualitative research, sample size is not usually determined by statistical representativeness. It depends on the purpose of the study, the specificity and diversity of the participants or cases, the depth of data required, the analytical approach, and practical constraints. Researchers may consider information power or whether further data collection is adding meaningful insight; they should explain and justify the decision rather than rely on a universal numerical rule.

4.2 Data Collection

✎ Activity 22. What is data collection?

Data collection is the process of gathering and measuring information on variables of interest, in an established systematic fashion that enables one to answer the stated research questions, test hypotheses, and evaluate outcomes. The data collection component of research is common to all fields of study including physical and social sciences, humanities, business, etc. While methods vary by discipline, the emphasis on ensuring accurate and honest collection remains the same.

Report preparation begins with planning the research program. An orderly investigation is a requisite for an orderly report. Report planning and program planning should be considered one and the same. To become a successful technical writer, you should develop the ability to foresee the general content of the report before the program begins. In most cases you should be able to prepare a preliminary report outline at the beginning of the program. Outlining should benefit both the report and the program, for obviously a well-prepared outline requires a carefully planned program.

Throughout the research process, keep the eventual report in mind. Maintain orderly records as the data are gathered. The little extra time required to record the results carefully can be of great value later. Report writing is difficult enough without having to recollect misplaced or unrecorded data. Write out your opinions as soon as the data are obtained. Comparing these opinions with those based on hindsight will often help you to interpret the data properly.

During the data-gathering stage consider how the data should be presented in the report and record the results in this manner. Any need for additional data will thus be revealed before the program is completed.

✎ Activity 23., what steps do you think are to be followed in order to collect reliable data?

The first step to be taken after you have decided upon a workable topic is to prepare yourself rather generally for the work you are to do. This means that you must familiarize yourself with the broad field of scholarship which includes the subject matter you intend to deal with, especially if it is one with which you are not already thoroughly acquainted. Even if you are familiar with the subject in general, you must acquire a picture of your special topic as a whole.

The next step would be to read materials which are helpful to you. For instance you can read text books. The advantage of textbooks is that their contents are ordinarily arranged either on a historical (chronological) basis or on a pattern which develops the subject matter gradually from simple to complex. Encyclopedias and textbooks customarily include selected bibliographies indicating suggested further readings covering the main aspects of the subject (topic) in the study. Thus, by consulting them, you have accomplished two useful things: you have acquired a background for your work and some helpful clues which will lead you to additional sources of information.

It will also be of immeasurably helpful to you if, wherever possible, you get to know the names of the authorities and the classic works in the field in which you are working. For instance, the names of Chomsky in Language teaching, James, Watson, and Thorndike in psychology: of Dewey. Terman, and Monroe in education: of Kuhlman, Johson, and Seligman in the social sciences: or of Van Doren. Ward, and Kittedge in English literature, when they signify the author or editor of a book, are an indication of an authoritative work in that particular field. These are but a few; there are dozens more in every department of scholarship.

In addition to books and encyclopedia, dictionaries are also of great importance. A good dictionary has many uses. And because it is a reference book with which we are so familiar, most of us are prone to overlook the wealth of varied material which it contains. Among the kinds of information which we customarily seek in a dictionary are: spelling, definition, synonyms, pronunciation, syllabication, and the origins and derivations of words. But in addition to these are others, such as: explanations and descriptions of things (sometimes accompanied by illustrations), rules of grammar and diction, slang and dialect terms, and tables of equivalent measurements. A dictionary usually has an appendix containing such items as a pronouncing gazetteer of geographical names, alphabetical lists of important characters from fiction and mythology, and biographical dictionary of important historical

personages, with their dates of birth and death and a guide to the pronunciation of the more unusual names. Many dictionaries also include a short history of the English language, lists of common foreign expression, standard abbreviations and their meanings, and the conventional symbols used by copyreaders and proofreaders.

Once you have identified resources, begin to evaluate whether they will be useful to you. General questions to help you ascertain this are:

Does the information pertain to your topic? Does it provide the type of information you need? Titles can sometimes obscure the contents of a chapter or book. Does the resource provide information at the right level of detail? Is it too general, too technical, or too theoretical? Is the reference a primary source or a secondary source? A primary source is an original document or manuscript. A secondary source is based upon the work of primary sources. For example, the report of a scientific study would be a primary source and an article synthesizing the results of several studies would be considered a secondary source.

Is the author someone who is recognized in the field as an expert? Authors who have written more than one book or article on the topic and are frequently cited by other scholars are likely candidates for the title of "expert." Does the author cite high quality sources to support his or her ideas? A writer who cites other high quality sources demonstrates that he or she has researched the topic.

Was the article published by a popular magazine or a scholarly journal? Popular magazines typically provide very general information about a topic, while scholarly journals and books provide greater substance and depth to the topic.

Is the writing objective and unbiased? Does the author present both sides (and all sides) accurately? As you know, if you read a position or argumentative paper on a topic, the author has adopted a stance. Therefore, you need to ensure that you read other views on the topic and understand the opinions of other authorities.

Dear Colleague, after you have evaluated the various resources on your topic, begin reading the information and take notes on what you read. Eventually you will organize these notes to create the structure of your report.

✎ Activity 24., how many ways of citing, from books, magazines and other written sources are you familiar with?

There are three basic ways to take notes when researching a topic. In each case, you will want to write down the specifics about where you found the information as well as the corresponding page numbers so that you can cite the author appropriately. Good records about references and citations help to prevent plagiarizing information.

Summarize. When you summarize information, you condense a lengthy description or argument into a few sentences so that you communicate only the main ideas.

Paraphrase. When you paraphrase information, you explain someone else's ideas in your own words without condensing it substantially.

Direct Quote. When you quote someone directly, you use his or her exact words. Typically, direct quotes are used when you feel the original author's exact words are important.

Specific page numbers for paraphrased or quoted material appear within the parenthetical citation following the abbreviation for page (p.). The location of the parenthetical citation for a quote depends upon the placement of quoted material within the sentence:

- If the quotation appears in mid-sentence, insert the final quotation mark, followed by the parenthetical citation; then complete the sentence.

Example: Specific page numbers for paraphrased or quoted material appear within the parenthetical citation following the abbreviation for page (p.). The location of the parenthetical citation for a quote depends upon the placement of quoted material within the sentence:

- If the quotation appears in mid-sentence, insert the final quotation mark, followed by the parenthetical citation; then complete the sentence.
- If the quotation appears at the end of the sentence, insert the final quotation mark, followed by the parenthetical citation and the end punctuation:
- If the quotation is long, it should be formatted as a block quotation, and the parentheses should appear after the final punctuation mark:
- If the quotation is long, it should be formatted as a block quotation, and the parentheses should appear after the final punctuation mark:

Example:

✏️ Activity 25. What do you think is the importance of ensuring accurate and appropriate data collection?

Regardless of the field of study or preference for defining data (quantitative, qualitative), accurate data collection is essential to maintaining the integrity of research. Both the selection of appropriate data collection instruments (existing, modified, or newly developed) and clearly delineated instructions for their correct use reduce the likelihood of errors occurring.

Consequences from improperly collected data include

- inability to answer research questions accurately
- inability to repeat and validate the study
- distorted findings resulting in wasted resources
- misleading other researchers to pursue fruitless avenues of investigation
- causing harm to human participants and animal subjects

While the degree of impact from faulty data collection may vary by **discipline** and the nature of investigation, there is the potential to cause disproportionate harm when these research results are used to support public policy recommendations.

Issues related to maintaining integrity of data collection:

The primary rationale for preserving data integrity is to support the detection of errors in the data collection process, whether they are made intentionally (deliberate falsifications) or not (systematic or random errors).

Scholars describe ‘quality assurance’ and ‘quality control’ as two approaches that can preserve data integrity and ensure the scientific **validity** of study results. Each approach is implemented at different points in the research timeline

One major point that can help you in maintaining integrity of data collection is designing and using good data collection instruments. Triangulating is studying the research question from at least three separate pieces of data and three points of view.

A good data collection instrument, particularly for interview and questionnaire, can be analyzed by its:

Purpose

A good data collection instrument must serve two purposes. First, it must translate the objectives of an investigation into specific questions, the answers to which will provide the data necessary to test the hypotheses and explore the area defined by the objectives. Each question should relate the corresponding objective so that the response obtained can be analyzed and interpreted accordingly. The research objectives and specifications of the data required must precede the construction of questionnaire. Secondly, the instrument must motivate the respondents to communicate the required information. It is essential to include a courteous and carefully constructed covering letter to explain the purpose and importance of the study. The covering letter should assure the respondent that delicate information will be held in strict confidence.

Language

The language of a good data collection instrument should be concise and directed towards producing uniformity of understanding among the respondents. The vocabulary should be simple and within the easy grasp of the least intelligent of the group under study.

Vague phrases and expressions should be avoided. Technical expressions should be used only if the inquiry is directed to a select group which is well- versed in the technical language used. Proverbs and quotations should be avoided. Subjective words, such as ‘bad’, ‘good’, ‘fair’ and the like do not lend themselves to quantitative measurements nor qualitative analysis unless they are used for comparisons on a rating scale.

Frame of Reference

The respondent’s frame of reference influences his/her answers. Complex questions that require the respondent to go through several steps of reasoning before answering are undesirable and have often resulted in misleading information. For example, in a question like ‘should study centers modernize their teaching?’, the word ‘modernize’ may have different connotations for different people.

Questions on controversial issues should be broken down into components, so that the researcher can determine the respondent’s feelings about various aspects of the problem, including those which he/she refuses to comment upon. A series of specific questions is needed so as to uncover degrees of intensity of feeling or conviction. For example, questions to understand people’s attitude towards the reservation policy have to be broken into issues like equality, equity, equal opportunities, etc.

The required answers should be within the informational domain of the respondents. For example a question ‘Do you read’ ‘Times’ may not bring the desired response as many respondents may not know about this magazine. The length of the questions and statements used should be governed by a reliable estimate of the respondents’ comprehension level.

Arrangement of Questions

The arrangement or ordering of questions should receive special attention. It should appear logical to the respondents. The questions placed first in the questionnaire should be the easiest to answer. 'Interest-generating' questions should be asked at the beginning. A proper sequence of questions proceeds from the general to specific, from simple to complex ones, from those that will create favorable attitude to those that may be somewhat delicate or sensitive.

Length of the questionnaire and interview question

A questionnaire and an interview question should not be longer than necessary. The total number of questions must not be too many to tire or bore the respondents. If too many questions are asked and the respondent becomes tired, the questions at the end of the series may not be well answered. If it is necessary to include a large number of questions, it is advisable to have separate questionnaires.

Form of response

The form in which the responses are recorded must be integrated with the form of the questions. There should be no hesitation in asking for responses in different forms in the same questionnaire, since it is frequently found that one form is better than another for questions about different aspects of the same subject. Questions requiring answers like 'yes' or 'no' are subject to least bias. These responses are easy to tabulate. However, they do not always yield sufficient information on the subject under study. In such cases, the use of multiple choice responses is desirable. Questions that present multiple choices to the respondent are effective when the choices are few and easy to follow.

4.2.1 Data Collection Instruments

✎ ***Activity 26. What are Main Types of Data Collection?***

while sample selection is obviously part of the data collection process, the methods of data collection section must provide readers with specific information about the rest of the data-collection techniques. For instance, whether the the research involves a survey or some other method of observation, whether the research used questionnaires or interviews, what conditions existed when data are collected, whether certain dimensions of the research design have to be compromised, whether the research experienced data loss from its beginnings to completion, and so on. A detailed statement of data-collection techniques is essential if readers are to develop confidence in the study.

Researchers must disclose known defects as well as positive aspects of the data-collection procedures used. This attention to strengths and liabilities applies to your collection of original data or to your use of secondary sources. Most of the time secondary sources will contain all the details about how the information was collected and manipulated. You need not to report all of this information, but be sure to include a citation so readers can find it if they so desire.

Primary and Secondary sources of Information.

It is convenient to think of all information as being classified generally into two kinds. We can call them primary and secondary, or firsthand and secondhand. Their value, like that of an automobile, decreases materially in direct proportion to the number of owners.

Remember that information which comes to us from primary sources is usually the best and the purest, in much the same manner as water drawn from a mountain spring. The closer you are to its point of origin, the less chance it has to become contaminated.

The following is a list of such primary sources arranged approximately in order of their purity and value:

1. Experimentation
2. Firsthand investigation: the interview and the questionnaire

Interactive interviewing	People asked to verbally describe their experiences of phenomenon.
questionnaire	People asked to write descriptions of their experiences of phenomenon.
Observation	Descriptive observations of verbal and non-verbal behavior.

4.2.1.1 Observation

4.2.1.1 Observation

Observation may be defined as a process in which one or more persons observe some real-life situation and record pertinent occurrences. Observation involves quick noticing of things. The researcher is expected to see, watch carefully and notice for the information (data) required in the study. Observation can be considered as a first hand or primary source. It is used to evaluate the overt behaviour of the individuals in controlled and uncontrolled situations.

Types of observation

Observations may be classified into two types:

a) Participant observation

b) Non-participant observation

Participant observation: In the process of 'participant observation' the observer becomes more or less one of the group members and may actually participate in some activity or the other of the group. The observer may play any one of the several roles in observation, with varying degrees of participation, as a visitor, an attentive listener, an eager learner, or as a participant observer.

Non-participant observation: In the process of 'non-participant observation', the observer takes a position where his/her presence is not felt by the group. He/she may follow closely the behavior of an individual or characteristics of one or more groups. In this type of observation, a one-way 'vision screen' permits the observer to see the subject but prevents the subject form seeing the observer.

Some reports will be based on your personal observation. **For example**, you may be required to write a report on an experiment you conducted in the laboratory, a job you performed or an event to which you were an eyewitness. Since in these cases you will have to rely on your sensory perceptions and memory you must develop the habits of careful observation and accurate recording-traits, which even, otherwise are helpful. Lest you should be caught unaware, it is advisable to keep a notebook and a pencil ready for writing down notes.

Planning for observation

Planning for observation includes definition of specific activities or units of behavior to be observed; the nature of the groups of subjects to be observed; the scope of observation individual or group;

determination of the length of each observation period; and deciding about the tools to be used in making the observation and recording.

Execution of observation

The expert execution of observation includes:

a) proper arrangement of specific conditions for the subject or subjects to be observed,

b) assuming proper role or physical positions for observing,

c) focusing attention on the specific activities or units of behavior under observation,

d) proper handling of recording instruments to be used, and

e) utilizing one's training and experience fairly effectively in terms of making the observation and recording the facts.

Recording and interpreting the observation

Recording of the observation data should take place either simultaneously or soon after the observation. In the former case, the observer goes on recording his/her observation data simultaneously with the occurrence of the phenomenon observed. In the latter case, the observer undertakes to record his/her observations not simultaneously with the actual event, but immediately after he/she has observed for a certain period of time while the details are still fresh in his/her mind. In viewing, classifying and recording behavior, the observer must take utmost care to minimize the influence of his/her biases, attitudes and values on the observation report. The observer should know what he/she is looking for in a given situation and should carefully and objectively record the relevant data. Subjectivity on the part of an observer may partly be due to his/her emotional involvement, his/her selective perceptions and memory. In order to overcome these biases, various mechanical instruments are used to obtain more accurate records of events. The use of cameras, tape-recorders, stop-watches, binoculars, audiometer, one-way vision screens, mirrors, etc., allows behaviour to be measured to a degree of accuracy with cannot be achieved by the unaided human observer. It is worthwhile to develop an "observation schedule" like a question schedule for making and recording observations. The specific behaviours to be observed and recorded should be listed in this schedule.

Uses and limitations of observation

Uses

i) Observation provides a direct method for studying various aspects of human behavior. Indeed, it may be the only effective way to gather data in a particular situation e.g., behaviors of a counselors in actual counseling sessions.

ii) Observation enables the researcher to record behavior at the time of occurrence.

Limitations

i) A subject may intentionally attempt to exhibit artificial behaviour when he/she knows that he/she is being observed.

ii) It is time consuming and costly.

4.2.1.2 The Questionnaire

‘Questionnaire’ is a commonly used and frequently abused tool for gathering a variety of data. A questionnaire may include of a series of questions pertaining to psychological, social, educational, or any such issues which are sent to an individual or a group, with the aim of obtaining relevant data on the topic of research.

It has to be short; there should be only one possible interpretation for each question. Design it so that it is simple and quick to fill out and so that it will draw out the specific information you are seeking. Avoid leading questions.

Types of questionnaires

Questionnaires can be classified in various ways. Here we confine ourselves to structured and unstructured questionnaires.

Structured questionnaires are those which pose definite and concrete questions. They are prepared well in advance and not on the spot. Additional questions may be used only when there is a need to clarify vague or inadequate replies by respondents or when further details are needed. The form of questions may require responses which are either closed or open.

Closed-form of questionnaires are used when categorized data are required. They include a set of questions to which a respondent can reply in a limited number of ways ‘yes’, ‘no’, ‘no-opinion’, or an answer from a short list of possible responses. He/she is asked to put a tick ( ) mark in a space provided on the answer sheet or is requested to underline a response. Sometimes he/she is asked to insert brief answers of his/her own.

The open ended responses on the other hand are free and spontaneous expressions by the respondent to the questions posed to him/her. The open-ended responses are used mainly for intensive study of a limited number of cases or preliminary exploration of new problems and situations. At times, the respondent is asked to write a descriptive essay and express his/her viewpoints or report on details and events, without restrictions imposed as in the case of closed questions.

Unstructured questionnaires are frequently referred to as interview guides. They also aim at precision and contain definite issues that are covered while conducting an interview. Flexibility is the chief advantage of the unstructured questionnaire. It is designed to obtain viewpoints, opinions, attitudes and to show relationships between various types of information which might escape notice under more mechanical types of interrogation. No predetermined responses are provided; instead, free responses are solicited.

Uses of questionnaire

1. A questionnaire is a popular means of collecting different kinds of data in research. It is widely used in educational research to obtain information about certain conditions and practices, and to inquire into the opinions of an individual or a group.

2. A questionnaire is administered personally either individually or to a group of individuals or is mailed to them to save a great deal of time and money in travel. In the former situation, the person administering the tool has an opportunity to establish rapport with the respondents, to explain the purpose of the study to the respondents and to explain the meaning of questions which may not be clear to them. In the latter situation, mailed questionnaire is mostly used when the individuals cannot be contracted personally. The range of administration of a mailed questionnaire may be national or international.

3. Questionnaires are used both to initiate a formal inquiry and also to supplement and check data previously accumulated. They may pertain to studies of economic or social problems, measurement of opinion on public issues or events, studies of administrative policies and changes, studies on the cost of living, consumer expenditure, child welfare, and numerous other issues.

Limitations of questionnaire

1. A questionnaire cannot be used with children and illiterates.

2. The return of the mailed questionnaire is often as low as 40 percent to 50 percent. As a result of this poor response, the data obtained are sometimes of limited validity. The respondents who return the questionnaires may not be representative of the entire group. It will make the sample a biased one and thus vitiate the findings.

3. Sometimes respondents may not like to respond in writing to questions of intimate and confidential nature or to questions involving controversial issues. For example, it has been experienced that people avoid questions related to marriage and government servants avoid answering question about policy matters of the government.

4. it is sometimes difficult to formulate and phrase questions on certain complex and delicate problems.

There is no check on the respondent who misinterprets a question or gives incomplete or indefinite responses.

4.2.1.3 Interview

Interview is a process of communication or interaction in which the subject or interviewee gives the needed information verbally in a face-to-face situation. In a sense, it is an oral questionnaire. In a research situation it may be seen as an effective, informal, conversation, initiated for a specific purpose as it focuses on certain areas. The main objective may be the exchange of ideas and experiences and eliciting of information. It is, in most respects, similar to the questionnaire. However, it is considerably less formal and a great deal more flexible. Much of its value depends upon the personality of the interviewer and his skill in presenting his questions. In many situations the objective of interview is to secure information about the individual's problems, his/her past history, job or family adjustments. In such situations, the major purposes of interviews are diagnosis and treatment. Interviewing is an art that demands intelligence, patience, tact and courtesy on your part. Even though you need one type of data, remember that all individuals cannot be dealt within an identical manner. You should be quick to readjust your approach and attitude to suit each case. You must first secure a person's attention, excite his interest and establish a report. He must open out to you and cooperate in giving you the information you want. And all this depends on your own knack of handling people.

✎ Activity 27. What major things do you need to consider before you conduct an interview?

First, you must do your own homework well. You must first of all break up your problem into significant components. It becomes easier to handle it in this manner. Then carefully prepare a set of questions covering all these aspects. Next, you must fix an appointment before calling on a person. Tell him beforehand the amount of time you are likely to take. Before starting the interview, give him a brief introduction about yourself, your organization and the purpose for which you are collection data. Do not embarrass him by asking personal questions or by seeking confidential information. If a person is hesitant about imparting a piece of information, do not press him, if he would like to be anonymous, respect his wish. Lead the interview but do not talk more than is essential to elicit the information.

Maintain a comfortable atmosphere throughout the interview and end the meeting gracefully. Do not prolong the interview unnecessarily.

Sometimes this method of collecting data is used to supplement information collected through questionnaires, through this method you can get qualitative data, which cannot be obtained, from yes/no answers or the choice of alternatives given in the questionnaire. Sometimes the answer given may not be clear. Through personal interview you can put supplementary questions and get the exact information you need. You get not only what is being said but also the exact feel of it. This will help you to draw your own conclusions better.

A list of questions, sometimes referred to as “the interview schedule,” should be prepared in advance, keeping in mind the same cautions that were presented above in connection with the questionnaire.

In order to establish a good rapport, the interviewer should greet the interviewee in a friendly manner so as to get settled in a relaxed manner. As an interviewer you should observe the following rules in order to elicit effective responses:

- Ask only one question at a time
- Repeat a question if necessary
- Try to make sure that the interviewee understands the questions.
- Listen carefully to the interviewee’s answer
- Observe the interviewee’s facial expressions, gestures, and tone or voice so as to derive meanings from his/her body language.
- Allow the interviewee sufficient time to answer the question, but do not let the interview drag on and on.
- Avoid suggesting answers to questions.
- Do not show signs of surprise, shock, anger, or other emotions if unexpected answers are given.
- Maintain a neutral attitude with respect to controversial issues during the interviews.
- Take note of answers that seem to be vague, ambiguous, or evasive.
- Use tact and skill in getting the subject back to an area of inquiry if he/she has strayed too far from the original question.
- In the unstructured interview, ask additional questions to follow up clues or to obtain additional information.

The advantage of the interview lies in the fact that you can modify your plans and change you questions to follow up leads brought out in the initial responses of your subject. You should try to redirect the interview to more fruitful topics when you feel that the required information is not sufficient. You should wind up the interview before the interviewee becomes tired.

Focused Group Interview

The focused group is a special type of group in terms of purpose, size, composition, and procedures. A focused group is typically composed of seven to twelve participants who are unfamiliar with each

other and conducted by a trained interviewer. These participants are selected because they have certain characteristics in common that relate to the topic of the focused group.

The researcher creates a permissive environment in the focused group that nurtures different perceptions and points of view, without pressuring participants to vote, plan, or reach consensus. The group discussion is conducted several times with similar types of participants to identify trends and patterns in perceptions. Careful and systematic analyses of the discussions provide clues and insights as to how a product, service, or opportunity is perceived.

A focused group interview can be defined as a carefully planned discussion designed to obtain perceptions on a defined area of interest in a permissive, non-threatening environment. It is conducted with approximately seven to twelve people by a skilled interviewer. The discussion is relaxed, comfortable, and often enjoyable for participants as they share their ideas and perceptions. Group members influence each other by responding to ideas and comments in the discussion.

CHARACTERISTICS OF FOCUSED GROUPS

Focused group interviews typically have four characteristics:

1. Identify the target (people who possess certain characteristics);
2. Provide a short introduction and background on the issue to be discussed;
3. Have focused group members write their responses to the issue(s);
4. Facilitate group discussion;
5. Provide a summary of the focus group issues at the end of the meeting.

Recording of the interview

The recording of the interview is obviously an essential step in interviewing. The interviewer may use a schedule, a structured format, rating scale or a tape recorder to record the responses of the interviewee. The use of a tape recorder during the conduct of the interview not only eliminates the omissions, distortions, elaborations and other modifications of data usually found in written interview responses, but it also provides an objective basis for evaluating the adequacy of the interview data in relation to the performance of the interviewee.

The use of a tape-recorder also permits the interviewer to devote full attention to the interviewee and save much of the time which he/she would otherwise use in writing down the responses during or after the interview. However, if a tape-recorder is not available, the interviewer has to take notes to record the responses.

Uses of the interview

- An interview provides an opportunity to the interviewer to ask questions on various areas of inquiry. It permits greater depth in responses which is not possible through any other means.

- An interview is not an entirely independent tool of research for gathering information pertaining to feelings, attitudes or emotions. It is supplementary to other tools and techniques. A combination of interviewing, observations, and statistical techniques often yield the best results, but the balance of emphasis shifts with the frame of reference and objectives of the study. Since an interview is a highly flexible tool in the hands of skillful interviewers, it allows a more liberal atmosphere than in the use of other techniques of investigation. Questions not readily grasped by interviewees can be rephrased or repeated with proper emphasis and explanations when necessary.

- An interview is an effective tool for a social scientist in the study of human behavior. Through this technique, a researcher can secure very intimate and personal knowledge about the subject of his/her study, which is denied to the natural scientist, who cannot communicate with the subjects despite all the instruments of precision.

Limitations of the interview

In spite of many uses of the interview method, it is not without limitations that jeopardize its value, even when it is used as a supplementary research technique.

- Interview is a time consuming technique.

- The effectiveness of the interview depends greatly upon the skill of the interviewer which everyone does not ordinarily possess. It takes time to master this skill.

- There is a constant danger of subjectivity on the part of the interviewer.

- An interview is very difficult to employ successfully because even in the presence of a skilled interviewer some interviewees do not respond freely, frankly and accurately.

- Since memory and retention are highly selective processes, interviewees generally provide accurate and vivid accounts of the most recent or intense experiences, or of situations that they encounter most frequently. Painful or embarrassing experiences are forgotten or consciously avoided by the interviewees. In such cases the responses lack accuracy.

✎ Activity 28. Dear Participant; let us see the following example of Data Collection

Imagine the data is collected from “Bishoftu” secondary school. It is about the reading program in teaching English language.

"In conjunction with administrators from “Bishoftu” secondary school, the researcher developed a set of questions and issues to discuss at group and individual interviews, wrote protocols to ensure consistency in data collection activities during classroom observations, and developed a teacher/staff questionnaire related to reading practices and instructional approaches.

Six primary research questions guided us in creating the data collection instruments and in analyzing the data. The questions are as follows:

1. What are the critical features of Bishoftu secondary school reading programs?
2. What contextual factors have impact on the implementation of these programs?
3. What strategies within the school are available for supporting the implementation of the programs?
4. How consistently are programs being implemented within and across grade levels?
5. What is the impact of various program components on student achievement in reading?
6. How well is reading achievement currently being measured?

The example shows the preplanning involved in data collection and indicates how to ensure that this phase of the process is directly related to research questions.

Data Collection Instruments. The following data collection instruments were developed for this study:

Group Interviews with Teachers. To identify the major issues surrounding the planning and implementation of reading instruction in the school, group interviews were conducted with reading/language teachers and staff. An interview protocol was used to facilitate discussion. Responses to the uniform standard questions contributed to the final analysis of the study findings.

Individual Interviews with Teachers. To investigate all of the major research questions presented earlier, individual 20-minute interviews were conducted with teachers and staff. An interview protocol was used to facilitate information gathering. Responses to the questions contributed to the final analysis of the study findings.

Teacher/Staff Questionnaire. A teacher/staff questionnaire designed to gather information relevant to the first two research questions was administered to all. The first part of the questionnaire focuses on general information related to school climate. The second part of the questionnaire focuses on program-specific information by asking reading/language teachers to indicate whether or not certain program characteristics are present in their instruction, and the extent to which they are present (amount of time spent on specific tasks). The researcher also gathered information about the extent to which reading is part of content-area instruction.

Classroom Observation Protocol. Reading instruction was observed in order to understand how instruction is actually delivered. Observation notes were analyzed according to major categories contained in the protocol to reveal themes. These themes were categorized initially as strengths, issues, and opportunities for growth. Additional data collection and review occurred during the observations. Researchers summarized notes after each observation.

The above example in general, shows some data collecting procedure protocols. Implementing the three modes of data collection instruments increases the data collection integrity

4.2.2 Quality Assurance

Since quality assurance precedes data collection, its main focus is **'prevention'** (i.e., forestalling problems with data collection). Prevention is the most cost-effective activity to ensure the integrity of data collection. Poorly written manuals increase the risk of failing to identify problems and errors early in the research endeavor. These failures may be demonstrated in a number of ways:

- Uncertainty about the timing, methods, and identify of person(s) responsible for reviewing data
- Partial listing of items to be collected
- Vague description of data collection instruments to be used in lieu of rigorous step-by-step instructions on administering tests
- Failure to identify specific content and strategies for training or retraining staff members responsible for data collection
- Obscure instructions for using, making adjustments to, and calibrating data collection equipment (if appropriate)
- No identified mechanism to document changes in procedures that may evolve over this book of the investigation.

Dear, an important component of quality assurance is developing a rigorous and frequent visit to the research advisor(s). Consulting research advisors is the need to effectively communicate the value of accurate data collection to your findings readers. The consulting aspect is particularly important to address the potential problem of the researcher who may unintentionally deviate from the original protocol. This phenomenon, known as **'drift'**, should be corrected with the discussions made between the researcher and the advisor(s).

Given the range of qualitative research strategies (non-participant/ **participant observation**, interview, archival, field study, ethnography, **content analysis**, oral history, biography, unobtrusive research) it is difficult to make generalized statements about how one should establish a research protocol in order to facilitate quality assurance. Certainly, researchers conducting non-participant/participant observation may have only the broadest research questions to guide the initial research efforts. Since the researcher is the main measurement device in a study, many times there are little or no other data collecting instruments. Indeed, instruments may need to be developed on the spot to accommodate unanticipated findings.

4.2.3 Quality Control

While quality control activities (detection/monitoring and action) occur during and after data collection, the details should be carefully **documented**. A clearly defined communication structure is a necessary pre-condition for establishing monitoring systems. There should not be any uncertainty about the flow of information between principal researcher and the readers following the detection of errors in data collection. A poorly developed communication structure encourages lax monitoring and limits opportunities for detecting errors.

Quality control also identifies the required responses, or 'actions' necessary to correct faulty data collection practices and also minimize future occurrences. Examples of data collection problems that require prompt action include:

- errors in individual data items
- systematic errors
- violation of protocol
- problems withindividual staff or site performance
- fraud or scientific misconduct

in the social/behavioral sciences where primary data collection involves human subjects, researchers are taught to incorporate one or more secondary measures that can be used to verify the quality of information being collected from the human subject. ***For example***, a researcher conducting a survey might be interested in gaining a better insight into the *occurrence of risky behaviors among young adult as well as the social conditions that increase the likelihood and frequency of these risky behaviors.*

To verify data quality, respondents might be queried about the same information but asked at different points of the survey and in a number of different ways. Measures of **'Social Desirability'** might also be used to get a measure of the honesty of responses.

There are two points that need to be raised here, 1) cross-checks within the data collection process and 2) data quality being as much an observation-level issue as it is a complete data set issue. Thus, data quality should be addressed for each individual measurement, for each individual

observation, and for the entire data set.

Each field of study has its preferred set of data collection instruments. The hallmark of laboratory sciences is the meticulous documentation of the lab notebook while social sciences such as sociology and cultural anthropology may prefer the use of detailed field notes. Regardless of the discipline, comprehensive documentation of the collection process before, during and after the activity is essential to preserve data integrity.

4.3 4.3 Data Analysis

The word "analyze" means to break something down into its parts. A meaningful analysis identifies the parts and demonstrates how they relate to each other. You may have information from different sources which examines different aspects of your topic. By breaking down the information, you may be able to see relationships between the different sources and form them into a whole concept.

When you're trying to make sense of the information coming out of your research process, you often have to look at it from different perspectives and sometimes have to step back and try to get a "big picture" view. Some ways to do this are to try out different organization patterns: compare and contrast, advantages and disadvantages, starting from a narrow premise and building on it, cause and effect, logical sequence.

After looking into the preliminary data, you, as a researcher, will want to consider what appears to be emerging. Your description of analysis techniques provides readers with information about specific techniques you intend to use. These are important considerations when analyzing data. For example, your analysis may involve using percentages, ratios, cross-tabulations, measures of association, or chi-squares. If you intend to use a complicated or unfamiliar technique of analysis, readers should be informed as to literature sources so they may consult these for clarification. This permits them to review or learn more about the techniques you will employ.

Qualitative research commonly generates substantial textual and other non-numerical material, which requires systematic organization, interpretation, and analysis. Researchers therefore need to plan carefully for the volume and complexity of data that may emerge during a qualitative study (Creswell & Poth, 2018).

Ways to Analyze Data

You need to be familiar with several ways of analyzing the data you have collected. They, in one way or another, contribute to the reliability and validity of your research findings. In that case, you have to apply your knowledge of triangulation.

Triangulating is studying the research question from at least three separate pieces of data and three points of view. Those three pieces of data can be, **for example,** the researcher's observations, recorded comments by the participant subjects/respondents (such as the tape recorded comments or quotes you, as a researcher, has noted) and examples of participants' work. Does the research question still fit the data that is emerging from the study?

Compare. Look through the data and keep comparing the data that was collected earlier in the study with data collected later in the study. Use different bases for comparison. **For example**, if you, as a researcher, have compared the students' academic performances in writing and speaking, in

October with what they did in January, you may try comparing the students' written work with their oral work.

Categorize and sort. Set up charts, columns, outlines, and ways of counting occurrences. You can make up different categories that fit the situation(s) or use categories developed by another researcher. Then, dear researcher, you have to decide on a way to order the data findings such as data chronologically, by importance and by frequency (how often an occurrence occurs).

Contrast. Look for what doesn't fit the assumptions or theories of other researchers and note what sticks out, goes against the research topic. Then after, try out different hunches about what the data means. Make an educated guess and then see if it's supported by the data. At this time, don't stick rigidly to an assumption or hypothesis that was originally held.

Restate the question. Rewrite the question many times; change it when necessary to fit what is important from the data that has been collected. What is it that you, as a researcher, really want to figure out? Sometimes you want to make the question more global, sometimes the question may become more tightly focused.

Visualize. Create a visual representation for what you have collected. Map out your data; draw it all on one page. Sketch the metaphors that come to mind when thinking of the data and what it all means. Use diagrams, sketches of things, people, happenings to show different ideas and groupings

Talk and validate. Dear Colleague, talk to others about your research. Explain the data interpretations to others; see if they can see the same things. Consider their different interpretations and use them to clarify, broaden and otherwise validate the findings.

Take a break. Dear Colleague, a research work is not a personal essay. It is rather a critical analysis of hypothesis proposition about presumed relations. There for,

sometimes, it is advisable to step back from the research process in order to clear your mind and give yourself a rest. Coming back to the process with a refreshed outlook will often lead to new understandings and perspectives.

4.4 Data Presentation

Because most technical reports rely on figures and tables for the presentation of data, the form and quality of the figures and tables are important in establishing the style and readability for the report. Good judgment should be used in selecting both the data to be presented and the method of presentation. Use only figures and tables that add to the value of your report. Present the data as simply and straightforwardly as possible so that your readers can easily grasp the significant points. Present data in the text, or in a figure, or in a table-but never in more than one way.

Before beginning to write the report, carefully select the data to include. Most carefully prepared programs yield more data than are needed to support the conclusions. Including all your data in the report is unnecessary. Use only data that are directly pertinent to your conclusions, and do not try impress readers with how much data you have collected. Quantity is no substituted for quality in presenting technical results.

Once you have selected the data to be included in your report, decide how they can best be presented. Should they be tabulated or potted? To answer this question, consider you readers needs. Do they need to know exact values? If so, tabulate your results. If relative trends are more important, use

graphs. Both the figures and tables should be as self – explanatory as possible and arranged logically to tell the main points of you story without reference to the text.

Figures

The figures used in technical reports generally are of three types-graphs, Drawings, and photographs. Figures are numbered with Arabic numerals in the order of their appearance. In the final report they are either inserted in the text near (preferable following) their first mention or grouped together at the back. Sketches are lettered consecutively ((a), (b), (c), etc.) if they are referred to more than once. Under no circumstances should the arrangement of black and white figures or the parts of one figure be out of sequence. Figures arranged in a group are in sequence form top to bottom or form left to right. Exceptions are sometimes made for color figures to reduce the number of pages printed in color.

Prepare figures with consideration for their appearance in the final printed documents the size of the printed figure including the legend (title) cannot exceed the dimensions of the report image area (71/8 by 91/8 in. in NASA reports). Within these limits various sizes, proportions, and arrangements of figures are possible. (A large, complex figure may be reproduced on facing pages.)

All figures must have legends; if a if a figure has parts ((a), (b), (c), etc.), it must have corresponding sub legends. Use similar wording in the legends of related figures. After you have assembled the rough draft of the report, thumb through the figures and tables, reading merely the title of each to make certain that the format and the nomenclature are consistent. Conditions applying to the entire figure or to a part are normally stated as part of the legend sub legend. But when the same conditions apply, for example, to every graph in a report, they are best stated once in the text.

Graphs

Graphs should be clear and simple with as few data curves as possible. It is usually best to have no more than six types of lines. Try to avoid interlaced or unrelated curves. As few words (labels) as possible should be inserted directly on the figure. Equations should be placed in the text, lengthy tabular material should be presented in a separate numbered table, and explanations and conditions should be added to the legends or placed in the text.

Choose coordinates that will give your readers a physical feel for the variables being presented. Clearly label what is plotted and the units used. Whenever possible plot all parts of any one figure or related figures on scales with the same increments. Add auxiliary scales at the left and bottom of the figure if tree are four or fewer scales place additional scales at the right or top. For ease in interpolation divide scales anthological, consistent increments. For example, when both U.S. customary and SI units are used, each scale must stand alone. Do not simply convert the values on one scale into the other system of units. Such a scale is useless to the reader.

Use the same data symbols and lines to represent the same conditions consistently thought the graphs of you report. The following data symbols and types of lines are commonly used:

Do not use the symbols + and x on figures with grid, and avoid solid or partly solid symbols if symbols overlap. The curves and data points may be identified by keys or labels. Keys are preferred when several curves must be distinguished or when several conditions are associated with each curve. Keys generally follow the format for tabular material and should be consistent throughout a set of figures.

Drawings

When you use drawings or sketches to illustrate test equipment, try to keep them simple. Include only those features of the equipment that are essential to your readers' understanding, and avoid unnecessary detail. Arrange with the Publishing Services Coordination Office for complex drawings to be prepared by a technical illustrator while the report is in the rough-draft stage, if possible, to allow adequate time for the illustration to be prepared.

Photographs

When your research project has "jelled," consult the Imaging Technology Center concerning the best way to record the data photographically, if applicable, and to show your apparatus and research facilities to the best advantage. Photographs of similar objects should be sized for compatibility. Glossy prints taken with back-and – white film reproduce best. Prints that have already been screened are not usable. The use of color in printing is discouraged because it greatly increases publishing costs.

Do not include a photograph of equipment which is so elementary that a sentence would describe it. Label the most important features being sown. Remember, equipment that seems simple to you may be complex to readers who are not familiar with it. Limit the labeling and the field of view to the main items discussed to avoid confusing reader's with extraneous items. Mark up a copy of the photograph rather than the glossy print.

If your photographs are Polaroid prints, have negatives and additional prints made before submitting them for use in use in report, for slides, etc. you are then protected in case of damage or loss, and prints are readily available of additional uses.

Include some object scale in the photograph to help your readers judge the size of the objects shown. Foe photomicrograph and electron micrographs use a scale instead of stating the magnification.

Tables

Tables are often included in technical reports to present data in an exact, highly concentrated form. But because tabulated data are so concentrated, many readers have difficulty grasping their significance. Tables are therefore the least preferred method of transmitting results to readers and should be used only when absolutely necessary. When you use tables, make them as brief and simple as possible. Otherwise your readers may not bother studying the detailed columns of figures, and you will have wanted your time in presenting the data. "Whenever a table, or columns within a table, can readily be put into words, do it" (ref.2)

Tables are numbered in the order of their mention, in Roman numerals except when a report contains 20 or more tables. Then Arabic numerals are used. Similar data at different conditions are organized into parts ((a), (b), (c), etc.) of the same table with subtitles. Numbered tables must have titles.

Present tabulated material in an organized manner. Like elements should read down not across. Variables are usually given in columns topped by box heads, with the constants given in the first, or stub, column. Box heads should be brief; if necessary, they may be amplified by footnotes. Box heads usually contain a word description of a concept or quantity, this symbol, and its unit, separated by commas; symbols must be defined when they are used. Arrange tabulated data in a logical order that you riders can easily recognize. Usually this arrangement is an ascending or descending order of values for the prime parameter. The order is necessary to clarify trends. You can also help your readers see relations and comparisons of data by carefully working the box heads and the stub column. Put items to be compared in adjacent columns. Generally numbers in columns are more easily

compared than numbers in rows. Another type of table is the leader work table, in which dissimilar data are listed in rows with leader dots connecting each parameter with the corresponding value.

Raw Data Chart		Condition A (1-3)			Condition B (4-6)	
Participant	Word List 1	Word List 2	Word List 3	Word List 4	Word List 5	Word List 6
1	12 secs	13 secs	12 secs	15 secs	14 secs	16 secs
2	12 secs					
3	12 secs					
4 etc	12 secs					

Give conditions that apply to an entire table in a head note. Indicate footnote citations by lower-case letters (superscripts) ordered across the table form left to right and top to bottom.

Technical Film or Videotape Supplements

Where dates can be more efficiently presented or concepts explained through motion pictures, consider the use of a technical film or videotape supplement to your report. For expert production assistance consults the Imaging Technology Center's video department. These films and videotapes are described in a catalog (ref.7) and made available on loan for nonprofit, noncommercial screening. (Catalogs of recent NASA Glenn and NASA- WIDE videos are available on the WWW.) Many requests are received form universities, industrial firms, and Government research organizations.

Microfiche Supplements

A microfiche supplement, like a technical film or videotape supplement, can make available considerably more graphic and textual information than the basic research document. Microfiche supplements are typically used for extensive sets of tables or figures and comprehensive bibliographies.

CD Supplements

CD Supplements, to document lengthy material electronically instead of in printed for can be ordered through the imaging Technology center.

Computer Program

Reports and papers should not include computer programs. Computer programs are to be distributed by the Computer Software Management and Information center (COSMIC), a NASA facility at the university of Georgia, In order to erasure and expedite the availability f a computer program to U.S. users, obtain a LEW-number form the Commercial Technology Office before submitting the publication to publishing Services for processing. (Reference the LEW number in the report.) The Commercial Technology Office will prepare a Tech Brief and send it and the program to COSMIC.

Trade Names

Use of trade names is discouraged because NASA considers it improper to advertise, endorse, or criticize commercial products in its publications. Use generic names whenever possible. Trade names may be used if their use is the only way to specify material or equipment that is necessary to reproduce the results. The first appearance of a trade name in the text must be accompanied by the name of its registered wormer (e.g. International Nickel Co., Inc., IN738). But the symbol for a registered

trademark (a Ringside a superscript circle) is not used. Those reports using trade names must include a trade name disclaimer:

Trade names or manufacturers' names are used in this report for identification only. this usage does not constitute an official endorsement, either expressed or implied, by the National Aeronautics and Space Administration.

The disclaimer is printed on the back of the title page of a NASA-series technical report or as a footnote to the title of a journal article or technical society presentation

A sample is a finite part of a statistical population whose properties are studied to gain information about the whole. And Sampling is the act, process, or technique of selecting a suitable population, or a representative part of a population for the purpose of determining parameters or characteristics of the whole population.

Sampling error comprises the differences between the sample and the population that are due solely to the particular units that happen to have been selected. There are two basic causes for sampling error. *Chance* and *sampling bias*

The other main cause of unrepresentative samples is non-sampling error. A non-sampling error is an error that results solely from the manner in which the observations are made

Like sampling error, non-sampling error may either be produced by participants in the statistical study or be an innocent by-product of the sampling plans and procedures.

There are three primary kinds of samples: the convenience, the judgment sample, and the random sample.

A systematic random sample is obtained by selecting one unit on a random basis and choosing additional elementary units at evenly spaced intervals until the desired number of units is obtained.

A stratified sample is obtained by independently selecting a separate simple random sample from each population stratum.

A cluster sample is obtained by selecting clusters from the population on the basis of simple random sampling.

Regardless of the field of study or preference for defining data (quantitative, qualitative), accurate data collection is essential to maintaining the integrity of research.

There are three modes of data collection. These are: Observation,interview and questionnaire.

Observation may be defined as a process in which one or more persons observe some real-life situation and record pertinent occurrences.

It classified into two types:

- ✓ Participant observation
- ✓ Non-participant observation

Uses

❖ Observation provides a direct method for studying various aspects of human behavior. Indeed, it may be the only effective way to gather data in a particular situation e.g., behaviors of a counselors in actual counseling sessions.

❖ Observation enables the researcher to record behavior at the time of occurrence.

Limitations

❖ A subject may intentionally attempt to exhibit artificial behaviour when he/she knows that he/she is being observed.

❖ It is time consuming and costly.

A questionnaire may include of a series of questions pertaining to psychological, social, educational, or any such issues which are sent to an individual or a group, with the aim of obtaining relevant data on the topic of research.

Types of questionnaires

✓ *Structured questionnaires*

✓ Unstructured questionnaires

Uses of questionnaire

It is widely used in educational research to obtain information about certain conditions and practices, and to inquire into the opinions of an individual or a group

The range of administration of a mailed questionnaire may be national or international.

Questionnaires are used both to initiate a formal inquiry and also to supplement and check data previously accumulated

Some Limitations of questionnaire are: it cannot be used with children and illiterates, sometimes respondents may not like to respond in writing to questions of intimate and confidential nature or to questions involving controversial issues, and it is sometimes difficult to formulate and phrase questions on certain complex and delicate problems

Interview is a process of communication or interaction in which the subject or interviewee gives the needed information verbally in a face-to-face situation.

The advantage of the interview lies in the fact that you can modify your plans and change you questions to follow up leads brought out in the initial responses of your subject.

Advantages

- High response rate
- Can control order of responding
- Respondent can ask for clarification
- Interviewer can ask follow-up questions

Disadvantages

- May suggest the desired response (nonverbally)
- People may give socially desirable responses They may not want to express their true beliefs to the interviewer
- Difficult to ask about embarrassing topics
- No anonymity for the respondent

The second stage of report preparation, data analysis and sorting, is probably the most difficult because it requires considerable mental effort to decide what you want to tell your readers. The beginning of this stage overlaps the data-gathering stage, for data analysis should begin as the data are collected. But the bulk of data analysis must be done near completion of the program. Your description of analysis techniques provides readers with information about specific techniques you intend to use. At this time reexamine the pertinent data and review your earlier opinions with respect to subsequent results.

During this data review the program conclusions should be drawn. This is the most important step in report preparation because the conclusions are the reason for the report and the basis for report preparation: They dictate what to include in a report and how to organize it. Trying to organize and write a report without knowing the conclusions is like starting an automobile trip without knowing the destination. You will not know where you are going, and you will never know when you get far enough to stop. Therefore choose your report destination early by drawing and clearly defining the program conclusions before you begin to organize and write the report. This is best done by first writing down all significant results in no particular order and then sorting them so that the results pertaining to a common factor are grouped together. Once the conclusions are drawn, list them in descending order of importance.

Selection of the data to be used in the report is another important part of this step. Choose only the data necessary to help your readers reach the conclusions you are drawing. Excessive data or data only loosely related to the conclusions will obscure them and confuse your readers. Of course, do not hide contradictory results. When definite contradictions exist, clearly alert your readers to this fact.

The next step in data analysis involves organizing the selected data into illustrations for the report. Sometimes the figures and tables prepared during the program can be used with only minor modifications. But usually these data-book illustrations contain extraneous information. And they seldom are arranged to emphasize the significance of the data and the corresponding results. For example, although data tabulated during a program are commonly arranged chronologically, that is not necessarily the best way to present the data to the reader.

New figures and tables usually must be prepared. Their organization should be carefully considered because illustrations are one of the best means of emphasizing and supporting conclusions.

After the illustrations have been prepared, write the significant points about each on an attached sheet of paper. What is the figure (table) supposed to show? How were the data obtained? Are there any qualifications to the figure (table)? This information will be useful when you begin writing the report. Because most technical reports rely on figures and tables for the presentation of data, the form and quality of the figures and tables are important in establishing the style and readability for the report.

The quality of a study depends on the fit between its research questions, sampling strategy, data-collection methods, and analysis. Researchers should recognize possible sampling and non-sampling errors, select appropriate instruments, and organize evidence carefully. Clear tables, figures, and explanations help readers understand how the evidence supports the report’s conclusions.

- Conclusion

- **Summary**

Data Collection refers to the appropriate choice of the procedures and use of materials from primary and secondary sources. Organizing the material is another common problem that people have when they are writing research report. Proper data collection will allow you to see the major categories of your report, but will not impose an order on them. This will allow you to place your ideas in a sequence most applicable to your purposes.

Aside from organizing materials, always feel free to use proper data collection instruments to help you think, when you are working on a research. Use data collection instruments like: questionnaire, interview and observation as often as you like, particularly when you are stuck, and you will find it more and more useful. In the research work you can control the quality of your work through quality assurance and quality control mechanisms. Quality assurance is an activity that takes place before data collection begins whereas Quality control is an activity that takes place during and after data collection.

the preceding section has covered the most common problems associated with statistical studies. The desirability of a sampling procedure depends on both its vulnerability to error and its cost. However, economy and reliability are competing ends, because, to reduce error often requires an increased expenditure of resources. Of the two types of statistical errors, only sampling error can be controlled by exercising care in determining the method for choosing the sample. The previous section has shown that sampling error may be due to either bias or chance. The chance component (sometimes called random error) exists no matter how carefully the selection procedures are implemented, and the only way to minimize chance sampling errors is to select a sufficiently large sample. Sampling bias on the other hand may be minimized by the wise choice of a sampling procedure.

in short sampling is a systematic way of selecting smaller group from the whole group. It can be said that using a sample in research saves mainly money and time, if a suitable sampling strategy is used, appropriate sample size selected and necessary precautions taken to reduce on sampling and measurement errors, then a sample should yield valid and reliable information.

After data has been properly collected, the next step is interpreting or analyzing which in other words means implying the real meanings of the data collected. A research paper may highlight a particular issue or problem in a field of study. The paper may focus on analysis of the issue and its solutions, possibly from both historical and current perspectives. The success of the paper is often based on how completely and clearly the writer has identified the key aspects of the issue and their significance to the field to which they relate.

✓ Chapter Checklist

Put a ‘ ’ mark against each of the following statements that you are able to answer.

I can:-

- Write steps for data collection.
- List ways of note taking.
- Define Data collection.
- Explain the two approaches of data integrity?
- Define sampling

- Write the purpose of sampling.
- List ways of note taking.
- Explain the two basic sampling errors
- Write steps for data collection.
- List ways of note taking.
- Define Data collection.
- Explain the two approaches of data integrity?
- State about data analysis.
- Explain ways of analyzing data.

◆ Self-Test

Part one Instruction: write a complete answer for the following questions.

1. What is sampling?

2. Write the purpose of sampling

3. Write the two basic sampling errors.

4. What is data collection?

5. What are some questions which are important to ascertain whether the data is useful to your research work?

6.What are the ways of note taking?

7.What is the implication of accurate data collection?

8.What are the two approaches to data integrity?

9.What is data analysis?

10. List some ways of analyzing data.

Part Two: Write true if the statement is correct and false if it is wrong.

1. Your description of analysis techniques provides readers with information about specific techniques you intend to use.

2. One of the limitations of interview is it cannot be used for children.

3. The advantage of the questionnaire lies in the fact that you can modify your plans and change you questions to follow up leads brought out in the initial responses of your subject.

4. There are three primary kinds of samples: the convenience, the judgment sample, and the random sample.

5. A stratified random sample is obtained by selecting one unit on a random basis and choosing additional elementary units at evenly spaced intervals until the desired number of units is obtained.

[are you tiered? If yes, please take a break and come back later, refreshed.]

CHAPTER 5

Research Proposal and Report Writing

Introduction

Research proposals explain what a researcher plans to investigate and how the study will be carried out. Research reports explain what was done, present the findings, and discuss their meaning for the research questions and intended audience. Both require a clear purpose, logical organization, appropriate evidence, and readable language.

This chapter introduces the main parts of proposals and reports and considers planning, drafting, organizing, revising, and editing a research report.

★ Learning Objectives

By the end of the chapter you will be able to:

- Explain the differences between academic reports and other types of reports.
- list the steps of report writing.
- describe some ways of structuring within the report.
- identify the main parts of a research report

5.1 Research Proposal

✎ Activity 29. What is the purpose of a Research Proposal?

A research proposal is a plan for a study. It normally explains the research problem, purpose or questions, relevant literature, proposed design and methods, anticipated ethical issues, and—where required—a timetable and budget. Hypotheses and variables are appropriate for some designs but are not required in every qualitative or exploratory study. The proposal describes work that is planned; a completed research report describes what was actually done, what was found, and what the findings mean.

A research proposal, then, is a written plan for a study. It spells out in detail what the researcher intends to do. It permits others to learn about the intended research and to offer suggestions for improving the study. It helps the researcher clarify what needs to be done and avoid unintentional pitfalls or unknown problems. A research report follows much the same format as a proposal, with two main differences: 1. it states what was done rather than what will be done (some alterations are almost inevitable), and 2. it includes the actual results of the study, along with a discussion of them

Steps in the development of a research proposal

5.2 Report Writing

✎ **Activity 30**.what is report?

A Report is an exploration of an idea which needs to be defined before it is developed. A report always relates to this central topic, it is necessary for you to introduce the research and make the reader aware of its importance and relevance. The introduction is the place where the research has

to make a good impression, informing the reader what is to come and encouraging him or her to read further. If the introduction is tedious or fails to make the rest of the research sound interesting, the reader will not wish to continue.

In the practical world of business or government, a report conveys information and (sometimes) recommendations from a researcher who has investigated a topic in detail. A report like this will usually be requested by people who need the information for a specific purpose and their request may be written in *terms of reference* or the *brief*. Whatever the report, it is important to look at the *instructions* for what is wanted.

Notice that, a report like this differs from an essay in that it is designed to provide information which will be acted on, rather than to be read by people interested in the ideas for their own sake. Because of this, it has a different structure and layout

Report Style

This section deals with the format of a research paper and the style in which it should be written. Report writers usually use a more formal writing style than do no technical writers. A degree of formality is required because the personal style of the writer must be secondary to the clear and objective transmission of information. Any injection of personality that obscures the exact meaning is undesirable. But this does not mean that report writing has to be dull and rigidly stereotyped. All writers should strive to make their writing enjoyable to read. Therefore attempt to develop a writing style that is both clear and interesting.

This section includes some specific suggestions for developing and improving your writing style. There is no satisfactory explanation of style, no infallible guide to good writing, no assurance that a person who thinks clearly will be able to write clearly, no key that unlocks a door, no inflexible rules by which the young writer may shape his course.

Although difficult to define, style establishes the readability of reports in effect, the style of the report sells the report. If your style of writing and presentation is not acceptable to your intended readers, they may not read your report.

A writing style is acquired only with diligent study and practice in writing. This chapter comments on general report requirements that must be met by any writing style. It then makes specific suggestions for developing your own writing style and for preparing figures and tables.

The preliminaries

By preliminaries is meant what comes before the text, which starts with the introduction. For a long dissertation or thesis, it includes the title page, the preface, the table of contents, the list of tables (if any), the list of figures or illustrations (if any),in that order. For a short paper, the important preliminaries are the title page and the table of contents.

The title page

The title page of a thesis includes the title of the thesis, the faculty and the institution to which the thesis is submitted, the degree for which it is presented, the name of the candidate, and the date. The title page of a term paper can be reduced to the following: title, name of student, name of instructor, title of course and date.

The table of contents.

The table of contents lists the main and the subdivisions into which the paper is divided. For a paper, it is preferable not to use too many subdivisions. However, the divisions should show the relationships among sections through indentation and a consistent system of numerals and letters:

Main Heading

Subheading

Run-in heading. — This heading is indented on the same line as the first line of the paragraph.

Below run-in heading: This heading is indented on the same line as the first line of the paragraph.

Three levels of headings should be sufficient; more may disrupt your readers' concentration. The typeface and placement of headings will vary with the type of publication.

Thorough outlining will make both writing and reading the report easier. Study your outline carefully to be certain that each item blends into a logical ordered presentation.

I

A.

B.

1.

2.

a)

b)

(1)

(2)

(a)

(b)

(i)

(ii)

II.

A.

B.

etc.

Another way of indicating divisions and subdivisions, often used in scientific journals, is the following, by numbers only: the main divisions (I, II,III) are numbered 1.2.3.; the first subdivision (A,B,C) becomes 1.1.,1.2.,2.3., each subsequent subdivision starts with 1., with or without the decimal point (1.11.,1.12.,1.13.,1.21.,1.22.,1.23.,1.31,1.32.,1.33.) or (1.1.1.,1.1.2.,1.1.3.,1.2.1.,1.2.2.,1.2.3.,1.3.1,1.3.2.,1.3.3.) there is no indentation in the text. The advantage of this system is that it facilitates reference to any part of the text. For example, 2.231. is a less complicated reference than II-B-3-a. the table of contents of this guide is numbered in this way.

5.3 Academic Reports

Activity 31. What makes the academic report different from other types of report?

A report written for an academic course can be thought of as a simulation. Students practice report writing at Colleges and Universities. And the experiences and skill of report writing helps them in the real life report writing. We can imagine that someone wants the report for a practical purpose, although we are really writing the report as an academic exercise for assessment. Theoretical ideas will be more to the front in an academic report than in a practical one.

Sometimes a report seems to serve academic and practical purposes. Students on placement with organizations often have to produce a report for the organization and for assessment on this book. Although the background work for both will be related, in practice, the report the student produces for academic assessment will be different from the report produced for the organization, because the purpose of each are different. When writing any kind of report you have to consider the following point: the purpose of the report and audience.

The answers to these questions will help you to decide what to put in the report and what style to write it in:

- Who is (are) your audience? Who are you writing for?

What do they know already?
What do they need to know?
What do they want to know?

Regardless of the specific style used to prepare reports, four general requirements must be met to produce good reports: clarity, conciseness continuity, and objectivity.

Clarity

The purpose of a report is to transmit conclusions and their supporting evidence. To do this, your report must convey your *exact* meaning to the reader. The text must be clear and unambiguous, mathematical symbols must be fully defined, and the figures and tables must be easily understood.

Clarity must be met from the readers' point of view. What may be clear to you as the author may not be clear to your readers. Remember, you are intimately familiar with the work, but they are not. You must continually reexamine your rough drafts with a reader's critical eye. Readers will not tolerate confusion. They must never become uncertain about what you are discussing, why you are discussing it, or what your plan of presentation is. They will rebel if forced into these mental gymnastics. If there is any discontinuity without proper explanation, the average reader will lay aside the report for later reading.

Once this happens, the chances are slight that it will ever be read. You usually have just one chance to sell the reader on the report's objectives. And that requires a presentation that is logical, simple, and systematic.

Conciseness

A report should be sufficiently detailed to communicate the evidence and reasoning needed by its readers. Conciseness means removing repetition and irrelevant material, not simply making a report as short as possible. The appropriate length depends on the purpose, audience, discipline, and reporting requirements.

On the other hand, do not condense reports at the expense of your readers' understanding Give enough information to enable them to understand clearly what you are describing and why you are describing it. Include enough background information to make the context clear. Do not assume that they will remember details of a previous report – or have even read it. Include all details needed to understand the current report. In short, make your reports brief but comprehensible.

Continuity

Reports should tell a complete story as logically and interestingly as possible. This requires continuity between succeeding sentences, paragraphs, and sections and between the written text and the figures and table. Transitional words, phrases, sentences, or even paragraphs may be needed to lead you readers through the story. But overusing transitions can slow the pace of your narrative.

Carefully choose the places at which you refer to figures and tables to limit distraction. Making these references at the beginning or end of a discussion is usually preferable.

Objectivity

Reports should be objective and show restraint. Be honest with your readers they will become suspicious if they detect hidden meanings or any types of subterfuge and you will then have little chance of convincing them of your conclusions. They expect you to evaluate the data honestly. Do not try to hide deficiencies in your research. No technical report is better than the research on which it is based. Tell your readers frankly what your assumptions were, what your probable errors are, and what you may not understand about the result.

In addition to being honest, be tactful. If you are faced with the problem of presenting technical results that may conflict with previous results or with the personal prejudices of some readers, refrain form making dogmatic statements and avoid sounding egotistical. Your readers will be persuaded by facts, but they may become irritated if you attempt to impress them with your cleverness or to claim credit for accomplishments. Write to express, not to impress.

To achieve clarity and continuity in a report, you must carefully direct your readers' attention throughout the report. Many successful writers do this by using the three classic principles of presentation.

1. Tell readers what you plan to tell them (Introduction)
2. Then tell them(Main text)
3. Finally tell them what you told them (Summary of Results or Conclusions).

State your purpose or objective clearly and follow it with a concise description of the method you will use in presenting the subsequent discussion. Then proceed with your presentation, making certain that it is consistent in every respect with your plan. Finally summarize your conclusions and recommendation.

Getting to the Point

Reports are not mystery novels; get to the point as directly as possible. Do not lead your readers in and out of blind alleys before taking them to the final destination. Omit information that does not directly relate to the conclusions. Remember, readers are interested primarily in conclusions and supporting evidence.

If you must include some information or discussion that may be of interest but is not directly pertinent to your conclusions, put it in an appendix. Using an appendix allows you to bring up points that may be of interested to some of your readers without distracting the reader who is interested solely in your conclusions.

Emphasizing Major Ideas

Because the purpose of technical reports is to communicate ideas, emphasize the major ideas so that they are not overlooked. Clearly subordinate supporting information to the major ideas. The report outline is particularly useful here because it establishes the major and supporting points for each section of the report.

Your major ideas can also be emphasized by briefly stating them at the beginning of each section and then summarizing them at the enc of the section. Emphasis can also be aided by careful use of headings.

5.4 Steps in Report Writing

The stages below describe a practical process for developing a research report. They should not be confused with a research timetable, which sets out research tasks, their expected duration, milestones, and deadlines.

5.5 Stages in Report Writing

Stage one: Define the purpose, audience, scope, and plan for the report.

Stage two: Gather and organize the relevant information and evidence.

Stage three: Analyze and interpret the information in relation to the research questions.

Stage four: Prepare an outline that organizes the main points and supporting evidence.

Stage five: Write the first draft, following the outline while allowing for necessary adjustments.

Stage six: Revise the draft for argument, evidence, organization, clarity, and completeness; then edit for language and presentation.

Stage seven: Check references, tables, figures, formatting, and requirements before preparing the final report.

Research information has limited value unless it is collected and published in a usable form and presented to those who may apply it. For these reasons never assume that your job is finished when the experimental or analytical phase has been completed. It is also your responsibility as an engineer or scientist to show promptly that your results are worthwhile and that you have reason to believe the field will be advanced by your efforts. The only way to convey these thoughts is by writing a good report.

Writing the rough draft, with a logically organized outline and the necessary illustrations already prepared, should be much easier than you thought. But do not expect to write the final version in the

first attempt. The rough draft should be the last of several versions, each an improvement of the preceding one. This final version is considered a “rough” draft because it still must go through a series of technical and editorial reviews. But it should be as polished as you can make it. From your point of view it should be ready for printing and distribution to a critical audience.

Try to start writing the first version of the draft immediately after completing the outline while the ideas developed there are still fresh in your mind. Write this first version as rapidly as possible. Concentrate on *what* you want to say rather than how to say it. Keep writing down the thoughts as they flow into your mind, following your outline. Avoid going back over what you have written until you are through writing. Then review this version—but only for its technical content. Are all of the ideas you wanted to express

included? Have you included irrelevant ideas? Does the report organization still seem logical? Sometimes writing the first version will reveal some unexpected problems that require a change in the outline.

In the second version of the rough draft, writing style becomes important. With the technical content in a well-organized form from the first version, this is the time to concentrate on how you say it. Keep your readers in mind. Remember, your purpose in writing the report is to transmit the information needed to support your conclusions. To make sure your readers understand your conclusions, you must transmit your information clearly, logically, concisely, honestly, and tactfully.

5.6 Structuring a Report

A report has a different structure and layout to an essay. A report is used for reference and is often quite a long document. It has to be clearly structured for you and your readers to find quickly the information wanted.

Follow the steps that you already have used to write your research proposal. The purposes of Reports differ so much that an advisor for your particular report is very important.

A supervisor or advisor can help you shape and revise a research report. Keep the following qualities in mind when writing:

- completeness
- correctness
- clarity
- conciseness

There are a variety of ways that you may choose to structure your ideas within the report. A few structuring techniques are described below:

General to specific: begins by discussing a main point in general and then moving to more specific information.

Chronological: provides information in a date and/or time sequence.

Problem-solution: discusses a problem and then offers possible solutions for the problem.

Familiar to unfamiliar: discusses what is familiar to the reader prior to discussing unfamiliar or new information.

Comparison: compares two ideas with each other on the basis of similarities and differences.

Causation: explains a cause-and-effect relationship between two ideas.

5.5 Writing the report

it is not sensible to leave all your writings until the end. There is always the possibility that it will take much longer than you anticipate and you will not have enough time. It is wise to begin writing up some aspects of your research as you go along. Remember that you do not have to write your report in the order that it will be read. Often it is easiest to start with the method section. Leave the introduction and the abstract to last. Last but by no means least, asks someone to proofread your work.

Dear, some writers' believe that in any type of writing the starting seems difficult. Writers are big procrastinators. They find countless reasons not to get started. Even when they finally get themselves seated at their desks, they always seem to find diversions: make the coffee, sharpen the pencil, go to the toilet, thumb through more literature... Remember that you are never `ready' to write; writing is something you must make a conscious decision to do and then discipline yourself to follow through.

It is easier said than done, but do not keep on waiting until you are "in the mood." It will not happen. Make an early start and write up a section as soon as it is ready.

remember that, you should not leave all your report writing until after your research is completed. Instead, get into a habit of writing up sections while your research is still in progress. Start writing with a section about which you feel that you are reasonably confident. Do not sit and stare at a blank page, just get writing. Don't forget that this is only a first draft. It does not have to be perfect. Your literature review can be written up early on (and added to if you read more or as you discover more). The methodology section is often reasonably straightforward to write.
Dear, to help you write your report you can use different techniques. Some of these techniques are:

5.6.1 Write the First Draft

so far we have discussed what a report writer, you, need consider when you write a report. Once you have developed the basic structure for your report, you can begin writing. Your goal in writing the first draft is to get your ideas onto paper. Experts indicate that there is no set process to follow, so use the process that works for you.

Drafting the body of your report

Introduction

This should include the following:

• The purpose of your report – why you studied the particular topic. The statement of the problem is useful here.

• Background information – may include a brief review of the literature already available on the topic so that you are able to 'place' your research in the field.

• Some brief details of your methods.

• An outline of the structure of the report

Literature Review

If asked to do a separate literature review you must carefully structure your findings. It may be useful to do a chronological format where you discuss from the earliest to the latest research – placing your research appropriately in the chronology. Alternately you could write in a thematic way, outlining the various themes that you discovered in the research regarding the topic. Again you will need to state where your research fits.

Methodology

Here you clearly outline what methodology you used in your research – what you did and how you did it.

Describe the procedures clearly enough for readers to understand what was done and, where appropriate, for other researchers to assess or replicate the study.

Past tense is commonly used to describe completed procedures. Passive voice may be appropriate in some contexts, but active voice is also acceptable when it makes the writing clearer.

Use verb tense consistently and follow the conventions of your discipline and institution.

• Ensure you clearly reference any material you have used from another source.

• Clearly label and number any diagrams, charts, graphs that are used. Ensure that they are relevant to the research and add to the text rather than just duplicating what you have said.

• You do not have to include or discuss the results in this section.

Results

This is where you indicate what you found in your research.

• You give the results of your research.

Discussion

This is where you discuss the relevance of your results, and how your findings fit with other research in the area. It will relate back to your literature review and your introductory research statement.

This is a summary of the most significant results/findings. You should not include any new material in this section. Sometimes you could indicate some areas where your research has limits or where further research would be useful.

The Findings section of the report provides what most readers are reading to find out--the results of the research. This section therefore must provide a focused description of what has been found with regard to the questions and expectations raised by the title, the statement of the problem, and earlier sections of the report. Hence, you must be both selective and careful in stating the information.

The Findings section should not contain all that has been found in this book of the research. Many interesting observations and tables must be excluded because they are not sufficiently relevant to the thrust of the report. Remember, you do not have to overload the reader with so many details that he/she loses focus on the main issue(s).

In presentations featuring tables, the first of these usually gives an overview of the data. Often this consists of a tabular presentation of the characteristics of the population or a sample if sampling was used. It provides data in a form that allows the reader to develop an orientation for the analysis which is to follow. Subsequent tables focus on the results of the analysis itself, presenting more specific aspects of the research.

For all tables it is essential that readers be provided complete information on what is in the table. Not all readers will draw the same inferences from data or they may wish to analyze it in a different way. Consequently, it is essential that sufficient information be provided so that readers may make their own calculations. For the novice writer especially, it is usually best to provide information according to the most familiar and tested format.

As was implicitly hinted at above, the general rule in reporting your finding is to give the general ideas first and then specific details. This is true of the results section as a whole: Begin with the central findings, and then move to more peripheral ones. It is also true within subsections: State the basic finding first, and then elaborate or qualify it as necessary.

It is advisable to give attention to the following points while you are writing your research report:

a) Remind the readers the conceptual question you are asking

b) Remind the readers the actual operation performed or the actual behavior measured. Note that this is an operational statement of the question.

c) Tell the readers the answer immediately.

d) Now discuss in numbers--present your statistics. (The reader not versed in the methods and statistics presented can skip to the next finding. Those who want details can read through the specifics.)

e) You should not take elaboration to mean a license to unleash an avalanche of only statistics and test results without indicating their meaning in terms of substantive results. You have failed if a knowledgeable reader must ask why a particular statistics is being reported.

f) Use tables, figures, or other visual displays when they help readers understand important results. Not every finding requires a separate table or figure; avoid unnecessary duplication between the text and visuals.

By structuring the results section in this way, by moving from general to specific, by announcing each result clearly in prose before wading into numbers and statistics, and by summarizing frequently, you enable the reader to decide just how much detail he or she wants to pursue at each juncture and to skip ahead to the next main point whenever that seem desirable.

Where qualitative evidence is relevant, integrate it carefully with quantitative results. Explain how quotations, observations, or other qualitative materials support or qualify the findings, while protecting participants' confidentiality.

Remember that, in the Findings section, the researchers (you) are obligated to report data which contradict as well as support any hypotheses or ideas which were investigated. Only by reporting such information can scientific knowledge grow. Contradictory evidence may lead to important insights for future investigation. You must also report data that are inconclusive. The Findings section therefore will often contain a number of highly tentative and qualified findings, reflecting the efforts of the researcher to neither over interpret nor under interpret the meaningfulness of the data.

Recommendations

This includes suggestions for what needs to be done as a result of your findings.

This is usually a brief section designed to serve three functions. First, it refocuses the reader's attention on the main issues and findings and away from the myriad details which have been just presented. Second, when appropriate, it qualifies some of the findings by stressing methodological limitations or alternative interpretations of the research. If**, for example**, the empirical results are directly contrary to the original hypothesis, you will summarize the reasons why this may have been the case. Third, it may suggest directions for future research based upon the experience of the researcher. Such suggestions should not be broad statements that more research would be useful, but specific statements about the types and strategies of research that appear most promising. Recommendations are usually listed in order of priority.

References

This includes all references used in your report or referred to for background information. This must be done using the referencing convention specified.

Appendices may contain supplementary materials that are relevant to the report but too detailed or lengthy for the main text. Examples include instruments, additional tables, consent documents where appropriate, and technical details. Do not include confidential or identifiable participant information without appropriate authorization.

These should add extra information to the report. If you include appendices they must be referred to in the body of the report and must have a clear purpose for being included. Each appendix must be named and numbered.

Title of report

Make sure this is clear and indicates exactly what you are researching.

Table of contents

List all sections, sub headings tables/graphs appendices and give page numbers for each.

A few suggestions that might help you get started are provided below.

- Write a paragraph that describes what the report is about.
- Free write for a set period of time or amount of text.
- Re-read the information you have gathered.
- Write the sections that you feel most comfortable with first, then write the others.
- Divide your outline into sections and tackle one section at a time.
- When you write your first draft, it is not important to worry about grammar, spelling, or the most appropriate word choice. The trick is to keep writing and not get bogged down by trying to write a perfect first draft. During the editing and revision stages you will be able to make necessary changes.

5.5.2 Revise and Edit the Report

Revision is more than correcting surface errors. It may involve reorganizing sections, strengthening explanations, checking whether evidence supports claims, removing irrelevant material, and clarifying

the relationship between findings and conclusions. Editing and proofreading then address sentence-level clarity, grammar, punctuation, and presentation.

Before you can revise your rough draft, you must recognize that it is not perfect. Approach it with a critical attitude. This can best be done by setting the draft aside for a few days, or at least overnight. This time lag should give you a fresh viewpoint and allow you to change to the role of a reader. This change in roles is most important because you must try to see what is actually written rather than what you think you wrote.

A useful review process may include separate checks of content and evidence, organization and argument, and language and presentation. These checks can be carried out by the writer, supervisor, or peer reviewer as appropriate.

1. The first review is of the material in the report. In this check ask yourself these questions: Are the conclusions valid? Is sufficient information given to support the conclusions? Is enough background information given to explain the results? Have all irrelevant ideas been deleted? Are the illustrations pertinent and necessary?

2. The second review is of the mechanics and organization. Are the subject and purpose clearly stated? Does the report flow smoothly from topic to topic? Are the relations between topics clear? Is each illustration clear and properly labeled? Are all required parts of the report included?

Make sure you can truly answer yes to all of these questions before you consider your draft finished. Do not try to make one review do the work of three. Trying to cover too many categories in one review usually results in oversights and errors. Some common faults observed in rough drafts are

(1) faulty grammar;

(2) clusters of nouns and adjectives modifying a noun and conversely strings of prepositional phrases after a noun;

(3) use of abstract nouns instead of action verbs

(4) parallel construction of words, phrases, and sentences in enumerations; and

(5) more complicated phrasings than required_Carefully review your draft to make sure you have avoided these common faults.

So far we have discussed the preparations you have to make for research writing. Once you have completed the first draft of the report, revise and edit it. While the goal in writing the first draft was to put your ideas on paper, the goal of this step is to improve the manner in which you are communicating your ideas. Verify the following as you revise and edit the paper:

- Do the ideas communicated in the report support its purpose?
- Are the main ideas communicated in the report correctly structured?
- Are the main ideas communicated in the report supported with evidence and examples?
- Have you explained terms that the reader might not understand?
- Have you tailored the report to the audience who will be reading it?

- Does the report flow smoothly? Are transitions between paragraphs smooth?
- Do the words you have chosen convey what you are trying to say?
- Have you altered sentence length and structure to make them interesting to read?
- Is the spelling correct?
- Is the punctuation correct?
- Is the grammar correct?

Plagiarism is presenting another person's words, ideas, data, images, or other intellectual work as one's own without appropriate acknowledgement. It can occur through direct copying, inadequate paraphrasing, or failure to cite a source. Plagiarism is distinct from copyright infringement, although the two may overlap.

Remember that all ideas that you borrow from other authors must be cited. This includes ideas that you have summarized ideas that you have paraphrased, and direct quotations. In general, everything other than common knowledge or knowledge that is not traceable to a specific source should be cited.

- **Conclusion**

The paper is your final product, but a research paper involves an extensive process before you can generate the product. If you focus too quickly on the end product, you may miss some of the important research steps and find yourself writing a paper without enough understanding of the topic to do.

Report is a concise and comprehensible writing from which a reader can grasp the aim of the writer. So as to communicate your readers, you have to develop your report based on the following steps:

Step one: Analyzing the task

As with any assignment task you must first analyze what is expected of you. This involves careful reading of the assignment task as outlined in your course information book. You may find the following questions useful when analyzing the task:

• What is the purpose of the report? Are you, for example, analyzing or persuading or reporting on an investigation?

• Who is the audience for the report?

• What is the topic of the report?

• What is the expected format of the report?

Step two: Developing a rough plan

• Use the section headings (outlined above) to assist with your rough plan.

• Write a thesis statement that clarifies the overall purpose of your report.

• Jot down anything you already know about the topic in the relevant section.

Step three: Write the report

Steps one and two above will guide your research for this report. You may need to report on other research on a particular topic or do some research of your own. Keep referring to your analysis and rough plan while you are doing your research to ensure that you remain on track. Give yourself plenty of time for this step, as the research phase of your work will usually take the most time of any step in producing your report. Also ensure you keep correct bibliographic details for all of the material you may later use in your report.

Reports should clearly differentiate between facts and opinions. Many researchers are remiss in doing this, overlapping discussions of their experimental results and the conclusions they have draws. Carefully alert your readers when fact ends and opinion begins.

The statement of your opinions is an instance where the use of the first person is desirable. For example, if you follow the presentation of some specific results with "It is believed that....," your readers cannot be sure if this is your opinion or a generally accepted belief. To avoid this confusion, use the first-person pronoun to say, for example, "From these result I conclude that..."

Dear, when you write a research report you can use the following points as a guide line:

- You need to plan carefully to make sure that the information which you have gathered gets put under the correct headings.
- Decide on your headings and subheadings. Make a list of the main parts (as shown under parts of a report) that you will need for your report. Then add your own headings and subheadings as appropriate.
- Go through the material you have gathered and list all your points and any supporting information under the appropriate headings.
- Go through the points under each heading and underline the most important.
- Cross through any that seem irrelevant, or put them under another heading if they are out of place. Leave the points which you are unsure about. You can decide whether to include or reject them later.
- Arrange the headings into a logical sequence. Read through what you have planned and decide whether or not to include the points about which you were unsure.
- Decide what supporting information should go into the appendices and what should remain in the main body.
- Draft some interim conclusions by summarizing, analyzing and evaluating your findings.
- Consider what recommendations (if required) should be made.
- Write a full draft, taking account of the points on structure outlined above, and the points on layout outlined below.

In general, use the following pattern to write your report: THINK, PLAN, WRITE, REVISE AND REWRITE. Dear Colleague, remember that no one, however gifted, can produce a passable first draft. Writing means rewriting.

Read through the draft, checking for errors and making revisions. See if each idea, sentence, or phrase can be expressed more clearly and/or more concisely

A report should be logically organized and clearly written. A complete report includes the following parts:

Individual Sections	Content of Each Section
Title of report	Concise heading indicating what the report is about
Table of Contents	List of major sections and headings with page numbers
Abstract/Synopsis	Concise summary of main findings
Introduction	Why and what you researched
Literature Review	Other relevant research in this area
Methodology	What you did and how you did it
Results	What you found
Discussion	Relevance of your results, how it fits with other research in the area
Conclusion	Summary of results/findings
Recommendations	What needs to be done as a result of your findings
References or Bibliography	All references used in your report or referred to for background information
Appendices	Any additional material which will add to your report

All sections of a research proposal or a research report should be consistent with one another.

Normally, the final section of a report is the offering of some suggestions for further research.

The essential difference between a research proposal and a research report is that a research report states what was done rather than what will be done and includes the actual results of the study. Thus, in a report, a description of the findings pertinent to each of the research hypotheses or questions is presented, along with a discussion by the researcher of what the findings of the study imply for overall knowledge and current practice.

The main body is the largest section of a proposal or a report and generally includes the problem to be investigated (including the statement of the problem or question, the research hypotheses and variables, and the definition of terms); the review of the literature; the procedures (including a description of the sample, the instruments to be used, the research design, the procedures to be followed, and a description and a justification of the statistical procedures used); and a budget of expected costs.

A research report communicates what was actually done in a study, and what resulted.

A research proposal communicates a researcher’s plan for a study.

- Conclusion

Summary

Outlining is a necessary preliminary step to report writing, it involves the planning needed to prepare a clear report that is logically organized, concise, and easy to read. Without an outline most inexperienced authors write reports that are confusing and difficult to follow. The outlining stage is a natural progression from the analysis and sorting stage. In the sorting stage concentration is on what

results should he presented in a report. In the outlining stage attention is directed to *how* these results should be presented.

Often the preliminary outline prepared at the beginning of the program can be used as a starting point for the report outline. But it should be revised and expanded to emphasize the conclusions drawn in the analysis and sorting stage. The revised outline should contain descriptive headings of each significant part of the report. This expanded outline should show the complete scope of the report, the relation of the various parts of the work discussed, the amount of space to be given each part, the order of treatment, the places for inclusion of illustrations, and the conclusions. Remember, the more detailed the outline is, the more useful it will be to you. Each heading, subheading, sub subheading, etc., should have as much detail as you will need to trigger your thoughts when you later write the corresponding sentences and paragraphs.

NASA reports typically contain a number of commonly used headings:

- Summary
- **Introduction**
- Symbols
- Theory
- Analysis
- Apparatus
- Test Specimens
- Test Procedure
- Sample Calculation
- Results and Discussion
- Summary of Results
- Conclusions
- Concluding Remarks
- Appendix

- **References**

All these headings need not be used in any single report. And headings may be combined. For example, Apparatus, Test Specimens, and Test Procedure can be combined into Experimental Methods.

These commonly used headings may be replaced with more descriptive headings, particularly in program summary reports, where details may be subordinated to broad objectives and generalizations. In these reports the more descriptive headings provide a means of ready reference and aid clarity. Descriptive headings usually make a report more interesting to read. But you must exercise originality to make them brief but clear.

The final outline should show the exact form, wording, and value of the headings to be used in the report. The headings, although brief, must serve as a reliable guide to the included material. They should be consistent in grammatical structure and should not contain verbs. Headings are not an integral part of the text but are provided to assist the reader in finding information. Therefore the paragraph below a heading should begin with a topic sentence that does not depend on the heading for clarity.

The whole text of the report should be accounted for under the headings shown in your outline, except for short introductory or transition paragraphs included to make the presentation flow smoothly. Because a subject cannot be subdivided into less than two parts, an outline should have at least two subheadings under a main heading—or none.

Exceptions to this rule include an occasional short remark or a single example included to illustrate a method. In addition to headings the outline may also contain descriptive words and key phrases to serve as reminders.

Several methods can be used to arrange the subject matter that will be represented in an outline. One of the best ways to start is to write down all the points that you want to include without regard to their order. You can then more easily arrange them in a logical order. Some authors use an index card system in which each separate item of the proposed report is tabulated together with a paragraph describing the material that must be treated under that item. Other authors follow a similar method but use full sheets of paper for each subject and give a much fuller description of the material under the subject headings. The latter method puts you in a good position to complete the report in a short time: Each subject has been so fully expanded that the problem remaining is one of combining and rewriting the information contained on the separate sheets of paper. Others choose to do their outlining on a computer. Any of these methods allows you to note ideas that occur while writing one section of a report but belong in other sections. Cultivate the habit of going to the original outline to record thoughts for later consideration.

Another useful purpose of the outline is to indicate the relative importance of headings. This relation can be shown by using a numbering system. (The numbers appear in the final report only when there is extensive cross-referencing.) Headings of equal weight, or importance, must be written in the same form. The order and form of the various headings used in NASA reports are

✓ Chapter Checklist

Put a ‘ ’ mark against each of the following statements that you are able to answer.

I can:-

- Write the difference between academic and other types of research.
- List the steps of report writing.
- Describe some ways to structure ideas within the report..
- Identify the main parts of a research report.

◆ Self-Test

Part One Instruction: write a complete answer for the following questions.

1. What makes the academic report different from other types of reports?

2. What should a research report writer consider when writing a report?

_____
_____3. What are some of the ways to structure one's idea in a report?

_____
_____

4. Write the main parts of a report.

5. What is research proposal?

6. What is the difference between research proposal and report?

7. What are the four general qualities of a good report?

8. What are the steps for report writing?

9. What are some of the things that a writer should check during editing?

10. The conclusion section of a report contains_____

Part Two

Write true if the statement is correct and false if it is wrong.

1. A research report communicates the intentions of the researcher.

2. A research proposal is a written plan for a study.

3. The title page of a thesis includes the title of the thesis, the faculty and the institution to which the thesis is submitted, the degree for which it is presented, the name of the candidate, and the date.

4. Conclusion is a summary of the most significant results/findings.

5. Outlining is important for a reader to get the central message of the writer.

CHAPTER 6

Citation and Referencing

Introduction

Academic research depends on sources that readers can identify and consult. In-text citations acknowledge the ideas and evidence used in a paper, while the reference list provides the publication details needed to locate those sources. Accurate and consistent documentation also helps writers distinguish their own contributions from the work of others.

This chapter explains citation and referencing practices, including ways to cite different source types and principles for organizing and formatting reference entries.

★ Learning Objectives

By the end of this chapter you will be able to:

- write the kinds of citation in research works.
- show how to write reference materials.
- describe the different ways of citing different sources

6.1 Citation

One of the most frustrating things that happens when you begin to write your research report from your notes is to discover that you forgot to write down the page number of a key quotation.

Throughout your research, compile a working source list that records the materials you consult. Use it later to prepare the formal reference list in the required referencing style.

It is often useful to work this way:

- Start your research with preliminary reading of general reference material, like an encyclopedia, dictionary, or other reference book.
- Next, look up material on your subject by using the library catalogue and consulting sources listed in the general reference material.
- Read through the bibliographies and notes of everything you read for further sources.
- Consult periodical indexes for articles on your subject.

One effective way of keeping track is to record your bibliographical notes on file cards, one note per card; this way, you can sort the bibliography easily. Note all essential information, including the library call number or any other source identification for your own ease of locating the material. Once you read the source, verify the information on your card and correct any errors.

As you examine each source, make a separate note of each fact or quotation you might want to use in your paper. Unless you are really good at manipulating text with your computer or laptop, it might be wise to use index cards when preparing notes. Be sure to identify the source of the information on the listing (include the author's name and page number on which the information appears). Try to summarize the information in your own words (paraphrasing); use quotation marks if you copy the information exactly. (This rule should apply whether you are copying a great deal of material or only

a phrase.) Give each listing a simple descriptive heading as you transfer information to your bibliography card, be sure to capitalize the main words and to underline the title. Record the call number in a prominent corner of the card for easy reference and jot any useful information below the bibliographical details

Look at the following Example on how to record your information on your bibliography card:

For quotations which are quoted by other author

For example if John Wax wrote the idea and Quoted by Mary Smith

There are many approaches to research — an essential part of every business and profession — and many ways to document findings. The library has books which will help you, and most English composition textbooks contain chapters on research techniques and style. It is important to follow consistently and accurately a recommended format that is clear and concise and that has been approved by your advisor.

This chapter follows the Publication Manual of the American Psychological Association (APA, 7th ed.). Use APA 7 consistently for in-text citations and the reference list. Do not combine APA conventions with MLA documentation rules.

According to the American Psychological Association (APA, 7th ed.), source material is cited using an author–date system. In-text citations identify the author and year and direct readers to the corresponding entry in the reference list. Cite sources when you quote, paraphrase, summarize, or otherwise use another author's ideas, findings, data, or wording. Every work cited in the text should have a matching reference-list entry, and every reference-list entry should be cited in the text, except for items explicitly presented as further reading.

APA 7 in-text citation principles

APA uses two forms of author–date citation: narrative and parenthetical. In a narrative citation, the author's name forms part of the sentence and the year follows in parentheses; for example, Bolles (2000). In a parenthetical citation, place the author and year together in parentheses; for example, (Bolles, 2000).

For one author, cite the surname and year. For two authors, cite both surnames every time: use "and" in a narrative citation (Seliger and Shohamy, 1989) and "&" in a parenthetical citation (Seliger & Shohamy, 1989). For three or more authors, use the first author's surname followed by "et al." from the first citation, unless this would make sources ambiguous (Ary et al., 1979). Add a page number or other locator for a direct quotation.

For a direct quotation, reproduce the wording accurately and provide author, year, and page number. For a paraphrase, provide author and year; a page number is recommended when it helps readers locate a specific passage. For a source without page numbers, use a paragraph number or a section heading where appropriate.

When several works are cited in one parenthetical citation, arrange them alphabetically by first author and separate them with semicolons. Arrange multiple works by the same author chronologically, earliest first; use letters such as 2020a and 2020b for works by the same author in the same year.

APA 7 reference-list basics

Begin the reference list on a new page and use the heading References, centred and bold. Arrange entries alphabetically by the surname of the first author or the full name of a group author. Double-space the list and use a 0.5-inch hanging indent. Use sentence case for book, article, and chapter titles; use title case for journal titles. Italicize book titles, journal titles, and journal volume numbers. Do not include the publisher's city or state. Include a DOI as a URL (https://doi.org/...) when available. A retrieval date is generally unnecessary unless the content is designed to change over time.

For works with up to 20 authors, include all authors in the reference entry. For works with 21 or more authors, include the first 19 authors, an ellipsis (without an ampersand), and the final author.

APA 7 reference examples (formats)

Book: Author, A. A. (Year). Title of the book in sentence case (edition, if applicable). Publisher.

Journal article: Author, A. A., & Author, B. B. (Year). Title of the article in sentence case. Title of Journal, volume(issue), page–page. https://doi.org/xxxxx

Chapter in an edited book: Author, A. A. (Year). Title of chapter. In E. E. Editor (Ed.), Title of book (pp. xx–xx). Publisher.

Thesis or dissertation in a repository: Author, A. A. (Year). Title of thesis or dissertation [Master's thesis or Doctoral dissertation, Name of Institution]. Repository. URL

Webpage: Author, A. A. or Organization. (Year, Month Day). Title of webpage in sentence case. Site Name (omit if same as author). URL.

Older examples that contained incomplete or unverifiable publication details have been removed. Use the APA 7 templates in this chapter and complete them only with details checked against the original source or a reliable publisher or repository record.

6.2 References 167

A reference list gives the publication details of sources cited in a report. Follow one referencing style consistently so readers can identify and retrieve each source. In APA, list only works cited in the text; place uncited recommended reading in a separately labelled Further Reading section.

APA 7th edition reference examples

In APA 7, a book reference generally follows this pattern:

Author, A. A. (Year). Title of the book in sentence case (edition, if applicable). Publisher. DOI or URL (when available)

Brown, J. D. (1988). Understanding research in second language learning: A teacher's guide to statistics and research design. Cambridge University Press.

Accurate citation and referencing make the use of evidence transparent and allow readers to locate the sources behind a claim. Writers should cite borrowed ideas, quotations, paraphrases, and data, and ensure that every in-text citation corresponds to an appropriate reference entry. Consistent formatting and careful checking help prevent omissions and reduce the risk of misattribution.

- Conclusion

- **Summary**

A report writer acknowledges the sources used by citing them in the text and providing complete reference-list entries. Include only sources cited in the text in the reference list; background reading not cited in the report may be listed separately as further reading.

The following guidance summarizes selected APA 7th edition principles for ordering, capitalization, and punctuation in reference entries.

Ordering References

- Arrange reference-list entries alphabetically by the surname of the first author or by the full name of a group author.
- For works with multiple authors, retain the author order shown in the publication. Apply APA 7 rules for the number of authors rather than changing the order.
- When no author is identified, move the title to the author position and alphabetize by the first significant word of the title.
- For works by the same author, arrange entries chronologically, beginning with the earliest work.
- For works by the same author published in the same year, use lowercase letters after the year (for example, 2020a and 2020b) and use the same letters in the in-text citations.
- For works with the same first author but different co-authors, apply APA 7 alphabetical ordering rules to the author lists.

Capitalization

- Use sentence case for titles of books, articles, and book chapters: capitalize the first word, the first word after a colon or dash, and proper nouns.
- Use title case for names of journals and other periodicals.
- Do not capitalize publisher names as if they were titles; reproduce the publisher name accurately.

Punctuation and formatting

- Do not place quotation marks around titles of journal articles or book chapters.
- Italicize book titles, journal titles, and journal volume numbers. Do not italicize article titles or issue numbers.
- Use an ampersand (&) before the final author in a reference-list entry with two or more authors.
- Place the publication year in parentheses after the author. Include edition information in parentheses after the italicized book title; do not include a place of publication in APA 7 book references.
- Include a DOI as a URL when one is available. For online sources, provide a working URL when appropriate; do not insert a line break manually into a DOI or URL.

✓ Chapter Checklist

Put a ‘ ’ mark against each of the following statements that you are able to answer.

I can:-

- Write the kinds of citation in research works.
- Show how to write reference materials.
- Describe the different ways of citing different sources.

◆ Self-Test

Instruction: write a complete answer for the following questions.

1. Why is it important to write documentation?

2. What are the three most important things to be done when you write references?

_____3. Write one example of your on how to cite from unpublished dissertation.

_____

_____

4. Write one example of your own on how to cite from professional Website.

5. Write an example how to cite from unpublished paper presented at a conference or meeting_____

6. Write an example how to cite from Newspaper article_____

7. Write an example how to cite from unpublished thesis_____

8. Write an example how to cite from edited book_____

9. Write an example how to cite from Article in an encyclopedia

_____

_____

10 Write an example how to cite from Non-recoverable sources

_____

_____

For a work with three or more authors, use the first author's surname followed by "et al." in every in-text citation, unless this would make two or more sources ambiguous. In that case, include enough surnames to distinguish them.

12 Write an example how to cite when a work has two authors

13 Write an example how to cite from *Audio recording*

14 Write an example how to cite from Television broadcast

_____

_____

15 What considerations should a write take during writing the reference materials in the reference sections?_____

CHAPTER 7

Ethical Considerations in Research and Report Writing

Introduction

Ethical practice is essential throughout research and report writing. Researchers should protect participants, represent evidence honestly, acknowledge other people's work, and report methods and findings transparently. Ethical standards also support trust, accountability, fairness, and responsible collaboration.

This chapter introduces key ethical considerations in conducting research and preparing reports, with attention to participants' rights, responsible use of evidence, and principles of ethical writing.

★ Learning Objectives

- Explain the importance of ethics in research and report writing.
- Identify some ethical considerations.
- List the ethical principles relevant to research and report writing.

Ethical issues in research

Most researchers want to receive credit for their contributions and do not want to have their ideas stolen or disclosed prematurely. Ethical norms in research build public support for research. Many of the norms of research promote a variety of other important moral and social values, such as social responsibility, human rights, animal welfare, compliance with the law, and health and safety. Ethical lapses in research can significantly harm human and animal subjects, students, and the public. For example, a researcher who fabricates data in a clinical trial may harm or even kill patients; and a researcher who fails to abide by regulations and guidelines relating to radiation or biological safety may jeopardize his health and safety or the health and safety others.

7.1 Ethical considerations

Research and report writing, like most other disciplines, has ethical guidelines governing the way in which research is conducted. These guidelines apply to research involving both human participants and animals. Researchers should avoid unnecessary harm, humiliation, or degradation and should take reasonable steps to minimize discomfort and risk. As far as possible, a researcher (you) should avoid the use of deception and should try to avoid causing discomfort. Human participants have the right to be told the purpose and outcome of the studies. At the end of the study, the researcher (You) should tell the participants exactly what the research

Before collecting information from human participants, researchers should determine whether institutional ethical review or approval is required and obtain it before beginning the study. They should explain the purpose, procedures, foreseeable risks and benefits, data use, and confidentiality arrangements in language participants can understand.

Informed consent should be voluntary and based on adequate information. Participants should be told that they may decline to participate and, subject to clearly explained practical and ethical limits, may withdraw. Researchers should avoid pressure or coercion, especially when they have authority over potential participants, such as when lecturers invite their own students to take part in research.

Anonymity and confidentiality are different. Anonymity means that a participant's identity is not known or cannot be linked to the data. Confidentiality means that the researcher may know the identity but takes reasonable steps to prevent unauthorized disclosure. Interviews, recordings, photographs, and detailed case descriptions are not automatically anonymous.

Researchers should plan how data will be collected, stored, accessed, retained, shared, and securely disposed of. Data should be accessible only to people who are authorized to use it. Sharing data must be consistent with participants' consent, privacy, confidentiality, intellectual-property rights, and applicable institutional or legal requirements.

Researchers should assess foreseeable risks and take reasonable steps to minimize physical, psychological, social, economic, and reputational harm. Additional safeguards may be needed when research involves children, people with limited capacity to consent, dependent students, or other groups who may be vulnerable to pressure or exploitation.

Researchers should disclose relevant conflicts of interest and report methods, findings, limitations, and unexpected or contradictory results honestly. Authorship should reflect substantial scholarly contributions; contributors who do not meet authorship criteria should be acknowledged appropriately, with their permission where required.

was about (if they were not able to do so before the research began), and where it is practical, researchers should offer to inform them of the outcome. It is also important that

the participants be ensured that any sensitive information they provide will remain

confidential. They should also be assured that if the research is published their individual

contribution will be combined with that of all other participants.

7.2 Ethical principles in report writing

when you write your report you are strongly advised to following ethical principle:

Honesty

If researchers plan to share their data with others, they should note that in the consent process, specifying how they will be shared and whether data will be anonymous. For example, researchers could have difficulty sharing sensitive data they've collected in a study of adults with serious mental illnesses because they failed to ask participants for permission to share the data. Or developmental data collected on videotape may be a valuable resource for sharing, but unless a researcher asked permission back then to share videotapes; it would be unethical to do so. When sharing, psychologists should use established techniques when possible to protect confidentiality, such as coding data to hide identities

Strive for honesty in all scientific communications. Honestly report data, results, methods and procedures, and publication status. Do not fabricate, falsify, or misrepresent data. Do not deceive colleagues, granting agencies, or the public.

Objectivity

Strive to avoid bias in experimental design, data analysis, data interpretation, peer review, personnel decisions, grant writing, expert testimony, and other aspects of research where objectivity is expected or required. Avoid or minimize bias or self-deception. Disclose personal or financial interests that may affect research.

Integrity

Keep your promises and agreements; act with sincerity; strive for consistency of thought and action.

Carefulness

Avoid careless errors and negligence; carefully and critically examine your own work and the work of your peers. Keep good records of research activities, such as data collection, research design, and correspondence with agencies or journals.

Openness

Share data, results, ideas, tools, resources. Be open to criticism and new ideas.

Respect for Intellectual Property

Honor patents, copyrights, and other forms of intellectual property. Do not use unpublished data, methods, or results without permission. Give credit where credit is due. Give proper acknowledgement or credit for all contributions to research. Never plagiarize.

Confidentiality

Give participants information about how their data will be used, what will be done with case materials, photos and audio and video recordings, and secure their consent.

Protect confidential (secret) in communications, such as papers or grants submitted for publication, personnel records, trade or military secrets, and patient records.

be sure confidential records are stored in a secure area with limited access, and consider stripping them of identifying information, if feasible. Also, be aware of situations where confidentiality could inadvertently be breached, such as having confidential conversations in a room that's not soundproof or putting participants' names on bills paid by accounting departments.

Responsible Mentoring

Help to educate, mentor, and advise others if you are capable to do so. Promote your advisees' welfare and allow them to make their own decisions.

Respect for colleagues

Respect your colleagues and treat them fairly.

Social Responsibility

Strive to promote social good and prevent or mitigate social harms through research, public education, and advocacy.

Non-Discrimination

Avoid discrimination on the basis of sex, race, ethnicity, or other factors that are not related to their scientific competence and integrity.

Competence

Maintain and improve your own professional competence and expertise through lifelong education and learning; take steps to promote competence in science as a whole.

Legality

Know and obey relevant laws and institutional and governmental policies.

Animal Care

Show proper respect and care for animals when using them in research. Do not conduct unnecessary or poorly designed animal experiments.

Human Subjects Protection

When conducting research on human subjects minimize harms and risks and maximize benefits; respect human dignity, privacy, and autonomy; take special precautions with vulnerable populations; and strive to distribute the benefits and burdens of research fairly.

* Adapted from Shamoo A and Resnik D. 2003. *Responsible Conduct of Research* (New York: Oxford University Press).

Academicians were often cautious about airing the ethical dilemmas they faced in their research and academic work, but that environment is changing today. Researchers are more likely to seek out the advice of their colleagues on issues ranging from supervising graduate students to how to handle sensitive research data.

Indeed, researchers face an array of ethical requirements: They must meet professional ethics to conduct research with human participants.

One of the best ways researchers can avoid and resolve ethical dilemmas is to know both what their ethical obligations are and what resources are available to them.

● Conclusion

There are several reasons why it is important to adhere to ethical norms in research. First, some of these norms promote the aims of research, such as knowledge, truth, and avoidance of error. Second, since research often involves a great deal of cooperation and coordination among many different people in different disciplines and institutions, many of these ethical standards promote the values that are essential to collaborative work, such as trust, accountability, mutual respect, and fairness; and to ensure that researchers are accountable to the public. Moreover, most researchers want to receive credit for their contributions and do not want to have their ideas stolen or disclosed prematurely.

Many of the norms of research promote a variety of other important moral and social values, such as social responsibility, human rights, animal welfare, compliance with the law, and health and safety.

As far as possible, a researcher (you) should avoid the use of deception and should try to avoid causing discomfort. Human participants have the right to be told the purpose and outcome of the studies.

In general, a writer should respect the following ethical principles in report writing:

Honesty, objectivity, integrity, carefulness, respect for intellectual property, and confidentiality.

✓ Chapter Checklist

Put a ‘ ’ mark against each of the following statements that you are able to answer.

I can:-

- Explain the importance of ethics in research and report writing.

- Identify some ethical considerations.
- List the ethical principles relevant to research and report writing.

◆ Self-Test

Part one: write a complete answer for the following questions.

1. Why is ethics important in research and report writing? Write at least three reasons.

2. List down some ethical principles.

Part two: Match words and phrases under column "A" to expressions under column "B".

column "A" column "B"

1. Human Subjects Protection a) Protect secret in communications

2. Animal Care b) act with sincerity; strive for consistency

3. Legality c) Strive to avoid bias

4. Responsible Mentoring d) Honor patents, copyrights

5. Confidentiality e) respect human dignity, privacy, and autonomy

6. Respect for Intellectual Property f) Help to educate and advise others

7. Openness g) Do not fabricate, falsify, or misrepresent data

8. Integrity h) Be open to criticism and new ideas

9.Objectivity i) poorly designed animal experiments

10. Honesty j) institutional and governmental policies

✓ Answers to self test exercises

CHAPTER 1

◆ Self-Test

1. Research is a systematic approach to find/ seek solution to a problem.

2. To disclose fundamental truth or laws which leads to new knowledge.

3

- training.
- solving problem and
- generating truth or knowledge

4. Fundamental Research and Applied research

5. Fundamental Research**:** This type of research aims at obtaining the empirical data that can be used to formulate and /or evaluate a theory.

Applied research is directed towards the solution of an immediate, specific and practical problem.

6. A primary source is the original writings of a person whose thinking, investigation, discovery, life, or imagination is the basis for the material, for example, research reports, philosophical discourses, autobiography, fiction, poetry, analysis of an issue.

7. A **secondary source** is a work written about someone else's thinking, investigation, discovery, life, or imagination, usually a response to someone else's writing, for example, analysis of a poem, interpretation or explanation of a research project by someone other than the researcher, biography, literary criticism, book review

8. The primary purpose of the scientific method is to develop a mechanism of organizing the facts, as they accumulate, and become meaningful from the standpoint of their objectives

9. Positivism is — the original answer key is incomplete here; consult the chapter discussion.

10. The term **argument** can be defined as "a series of generalizations or propositions, supported by evidence or reasoning and connected in a logical manner, that lead to a justified conclusion.

Part Two

1. True 2. True 3. False 4. True 5. False

CHAPTER 2

◆ Self-Test

1.

a) Gathering materials: selecting a topic, background reading, preparation of tentative bibliography, preparation of tentative outline.

b) Writing the report: Think, Plan and write

2. Interest, Affordability, Cost, etc.

3. documents, records and literature of the field

4.. Library search can help in at least two broad ways:

c) To find out if the information which is the object of the proposed research is already available.

d) To acquire a broad general background in the area of research.

5. Professional experience of a research,Contacts and discussions with research oriented people and generalizations that are drawn from various theories.

6. A hypothesis is a testable statement.

7.

- . Select a topic that is of interest to you.
- Select a topic for which you have some background knowledge.
- Select a topic for which you will be able to find references.
- Understand the Audience
- Cost

8. It is a statement about one or two measures of the population under study.

9. The functions of the literature review is to prepare the reader to understand the purpose of the proposed research and within that function; the researcher use a literature review that persuades the readers that there is a need for the proposed research to be carried out.

10.Writing a broad topic.

Part Two

1. True 2. True 3. True 4. True 5. True

CHAPTER 3

◆ Self-Test

1. Research design is the overall plan that connects the research question with the evidence to be collected and the procedures for analyzing it.

2. Research design among other things includes the following important issues:

- Variables used
- Population and selection of sample
- Tools used for data gathering
- Data gathering procedures
- Methods of data analysis:- Quantitative, qualitative, etc.

3. Qualitative and quantitative approaches are two broad approaches to research. Qualitative research commonly explores meanings, experiences, and processes; quantitative research commonly measures variables and analyzes numerical data.

- The aim of **qualitative** analysis is a complete, detailed description.
- Recommended during earlier phases of research projects.
- Researcher may only know roughly in advance what he/she is looking for.
- The design emerges as the study unfolds.
- Researcher is the data gathering instrument.
- Data is in the form of words, pictures or objects.
- Qualitative data is more 'rich', time consuming, and less able to be generalized.
- Researcher tends to become subjectively immersed in the subject matter.

Quantitative

- In **quantitative** research we classify features, count them, and construct statistical models in an attempt to explain what is observed.
- Recommended during latter phases of research projects.
- Researcher knows clearly in advance what he/she is looking for.
- All aspects of the study are carefully designed before data is collected.
- Researcher uses tools, such as questionnaires or equipment to collect numerical data.
- Data is in the form of numbers and statistics.
- Quantitative data is more efficient, able to test hypotheses, but may miss contextual detail.
- Researcher tends to remain objectively separated from the subject matter.

4.

- By collecting reliable data
- Through a constant justification of your interpretation of the research data

5. Qualitative studies are tools used in understanding and describing the world of human experience. The ultimate aim of qualitative research is to offer a perspective of a situation and provide well-written research reports that reflect the researcher's ability to illustrate or describe the corresponding phenomenon.

6. Quantitative research generally converts observations into discrete units that can be compared to other units by statistical analysis

Part two

1. True 2. True 3. False 4. True 5. False

6. True 7. True 8. False

CHAPTER 4

Self-Test

1. Sampling is the act, process, or technique of selecting a suitable population, or a representative part of a population for the purpose of determining parameters or characteristics of the whole population.

2. The purpose of sampling in determining parameter is to save economy, time and to reach at the large size of many populations by reducing the problems of inaccessibility to some of the population so that we can collect reliable data.

3. The two basic sampling errors are: Chance and sampling biases

4. Data collection is the process of gathering and measuring information on variables of interest, in an established systematic way.

5. Does the information pertain to your topic? Does it provide the type of information you need? Titles can sometimes obscure the contents of a chapter or book. Does the resource provide information at the right level of detail? Is it too general, too technical, or too theoretical? Is the reference a primary source or a secondary source?

6. Summarizing, paraphrasing and quotation.

7. It is maintaining the integrity of research

8. Quality assurance and quality control.

9. Data analysis involves organizing, examining, summarizing, and interpreting collected data in relation to the research questions. The procedures differ for quantitative and qualitative data.

10. Some ways of data analysis are: triangulation, compare and contrast, categorize and sort, etc.

Part two

1. True 2. False 3. False 4. True 5. False

CHAPTER 5

◆ Self-Test

1. An academic report is a structured account prepared for an academic purpose. Its organization and content depend on the assignment, discipline, and intended readers; it should present relevant evidence and reasoned conclusions.

2 The purpose of the report and audiences.

3. Writing from general to specific, chronological, cause-effect, familiar to unfamiliar, comparison and contrast.

4. Introduction, main body and conclusion.

5. A research proposal is a written plan for a study.

6. A research proposal describes a study that is planned, including its problem, purpose, literature, proposed methods, and relevant ethical and practical arrangements. A research report describes a study that has been conducted, including its methods, findings, discussion, and conclusions.

7. completeness,correctness, clarity and conciseness

8. Stage one: Framing the issues and planning

Stage two: gathering information

Stage three: analyzing the information

Stage four: Outlining the report

Stage five: Writing the rough draft

Stage six: Revising rough draft

Stage seven: writing the final Report

9.

- faulty grammar;
- clusters of nouns and adjectives modifying a noun and conversely strings of prepositional phrases after a noun;
- use of abstract nouns instead of action verbs
- parallel construction of words, phrases, and sentences in enumerations; and
- more complicated phrasings than required Carefully review your draft to make sure you have avoided these common faults.

10. A summary of the most significant results/findings

Part two

1. False 2. True 3. True 4. True 5. False

CHAPTER 6

◆ Self-Test

1. To acknowledge the sources and not to forgot to write down the Author (s), year, page number of a key quotation.

2. Ordering, Capitalization and punctuation.

The following are model formats. The bracketed details are placeholders and must be replaced with verified information from a real source; they are not completed references.

3. Unpublished dissertation: Author, A. A. (Year). Title of dissertation [Unpublished doctoral dissertation]. Name of Institution.

4. Professional website: Author or Organization. (Year, Month Day). Title of webpage. Site Name, if different from the author. URL.

5. Conference presentation: Presenter, A. A. (Year, Month Day). Title of presentation [Paper presentation]. Name of Conference, Location. URL, if available.

6. Newspaper article: Author, A. A. (Year, Month Day). Title of article. Name of Newspaper. URL.

7. Unpublished thesis: Author, A. A. (Year). Title of thesis [Unpublished master's thesis]. Name of Institution.

8. Chapter in an edited book: Author, A. A. (Year). Title of chapter. In E. E. Editor (Ed.), Title of book (pp. xx–xx). Publisher.

9. Encyclopedia entry: Author, A. A. (Year). Title of entry. In E. E. Editor (Ed.), Title of encyclopedia (edition, volume, pages). Publisher.

10. Nonrecoverable source: cite a personal communication in the text only, for example (A. A. Author, personal communication, Month Day, Year); do not add it to the reference list.

11. Three or more authors: use the first author's surname followed by et al. from the first citation, for example (Author et al., Year).

12. Two authors: use "and" in a narrative citation, Author and Author (Year), and "&" in a parenthetical citation, (Author & Author, Year).

13. Audio recording: Creator, A. A. (Role). (Year). Title [Audio recording]. Publisher or Platform. URL.

14. Television episode or broadcast: Writer, A. A. (Writer), & Director, B. B. (Director). (Year, Month Day). Title of episode [TV series episode]. In Producer, C. C. (Executive Producer), Title of series. Production Company.

15. Order, capitalization, and punctuation of entries on the reference page.

CHAPTER 7

◆ Self-Test

Part one:

1

- To protect participants and respect their rights, privacy, dignity, and welfare.
- To promote the aims of research, including the pursuit of knowledge and truth and the avoidance of error.
- To support trust, fairness, accountability, and responsible collaboration among researchers.

2. Examples include honesty, objectivity, integrity, carefulness, openness, respect for intellectual property, confidentiality, responsible mentoring, social responsibility, legality, and protection of human and animal participants where relevant.

Part Two:

1 e 2 i 3 j 4 f 5 a

6 d 7 h 8 b 9 c 10 g

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